QUALIFICATION STANDARD AND PLAN (QSP)

Naval Combat Systems Engineering (NCS ENG) Basic Occupation Qualification
NQual (AIRX)

MITE ID
Ashore Phase: 109385
Afloat Phase: 109386
Qualification Board: 109387

Regular Force: Naval Combat Systems Engineer 00344
Training Authority (TA): Commander Naval Personnel and Training Group (CNPTG)
Curriculum Control Authority (CCA): HMCS Venture

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RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0 QSPRB	Initial QSP	17 Jun 13	<<<O.S.B.>>> Cdr D.J. Benoit CFNES Cmdt
1.1	N/A	Sept 18	CO NTDC(P) Cdr A. Sauvé
1.2	N/A	Feb 20	CO NTDC(P) Cdr A. Sauvé
2.0	Change QSP to new template	27 Jul 23	CO HMCS Venture

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Personnel and Training Group (CNPTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification;

   c. A-PD-055-002/PP-002 Non-Commissioned Member General Specification;

   d. A-PD-055-002/PP-001 NCS Eng Job Based Occupational Specification; and

   e. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

3. This QSP publication is effective on receipt and supersedes QSP revision 1.2, dated Feb 2020 (RDIMS #650234). This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 2003 at CFNES Halifax working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened 2004 at CFNES Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for personnel to perform the duties of a Naval Combat Systems Engineering (NCS ENG) Officer Basic Occupation Qualification (BOQ).

GRANTING OF NQUAL

2. A trainee will achieve the qualification NCS ENG Basic Occupation Qualification (MITE 109385, 109386, 109387, NQUAL AIRY) when the standards listed in the Performance Objectives (POs) have been met and PC 001 has been successfully passed.

CO-QUALIFICATION

3. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. Naval Engineering Common (Ashore 119913 / Afloat 119912);

b. Officer of the Day qualification – Ashore (108364);

c. Officer of the Day qualification – Afloat (109030);

d. Defence Resource Management Information System (DRMIS) (119096,119097, 119098)

e. Maritime Warfare Basic (108163);

f. A Qualification Board (PC 001), see Annex I,

g. Designated TEs incorporating material from both the NEC and NCS Eng OJPRs (109387).

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. Possess an acceptable Baccalaureate Degree in Science or Engineering IAW CFAO 6-1 if they are a Regular Officer Training Plan (ROTP), Direct Entry Officer (DEO), University Training Plan for Non-Commissioned Members (UTPNCM), or Special Commissioning Plan (SCP) entry officer;

b. Have completed or have been waived the applicable Academic Bridging course if they are a Commissioned From the Ranks Program (CFRP) entry officers; and

c. Have completed or have been waived the Naval Warfare Officer II course (119728); and

d. Have completed or have been waived the Naval Engineering Indoctrination (NEI) (109286).

CHAPTER 2 - PERFORMANCE OBJECTIVES

PO 001- MANAGE NAVAL WEAPON SYSTEMS

1. Performance: Manage Naval Weapon Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NCS ENG shall manage Naval Weapon Systems:

   a. IAW Refs, by:

      (1) Describing Control Theory;

      (2) Describing Drive Systems;

      (3) Describing Naval Gun Systems;

      (4) Describing Naval Missile Systems;

      (5) Describing Underwater Weapons Systems; and

      (6) Supporting Naval Weapons Systems onboard HMC Ships.

4. Reference Codes:

   a. A17, A18, A19. A20, A23, A24, A28, A36, A37, A38, A41, A42, A45, A46,A47, A50, A51, A52, A56, A166, A169, A170, A171, A172, A173, A174,A175, A176, A177, A178, A179, A180, A181, A182, A183, A193, A368, A369, A370, A374, A381, A382, A383, A384, A385, A386, A387, A388, A389, A390, A391, A392, A393, A394, A395, A396, A397, A398,

A399, A400, A401, A402, A403, A404, A405, A406, A407, A408, A409, A410, A411, A412, A413, C8, C14, and C15.

5. Task Numbers:

6. Remarks: Nil.

PO 002 - MANAGE NAVIGATION SYSTEMS

1. Performance: Manage Navigation Systems

2. Conditions:

    a. Given:

        (1) Operational Ship;

        (2) References; and

        (3) Engineering and Operational Staff Support.

    b. Denied: N/A

    c. Environment:

        (1) Ashore; and

        (2) Afloat.

3. Standard: The NCS ENG shall manage Navigation Systems:

    a. IAW Refs, by:

        (1) Describing Inertial Navigation Systems;

        (2) Describing Global Positioning Systems;

        (3) Describing Speed Logs;

        (4) Describing Navigation Distribution Systems;

        (5) Describing Electronic Chart Systems;

        (6) Describing Echo Sounders;

        (7) Describing the Stabilized Horizon Reference System;

        (8) Describing the Automatic Identification System;

        (9) Describing the Meteorological System; and

        (10) Supporting Navigation Systems onboard HMC Ships.

4. Reference Codes:

a. A111, A112, A113, A114, A115, A116, A117, A118, A119, A120, A121, A122, A123, A124, A125, A126, A127, A189, A190, A191, A192, A278, A279, A280, A281, A282, A283, A284, A285, A286, A259, A288, A289, A290, A291, A292, A293, A351, A352, A353, A354, and A355.

5. Task Numbers:

6. Remarks: Nil.

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

1. Performance: Manage Communications Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NCS ENG shall manage Communications Systems:

   a. IAW Refs, by:

      (1) Describing Naval Communication Systems;

      (2) Describing Internal Communication Systems;

      (3) Describing External Communication Systems;

      (4) Describing Satellite Communication System;

      (5) Describing Communication Control and Management System;

      (6) Describing the Message Handling Distribution System; and

      (7) Supporting Communications Systems onboard HMC Ships.

4. Reference Codes:

   a. A84, A85, A86, A87, A88, A89, A90, A91, A92, A93, A94, A95, A96, A97, A98, A99, A100, A101, A102, A103, A104, A105, A106, A107, A108, A109, A110, A152, A153, A184, A185, A186, A187, A188, A211, A212, A213, A214, A215, A216, A217, A218, A219, A220, A221, A222, A223, A224, A225, A226, A227, A228, A229, A230, A231, A232, A233, A234, A235, A236, A237, A238, A239, A240, A241, A242, A243, A244, A245,

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5. Task Numbers:

6. Remarks: Nil.

PO 004 - MANAGE ABOVE WATER SURVEILLANCE SYSTEMS

1. Performance: Manage Above Water Surveillance Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NCS ENG shall manage Naval Above Water Surveillance Systems:

   a. IAW Refs, by:

      (1) Describing Electromagnetic Theory;

      (2) Describing Antenna Systems Theory;

      (3) Describing Radar Systems Theory;

      (4) Describing Applications of RADAR Systems;

      (5) Describing IFF Systems;

      (6) Describing Electro-Optics/Infra-Red Systems;

      (7) Describing Naval Fire Control Systems; and

      (8) Supporting Naval Above Water Surveillance Systems onboard HMC Ships.

4. Reference Codes:

   a. A14, A15, A32, A33, A35, A37, A38, A49, A51, A55, A87, A107, A114, A115, A130, A141,A142, A143, A144, A145, A146, A147, A148, A149,

A150, A151, A152, A153, A157, A158, A159, A215, A219, A220, A221, A224, A225, A229, A231, A255, A278, A279, A299, A304, A305, A313, A314, A315, A317, A318, A319, A320, A321, A322, A324, A325, A326, A327, A328, A329, A330, A331, A332, A333, A334, A335, A336, A337, A338, A339, A340, A341, A342, A343, A344, A345, A346, A347, A349, A348, A351, A352, A355, A356, A362, A364, A385, A425, A428, A430, C8, C9, C11, and C16.

5. Task Numbers:

6. Remarks: Nil.

PO 005 - MANAGE UNDER WATER SENSOR SYSTEMS

1. Performance: Manage Under Water Surveillance Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NCS ENG shall manage Under Water Sensor Systems:

   a. IAW Refs, by:

      (1) Describing Digital Signal Processing;

      (2) Describing Sonar Theory;

      (3) Describing the Applications of Sonar Systems;

      (4) Describing Acoustic Countermeasure Systems; and

      (5) Supporting Underwater Sensor Systems onboard HMC Ships.

4. Reference Codes:

   a. A80, A116, A117, A121, A122, A123, A154, A155, A163, A164, A165, A167, A168, A189, A276, A282, A294, A312, A350, A366, A367, A371, A372, A373, A374, A376, A378, C6, C7, and C8.

5. Task Numbers:

6. Remarks: Nil.

PO 006 - MANAGE COMMAND AND CONTROL SYSTEMS

1. Performance: Manage Command and Control Systems
2. Conditions:
    a. Given:
        (1) Operational Ship;
        (2) References; and
        (3) Engineering and Operational Staff Support.
    b. Denied: N/A
    c. Environment:
        (1) Ashore; and
        (2) Afloat.
3. Standard: The NCS ENG shall manage Command and Control Systems:
    a. IAW Refs, by:
        (1) Describing Command and Control System Theory;
        (2) Describing In-Service Command and Control System Hardware;
        (3) Describing CCS Software;
        (4) Describing CCS Operations; and
        (5) Supporting CCS onboard HMC Ships.
4. Reference Codes:
    a. A446, A447, and A448.
5. Task Numbers:
6. Remarks: Nil.

PO 007 - MANAGE DATA PROCESSING SYSTEMS

1. Performance: Manage Data Processing Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NCS ENG shall manage Data Processing Systems:

   a. IAW Refs, by:

      (1) Describing Analog Electronics Theory;

      (2) Describing Digital Electronic Theory;

      (3) Describing Computer Networks System Theory;

      (4) Describing Tactical Data Link Systems; and

      (5) Supporting Data Processing Systems onboard HMC Ships.

4. Reference Codes:

   a. A128, A129, A130, A131, A132, A133, A134, A135, A136, A137, A138, A139, A140, A193, A295, A296, A297, A299, A300, A301, A302, A303, A304, A305, A306, A444, A445, C5, C8, C12, C13, and C16.

5. Task Numbers:

6. Remarks: Nil.

PO 008 - MANAGE ELECTRONIC WARFORE SYSTEMS

1. Performance: Manage Electronic Warfare Systems

2. Conditions:

    a. Given:

        (1) Operational Ship;

        (2) References; and

        (3) Engineering and Operational Staff Support.

    b. Denied: N/A

    c. Environment:

        (1) Ashore; and

        (2) Afloat.

3. Standard: The NCS ENG shall manage EW Systems:

    a. IAW Refs, by:

        (1) Describing EW Fundamentals;

        (2) Describing ESM Systems;

        (3) Describing EA Systems; and

        (4) Supporting EW Systems onboard HMC Ships.

4. Reference Codes:

    a. A128, A129, A130, A131, A132, A133, A134, A135, A136, A137, A138, A139, A140, A193, A295, A296, A297, A299, A300, A301, A302, A303, A304, A305, A306, and C8.

5. Task Numbers:

6. Remarks: Nil.

PO 009 - MANAGE AMMUNITION

1. Performance: Manage Ammunition

2. Conditions:

    a. Given:

        (1) Operational Ship;

        (2) References; and

        (3) Engineering and Operational Staff Support.

    b. Denied: N/A

    c. Environment:

        (1) Ashore; and

        (2) Afloat.

3. Standard: The NCS ENG shall manage Ammunition:

    a. IAW Refs, by:

        (1) Describing Explosive Theory;

        (2) Describing In-Service Naval Ammunition;

        (3) Describing Ammunition Accounting;

        (4) Describing the Ammunition Organization;

        (5) Describing the Ammunition and Explosive Safety Program; and

        (6) Participating in Ammunition Operations onboard HMC Ships.

4. Reference Codes:

    a. A17, A18, A19, A20, A24, A36, A37, A38, A41, A42, A45, A46, A47, A50, A51, and A52.

5. Task Numbers:

6. Remarks: Nil.

PO 010 - MANAGE SHIP'S SIGNATURE

1. Performance: Manage Ship's Signature
2. Conditions:
    a. Given:
        (1) Operational Ship;
        (2) References; and
        (3) Engineering and Operational Staff Support.
    b. Denied: N/A
    c. Environment:
        (1) Ashore; and
        (2) Afloat.
3. Standard: The NCS ENG shall manage Ship's Signature:
    a. IAW Refs, by:
        (1) Describing Ship's Signature Management; and
        (2) Supporting Ship's Signature Management onboard HMC Ships.
4. Reference Codes:
    a. A14, A15, A20, A29, A38, and A42.
5. Task Numbers:
6. Remarks: Nil.

PO 011 - MANAGE SHIPBOARD HAZARDS

1. Performance: Manage Shipboard Hazards

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NCS ENG shall manage Shipboard Hazards:

   a. IAW Refs, by:

      (1) Describing the Shipboard Risk Management Programs;

      (2) Describing the Shipboard Non-Ionizing Radiation Hazard Safety Program;

      (3) Describing Hazardous Material Clean-up Procedures; and

      (4) Participating in Risk Management Programs onboard HMC Ships.

4. Reference Codes:

   a. A13, A14, A15, A17, A38, A52, A55, A57, A58, A59, A60, A61, A62, A63, A73, A74, A75, A76, A77, A78, B01, B02, B04, B05, B06, B07, and B09.

5. Task Numbers:

6. Remarks: Nil.

PO 012 - DIRECT EMERGENCY RESPOPNSE TEAM (ERT) OPERATIONS

1. Performance: Direct Emergency Response Team (ERT) Operations

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NCS ENG shall manage Direct Emergency Response Team (ERT) Operations:

   a. IAW Refs, by:

      (1) Describing ERT Operations; and

      (2) Participating in ERT Operations onboard HMC Ships.

4. Reference Codes:

   a. A13, A14, A15, A17, A18, A20, A38, A52, A55, A57, A58, A59, A60, A61, A62, A63, A73, A74, A75, A76, A77, A78, B01, B02, B04, B05, B06, B07, B08, and B09.

5. Task Numbers:

6. Remarks: Nil.

TRAINING PLAN

CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Naval Combat Systems Engineering Application Course (NCSEAC), which is the ashore phase of training. NCSEAC will be delivered through a series of lectures, labs and equipment tours; and

   b. The afloat phase of training which require self-study and mentorship from the CSE department of the trainee's ship.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 115.5 training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
Lt(N)/LCdr	00344	1. A Baccalaureate Degree in Science or Engineering; 2. NCS ENG HOD qualified; 3. Experience in the subject matter as identified in training limitations below	5	
Civilian	N/A	1. A Baccalaureate Degree in Science or Engineering; 2. Experience in the subject matter as identified in training limitations below.	2	

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

EO#	Title	Required Qualification
004.01	Describe Electromagnetic Theory	Bachelor of Engineering (Electrical/Computer)
005.01	Describe Digital Signal Processing (DSP) Theory	Bachelor of Engineering (Electrical/Computer l) and background in DSP; or, Civilian/Military

		with related post graduate or specialized DSP training
007.01	Describe Analog Electronics Theory	Bachelor of Engineering (Electrical/Computer)
007.02	Describe Digital Electronics Theory	Bachelor of Engineering (Electrical/Computer)
007.03	Describe Computer Networks Theory	Bachelor of Engineering (Computer) or equivalent experience

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 15;

b. Maximum Load – 24; and

c. Minimum – 6.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager for career courses; or

b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. In order to achieve a passing mark for the NCS ENG Academic course (NCSEAC), trainees must pass all modules with a minimum mark of 60%. Also, trainees must achieve a minimum mark of 50% in all Critical ECs (CECs). The weighting of the marks will be calculated as 60% for CECs, 20% for formative work, 20% Written test (different than the one in the CECs) Where there are no ECs as written tests in the module, the marking will be weighted 70% for the CEC and 30% for formative work.

10. The final mark for each module is calculated using the following formulas:

a. PO 001: 30% for CEC 001 + 30% for CEC 002 + 20% for ECs written test + 20% for formative work;

b. PO 002: 70% for CEC 003 + 30% for formative work;

c. PO 003: 60% for CEC 004 + 20% for ECs written test + 20% for formative work;

d. PO 004: 30% for CEC 005 + 30% for CEC 006 + 20% for ECs written test + 20% for formative work;

e. PO 005: PO 005: 60% for CEC 007 + 20% for ECs written test + 20% for formative;

f. PO 006: 70% for CEC 008 + 30% for formative work;

g. PO 007: 15% for CEC 009 + 15% for CEC 10 + 15% for CEC 11 + 15% for CEC 12 + 20% for ECs written test + 20% for formative work;

h. PO 008: 70% for CEC 013 + 30% for formative work;

i. PO 009: 70% for CEC 014 + 30% for formative work;

j. PO 010: No CEC; assessed as part of OJPR;

k. PO 011: No CEC; assessed as part of OJPR; and

l. PO 012: No CEC; assessed as part of OJPR.

11. The final overall course mark is an average of all module marks. Full details of the breakdown of the marking scheme for each PO can be found in Annex H of the QSP.

12. The marking scheme for NCSEAC, including Annex H – Assessment Plan, shall be provided to trainees prior to the start of the NCSEAC.

13. During the ashore phase, any trainee who fails a CEC shall be given a supplemental. IAW Ref NPTORD 4500-54 – Training Review Board Process, a TRB shall be convened whenever a student fails a supplemental CEC or at any time a trainee fails a third CEC during the entire NCSEAC. A TRB may also be convened for other reasons if deemed necessary IAW NPTORD 4500-54 – Training Review Board Process.

14. Upon completion of the afloat phase (Annex J) and IAW NAVORD 4500-20, a Qualification Board (PC 001) will be convened for each NCS ENG trainee. Board examinable subjects are based on the NEC QSP, MITE 119912 and NCS ENG BOQ

QSPs. To be successful, the candidate shall achieve a minimum overall average mark of 60%, including a minimum of 60% for each of questions 1 and 2 (critical questions). Questions will be marked IAW the marking guide in Annex I – Board Assessment Guide. Details of the Qualification Board can be found in Annex I. Upon successful completion of the Qualification Board, the NCS ENG trainee will be awarded the NCS ENG occupation qualification (AIRX). Any trainee who fails PC 001 shall be given a supplemental PC. If the trainee fails the supplemental PC, a TRB shall be convened IAW NPTORD 4500-54 – Training Review Board Process.

15. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-51. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

16. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

17. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

18. There are no additional requirements to maintain currency in this qualification; or Currency.

LANGUAGE OF INSTRUCTION

19. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

20. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

21. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

22. Formal Course: A Canadian Forces Course Report, Form CF 377 upon completion of the ashore phase and CF289 Certificate of Achievement upon successful completion of Qualification Board, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

TRAINING ESTABLISHMENTS

23. The designated TE(s) for this training can be found in annex N.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – MANAGE NAVAL WEAPON SYSTEMS

EO 001.01 DESCRIBE CONTROL THEORY

1. Performance: Describe Control Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Control theory:

   a. IAW Refs, by completing the following:

      (1) Describe Control Systems;

      (2) Explain Terminology and Concepts;

      (3) Describe Feedback Mechanisms;

      (4) Describing Naval Missile Systems;

      (5) Describe Components of Control Systems; and

      (6) Describe Applications of Control Systems

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Control Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
1.1	Control System Definition			
1.2	Open-Loop Control			
1.3	Closed-Loop Control			
1.4	Advanced Closed Loop Control			
2	**Explain Terminology and Concepts**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	

Teaching Points		Instructional Strategies	Time (min)	References
2.1	Terminology			
2.2	Automatic Control			
2.3	Block Diagrams and transfer Functions			
2.4	System Linearity			
2.5	Load Changes			
2.6	Damping and Instability			
2.7	Objectives of Good Control			
2.8	Criteria of Good Control			
3	**Describe Feedback Mechanisms**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
3.1	Mathematics of Feedback			
3.2	Effect on Overall Gain			
3.3	Effects on System stability			
3.4	Effects on System Sensitivities			
3.5	Effects on I/P and O/P Impedance			
3.6	Effects on System Bandwidth			
4	**Describe Components of Control Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
4.1	Types of Sensors			
4.2	Types of Actuators			
5	**Describe Applications of Control Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
5.1	Synchros and Servos			
5.2	Control Systems Tests			

5. Total EO Time: Total: 21 x 50 minute periods as follows:

   a. 21 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Control Systems Lab.

7. Remarks: Nil.

PO 001 – MANAGE NAVAL WEAPON SYSTEMS

EO 001.02 DESCRIBE DRIVE SYSTEMS

1. Performance: Describe drive systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Drive Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Electrical Drive Theory;

      (2) Describe Hydraulic Drive Theory;

      (3) Describe Pneumatic Drive Theory; and

      (4) Describe Mechanical Principles;

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Electrical Drive Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**500**	
1.1	Pros/Cons			
1.2	Principles of Electric Motors			
1.3	Electrical Motors			
1.4	Types			
2	**Describe Hydraulic Drive Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**500**	
2.1	Fluid dynamics; 1. Bernoulli's Equation.			
2.2	Generic Hydraulic Systems			
2.3	Components; 1. Pumps;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Actuators; i. Pistons; and ii. Motors. 3. Pipes/hoses/tubes; 4. Fittings; 5. Valves (main types); 6. Reservoirs; and 7. Cooling.			
2.4	Symbology			
3	**Describe Pneumatic Drive Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**500**	
3.1	Difference between Hydraulic and pneumatics drive systems			
3.2	HP and LP air			
4	**Describe Mechanical Principles**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**500**	
4.1	Types/Purposes; 1. Ratios; and 2. Wear.			
4.2	Bearings; 1. Types/Purposes; and 2. Wear.			
4.3	Fluids and lubricants; 1. Types/Purposes.			
4.4	Corrosion; 1. Principles (voltage cell); 2. Types/Impact; and 3. Prevention.			
4.5	Ship's Stability			

5. Total EO Time: Total: 40 x 50 minute periods as follows:

   a. 40 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Control Systems Lab; and

   b. Hydraulic Systems Lab.

7. Remarks: Nil

PO 001 – MANAGE NAVAL WEAPON SYSTEMS

EO 001.03 DESCRIBE NAVAL GUN SYSTEMS

1. Performance: Describe Naval Gun Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Naval Gun Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Naval Gun Systems; and

      (2) Describe In-Service Naval Gun Systems

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Naval Gun Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**900**	
1.1	Historical Background			
1.2	Theory and Design Principles; 1. Layer Defence; i. Point Defence; and ii. Naval Gun Support (NGS). 2. Purpose (e.g. Hard skills) 3. Gyroscopic Principles; 4. Gun Firing sequence (Detect to engage); 5. Reference Co-ordinates; and 6. Barrel/Round Physical properties; i. Rifling ii. Calibre and dimension;			

Teaching Points		Instructional Strategies	Time (min)	References
	iii. Construction; and iv. Recoil.			
1.3	Types(advantages/disadvantages/ comparison); 1. Auto/Semi-Auto/Manual; 2. Air/Surface; 3. Calibre; and 4. Manned/Unmanned.			
1.4	Performance characteristics; 1. Range; 2. Rate of Fire; 3. PK (Probability of Kill); 4. Round Capacity; 5. Loading/Reloading Time; 6. Muzzle Velocity; 7. Accuracy; 8. Calibre; 9. Slew Rate; 10. Arcs of Fire; and 11. Performance Measurements; i. Missed Distance Indicator (MDI); ii. Continuous Aim Correction (CAC); and iii. Spot Correction.			
1.5	System components; 1. Basic Block Diagram; 2. Barrel; 3. Breech; 4. Cupola; 5. Feed System; 6. Loading System; 7. Cooling System; 8. Munitions; 9. Control System; and i. Elevating System; and ii. Laying System 10. Mounting Arrangement.			
1.6	Ammunition; 1. Types; i. Self-Propelled; ii. Ballistic; and			

Teaching Points		Instructional Strategies	Time (min)	References
	iii. Clearing Charge 2. Cartridge/Shell 3. Propellants & Primer (compositions & construction); 4. Fusing; i. Impact; ii. Time; iii. Proximity; and a) RF; b) Pressure; c) Magnetic; and d) Wiping. iv. Safe and Arming: a) Warheads; b) Blast; c) Fragmenting/non-fragmenting; d) Practice; i) Air; ii) Surface; and iii) PRAP e) Kinetic; and f) Shaped. 5. Sequence of Events/Explosive Train.			
1.7	Interfacing/Integration with other systems; 1. CCS/Link; 2. FCS; 3. Navigation Data Distribution System; 4. Electrical; 5. Environmental; and 6. Veto.			
1.8	Systems trials/alignments; 1. Gun to FCS alignment; 2. Datum Alignment; i. Purpose; and ii. Datum's/benchmarks. 3. Initial Velocity Test; 4. Gun Roller path test; 5. INO Inspections;			

Teaching Points		Instructional Strategies	Time (min)	References
	i. Purpose; ii. Elements; a) NDT; iii. Time based vs. rounds based, and; iv. Waivers 6. Gun dynamic tests; i. SHM; ii. Step; iii. Ramp; iv. Run-In; and v. Frequency response;			
1.9	Emerging Technologies; 1. Vertical Gun System (VGS); 2. Hypersonic Projectile; 3. SMART Munitions; and 4. Directed Energy.			
1.10	Future In-Service technologies; 1. CIWS 1B; and 2. 57mm Mk3.			
2	**Describe In-Service Naval Gun Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
2.1	Performance characteristics, capabilities and limitations for each in-service naval gun system;			
2.2	System components/signal flow/software for each in-service naval gun system; 1. System level block diagram; 2. Firing Sequence; and 3. Firing Criteria.			
2.3	Ammunition for each in-service naval gun system			
2.4	Modes of operations for each in-service naval gun system			
2.5	System redundancies for each in-service naval gun system			
2.6	System diagnostic capabilities/tools for each in-service naval gun system			

Teaching Points		Instructional Strategies	Time (min)	References
2.7	Interfacing/integration with other systems for each in-service naval gun system; 1. INS; 2. CCS; 3. VETO; 4. FCS; 5. Electrical; and 6. Environmental.			
2.8	Maintenance requirements for each in-service naval gun system;			
2.9	System trials/alignments for each in-service naval gun system			
2.10	Safety considerations for each in-service naval gun system			
2.11	Manning Requirements for each in-service naval gun system			
2.12	Operating policy and procedures for each in-service naval gun system; 1. Keys 2. Gun Operations; 3. Weapon/firing restrictions; and 4. Safety Considerations; i. Firing cut outs; ii. Pre-firing brief; and iii. Electrically or mechanically safe; 5. Misfire procedures/Hot gun; and 6. Identify NAVORD on Safety Firing			

5. Total EO Time: Total: 29 x 50 minute periods as follows:

   a. 29 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisite topics:

1) EO 001.01 – Describe Control Theory; and

2) EO 001.02 – Describe Drive Systems;

b. Related topics:

1) EO 004.07 – Describe Naval Fire Control Systems.

PO 001 – MANAGE NAVAL WEAPON SYSTEMS

EO 001.04 DESCRIBE NAVAL MISSILE SYSTEMS

1. Performance: Describe Naval Missile Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Naval Missile Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Naval Missile System;

      (2) Describe in-service Naval Missile Systems; and

      (3) Describe Standard Missile Systems;

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Naval Missile Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**850**	
1.1	Historical Background			
1.2	Theory and Design Principles; 1. Offensive/Defence; i. Layered Defence; ii. Point Defence; and iii. Area Defence. 2. Flight Phases; 3. Flight Dynamics; 4. Launching Sequence (detect to engage); 5. Guidance; 6. Propulsion; 7. Launching Systems;			

Teaching Points		Instructional Strategies	Time (min)	References
	8. Seeker Types – IR, EO, RADAR; 9. Safety and Arming; and 10. Telemetry; 11. Control of missile systems; i. Pre-Launch; ii. Launch; and iii. In-flight. 12. Missile safety (misfire, dud, restrained firing).			
1.3	Types (advantages/disadvantages/comparison); 1. Anti-Surface; 2. Anti-Air; 3. Land Attack; 4. Guidance; i. Control; ii. Homing; and iii. Pre-set; 5. Propulsion; i. Solid Fuel; ii. Liquid Fuel; iii. RAM Jet; and; iv. Turbo Jet. 6. Launcher; i. Rail; ii. Canister; iii. Vertical; iv. Platform; and v. Zero length.			
1.4	Performance characteristics; 1. Range 2. Trajectory/Flight Path; 3. Speed; 4. Manoeuvrability; 5. Accuracy; 6. Warhead; i. Type; and ii. TNT equivalency; 7. PK (Probability of Kill); i. Kill Radius 8. Rate of Fire; and 9. Firing Arcs;			

Teaching Points		Instructional Strategies	Time (min)	References
	10. Performance measurement; and ii. Kill assessment;			
1.5	System components (launcher and missile); 1. Launcher; 2. Missile; i. Guidance; ii. Warhead; iii. Structure/Body; iv. Propulsion; v. Control System; and vi. Fuse; 3. Associated Systems; i. Wire Guidance; ii. Illuminator; and iii. Simulators.			
1.6	Interfacing; 1. CCS/Link; 2. FCS; 3. Navigation System; 4. Electrical; and 5. VETO.			
1.7	System trials/alignments			
1.8	Emerging Technologies			
2	**Describe in-service Naval Missile Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow/software; 1. System Level Block Diagram; 2. Firing sequence; 3. Firing Criteria; 4. Missile types; and 5. Launcher characteristics.			
2.3	Modes of operation;			
2.4	System redundancies;			
2.5	System diagnostic capabilities/tools; 1. Test Sets/Simulators.			

Teaching Points		Instructional Strategies	Time (min)	References
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials; 1. Weapon Certification 2. MRC 3. MRI			
2.9	Safety considerations; 1. Alarms; 2. Protection Systems; and 3. HAZMAT.			
2.10	Manning requirements; and			
2.11	Operating policy and procedures; 1. Keys Organization; 2. Firing safety; and 3. Weapon Restriction/firing restrictions.			

5. Total EO Time: Total: 26 x 50 minute periods as follows:

   a. 26 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisite topics:

      1) EO 001.01 – Describe Control Theory; and

      2) EO 001.02 – Describe Drive Systems;

      3) EO 005.01 – Describe Digital Signal Processing;

   b. Related topics:

      1) EO 004.01 – Describe Electromagnetic Theory; and

      2) EO 004.07 – Describe Naval Fire Control Systems;

PO 001 – MANAGE NAVAL WEAPON SYSTEMS

EO 001.05 DESCRIBE UNDERWATER WEAPONS SYSTEMS

1. Performance: Describe Underwater Weapons Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Underwater Weapons systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Torpedo System; and

      (2) Describe in-service Torpedo Systems

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Torpedo Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**600**	
1.1	Historical background			
1.2	Theory and Design Principles; 1. Gyroscopic principles; 2. Layer Defence; 3. Search Patterns/torpedo path; 4. Under Water Explosions; 5. Propellants; 6. Hydrodynamics; 7. Magnetism; and 8. Torpedo Firing Sequence (detect to engage).			
1.3	Types (advantages/disadvantages/comparison); 1. Propulsion; i. Rocket; and			

Teaching Points		Instructional Strategies	Time (min)	References
	ii. Internal Combustion. 2. Torpedoes; i. Anti-ship; and ii. Anti-Submarine. 3. Guidance; i. Active; and ii. Passive; and iii. Wire guided. 4. Launcher; i. Surface launched; ii. Rocket launched; and iii. Air launched. 5. Mines; i. Moored; and ii. Floating/Drifting. 6. Fusing; i. Contact; and ii. Influence/proximity.			
1.4	Performance characteristics 1. Range; 2. Search Patterns and homing; 3. Manoeuvrability; 4. Speed; 5. Accuracy; 6. Search depth/operating depth/Max depth; 7. Warhead-Type, TNT equivalency; 8. PK (Probability of Kill); i. Kill Radius; ii. Firing Arcs; and iii. Performance measurements.			
1.5	System components; 1. Launcher (surface and air); and 2. Torpedo; i. Propulsion; ii. Control & guidance; iii. Warhead & fuse; and iv. Structure and Body. 3. Handling systems.			
1.6	Interfacing; 1. CCS/Link;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Electrical; 3. Environmental; 4. Navigation system; 5. VETO; and 6. HP air.			
1.7	System trials/alignments			
1.8	Emerging technologies			
1.9	Future In-Service Technologies			
2	**Describe In-Service Torpedo Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Performance characteristics, capabilities, and limitations			
2.2	System components/signal flow; 1. Firing sequence; and 2. Firing criteria.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools; 1. Torpedo Test Set.			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials; 1. Practice Torpedoes(HOTTORP, REXTORP, TM); and 2. Air Shot.			
2.9	Safety considerations; 1. Otto Fuel; 2. HP Air; and 3. Lifting appliances.			
2.10	Manning requirements; and			
2.11	Operating policy and procedures; 1. Keys Organizations; 2. Firing Safety; and 3. Weapon Restriction.			
2.12	Torpedo types			
2.13	Launcher types			

5. Total EO Time: Total: 15 x 50 minute periods as follows:

a. 15 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites:

1) EO 001.01 – Describe Control Systems; and

b. Related topics (Review):

1) EO 002.01 – Describe Inertial Navigation Systems;

2) EO 005.01 – Describe Digital Signal Processing; and

3) EO 005.02 – Describe SONAR Theory.

PO 001 – MANAGE NAVAL WEAPON SYSTEMS

EO 001.06 SUPPORT NAVAL WEAPONS SYSTEM ONBOARD HMC SHIPS

1. Performance: Support Naval Weapons Systems onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall describe Naval Weapons Systems as fitted onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard weapon systems to the satisfaction of the CSEO;

      (2) Conduct a walk-through of the major weapon systems with the senior maintainer;

      (3) Deliver a brief on the major weapon system diagrams to the satisfaction of the CSEO;

      (4) Stand eight (8) watches with the W Eng Tech (Armament); and

      (5) Submit a Technical Defect Report on a major problem with one of the fitted Naval Weapon Systems selected by the CSEO.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The NCS ENG shall, to the satisfaction of the CSEO, complete the following for the Gun System:**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Observe and participate in the following; 1. Operations; i. Pre and post firing checks; ii. Pre-fire brief; iii. Loading and unloading ammunition; and iv. Observe all Weapon Systems firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire crew). 2. Tests and Trials; i. INO (NDT for guns); ii. Initial Velocity (IV) determination; iii. Gun Dynamic Tests; iv. Weapon Systems alignment; v. Daily/Periodic System Operational Tests (DSOT/PSOT); vi. Missile Readiness Checks; and vii. Combat System Alignment Verficiation (CSAV). viii. Torpedo Readiness Checks 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
1.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
1.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
2	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for the Missile System:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Observe and participate in the following: 1. Operations; i. Pre and post firing checks; ii. Pre-fire brief; iii. Loading and unloading ammunition; and			

Teaching Points		Instructional Strategies	Time (min)	References
	iv. Observe all Weapon Systems firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire crew). 2. Tests and Trials; i. Weapon Systems alignment; ii. Daily/Periodic System Operational Tests (DSOT/PSOT); iii. Missile Readiness Checks; and iv. Combat System Alignment Tests (CSAT) v. Weapons Certification. 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
2.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all CSE related rounds; and 3. Identify MSE systems considerations.			
2.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components; 3. Modes of operation; 4. System redundancies;			

Teaching Points		Instructional Strategies	Time (min)	References
	5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
3	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for the Underwater Weapon Systems**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	Observe and participate in the following: 1. Operations: i. Pre and post firing checks ii. Pre-fire brief; iii. Loading and unloading ammunition; and iv. Observe all Weapons Systems firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire crew). 2. Tests and Trials: i. Weapons Systems certification; ii. Daily/Periodic System Operational Tests (DSOT/PSOT); and			

Teaching Points		Instructional Strategies	Time (min)	References
	iii. Torpedo Readiness Checks. 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
3.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; 3. Identify MSE systems considerations.			
3.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
4	**Present a system briefing to designated ship's staff on the major**	**Method:** OJT **Media:** N/A		

Teaching Points		Instructional Strategies	Time (min)	References
	Weapon Systems. This brief shall include:	**Environment:** Ship		
4.1	System description			
4.2	System overview			
4.3	Major system components			
4.4	Operating modes			
4.5	Capabilities and limitations			
4.6	Interfaces			
4.7	Redundancies/survivability			
4.8	Combat/CSE liaison considerations			
5	**Stand Department Watches to include the following:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5.1	Stand eight (8) watches with the W.ENG TECH (Armament)			
6	**Submit a technical report on a major problem with one of the fitted Naval Weapon Systems selected by the CSEO. (Note: Only one submission is required for PO 001) As a minimum, the report shall cover the following points:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6.1	Identification/discussion of the problem			
6.2	Operational impact of the problem			
6.3	Options to resolve the problem			
6.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.01 DESCRIBE INERTIAL NAVIGATION SYSTEMS

1. Performance: Describe Inertial Navigation Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Inertial Navigation Systems:

   a. IAW Refs, by completing the following:

      (1) Describe a Generic Inertial Navigation System; and

      (2) Describe In-service Inertial Navigation Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Inertial Navigation Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**600**	
1.1	Historical background;			
1.2	Theory and design principles; 1. Gyroscopic principles; 2. Theory of accelerometers; 3. Theory of inertial navigation; i. Filters; and ii. Periodic Update requirements. 4. Navigation principles and terminology; 5. Interferometric Theory (Ring Laser Gyro); and 6. Datum's/Benchmarks.			
1.3	Types (advantages/disadvantages/comparison);			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Gyroscopic; 2. Laser; and 3. Fibre Optic.			
1.4	Performance characteristics; 1. Accuracy; i. Positional; ii. Heading, pitch and roll; and iii. Coarse (1x) and Fine (36x) Signals. 2. Settling time; 3. Error recovery; 4. Input tolerance; and 5. Drift.			
1.5	System components; 1. Gyro and Gimbals; 2. Accelerometer; 3. Laser/mirror/prism; 4. Interferometer; 5. Amplifier; 6. Control Logic; 7. Processor; and 8. Display.			
1.6	Interfacing/integration with other systems; 1. Ship's Speed system; 2. Ship's Position system; 3. Navigation Distribution system; 4. Electrical; and 5. Environmental;			
1.7	System trials/alignments; and 1. Heading; 2. Pitch; and 3. Roll.			
1.8	Emerging Technologies			
2	**Describe In-Service Inertial Navigational Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow			

Teaching Points		Instructional Strategies	Time (min)	References
2.3	Modes of operation			
2.4	System redundancies			
2.5	System Diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems; 1. GPS; 2. Doppler Log; 3. Nav Processors/CCS; and 4. Nav Data Dist. System.			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Operating policy and procedures; 1) Log book; 2) Gyro Failure; and 3) Power Failure.			

5. Total EO Time: Total: 14.5 x 50 minute periods as follows:

   a. 14.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.02 DESCRIBE GLOBAL POSITIONING SYSTEMS

1. Performance: Describe Global Positioning Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Global Positioning Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Global Positioning Systems; and

      (2) Describe In-service GPS.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Global Positioning System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
1.1	Historical background; 1. Early radio fixing techniques (e.g. Loran-C); and 2. Alternate international implementations.			
1.2	Theory and design principles; 1. Spread Spectrum; 2. Satellites; i. Orbits; ii. Footprints; iii. Number; and iv. Coverage 3. GPS Fixing; i. Time accuracy; ii. Triangulation; and			

Teaching Points		Instructional Strategies	Time (min)	References
	iii. Position calculation. 4. GPS Signals; i. CA; ii. P-Code; iii. Y-Code; and iv. Messages. 5. Selective availability and anti-spoofing; 6. Ownership; 7. Speed calculation; 8. Heading calculation; and 9. Sources of error; i. Propagation; and ii. Differential GPS.			
1.3	Types (advantages/disadvantages/comparison); 1. Standard; 2. Differential; and 3. Space Based Augmentation.			
1.4	Performance characteristics; 1. Accuracy; i. Geometric Dilution of Precision (GDOP); ii. Positional Degrees of Precision (PDOP); iii. Horizontal Degrees of Precision (HDOP); and iv. Vertical Degrees of Precision (VDOP).			
1.5	System components/signal flow; 1. Antenna; 2. Satellites; 3. Rx/processor; 4. Display; and 5. Ground Control System NAV Processor/CCS.			
1.6	Interfacing/integration with other systems; 1. Nav Processors/CCS; 2. INS; 3. Electrical; 4. Users; and 5. Message Formatting Standard.			

Teaching Points		Instructional Strategies	Time (min)	References
1.7	Emerging Technologies			
2	**Describe In-Service GPS (Trimble, StarLink, FORCES)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**15**	
2.1	Performance characteristics			
2.2	Capabilities and limitations			

5. Total EO Time: Total: 6.6 x 50 minute periods as follows:

   a. 6.6 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.03 DESCRIBE SPEED LOGS

1. Performance: Describe Speed Logs

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Speed Logs:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Speed Logs; and

      (2) Describe In-service Speed Log.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Speed Logs**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Historical background;			
1.2	Theory and design principles; 1. Doppler Theory; and 2. Electromagnetic.			
1.3	Types (advantages/disadvantages/comparison); 1. Doppler; and 2. Electromagnetic.			
1.4	Performance characteristics; 1. Accuracy; 2. Dynamic Range; and 3. Frequency.			
1.5	System components/signal flow; 1. Sensing elements; 2. Transceiver;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Processor; 4. Digital Filter; and 5. Display/Repeats.			
1.6	Interfacing/integration with other systems; 1. Electrical; 2. Nav Data Dist. System; and 3. INS.			
1.7	System trials/alignments; 1. Calibration; and 2. Measured Mile.			
1.8	Emerging Technologies			
2	**Describe In-Service Speed Log**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**15**	
2.1	Performance characteristics; capabilities and limitations			
2.2	System components/sign flow			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Operating policy and procedures			

5. Total EO Time: Total: 1.5 x 50 minute periods as follows:

   a. 1.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.04 DESCRIBE NAVIGATION DISTRIBUTION SYSTEMS

1. Performance: Describe Navigation Distribution Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Navigation Distribution Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Navigation Distribution Systems; and

      (2) Describe In-service Navigation Distribution Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Navigation Distribution Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Historical background; 1. Repeaters; 2. Switchboards; and 3. Nav Bus (INDB).			
1.2	System Components; 1. Servers; 2. Network Switches; 3. Remote Terminal Units; 4. Control/Display Units; and 5. Isolation switches.			
1.3	Types (advantages/disadvantages); 1. Digital; 2. Analog; and			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Hybrid.			
1.4	Performance characteristics; 1. Capacity; 2. User/loading/degradation; 3. Accuracy; i. Heading; ii. Pitch; and iii. Roll.			
1.5	Interfacing/integration with other systems; 1. Weapons; 2. Sensors; 3. Nav Processor/CCS; 4. Electrical; 5. Communications; 6. Doppler Log; 7. INS; and 8. Misc.			
1.6	System trials/alignments			
1.7	Emerging Technologies			
2	**Describe In-service Navigation Data Distribution Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Performance characteristics; capabilities and limitations;			
2.2	System components/signal flow			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	System trials			
2.8	Maintenance requirements			
2.9	Operating Policy and Procedures; 1. Primary/Secondary feeds.			

5. Total EO Time: Total: 2.3 x 50 minute periods as follows:

   a. 2.3 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 001.01 – Describe Control Theory.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.05 DESCRIBE ELECTRONIC CHART SYSTEMS

1. Performance: Describe Electronic Chart Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Electronic Chart Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Electronic Chart Systems; and

      (2) Describe In-service Electronic Chart Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Electronic Chart Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Historical background			
1.2	Theory and design principles; 1. Digital Charts; i. Vector; and ii. Raster/Bitmap.			
1.3	Types (advantages/disadvantages); 1. Chart standards.			
1.4	Performance characteristics; 1. Resolution; and 2. Position accuracy.			
1.5	System Components; 1. PCs; 2. Electronic Charts; and 3. Displays			

Teaching Points		Instructional Strategies	Time (min)	References
1.6	Interfacing/Integration with other systems; 1. Nav Data Dist. System; 2. GPS; 3. DSL; 4. Nav RADAR; 5. Echo Sounder; 6. Electrical; and 7. Environmental.			
1.7	System trials/alignments			
1.8	Emerging technologies			
2	**Describe In-service Navigation Data Distribution Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow/software			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/ tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Operating policy and procedures			

5. Total EO Time: Total: 1.8 x 50 minute periods as follows:

   a. 1.8 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.06 DESCRIBE ECHO SOUNDERS

1. Performance: Describe Echo Sounders

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Echo Sounders:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Electronic Echo Sounders; and

      (2) Describe In-service Echo Sounder.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic electronic Echo Sounder**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
1.1	Historical background			
1.2	Theory and design principles; 1. Time/distance relationship; 2. Factors affecting accuracy; i. Velocity changes of sound due to depth, temperature and salinity; ii. Biomass; and iii. Bottom type.			
1.3	Types (advantages/disadvantages/comparison); 1. Multiple vs single frequency; and 2. Single vs multiple beam.			
1.4	Performance characteristics; 1. Accuracy; i. Dynamic range (min/max);			

Teaching Points		Instructional Strategies	Time (min)	References
	ii. Frequency; and iii. Output power; and iv. Breamwidth.			
1.5	System Components/signal flow; 1. Transducer; 2. Transceiver; 3. Processor; and 4. Displays/Repeats.			
1.6	Interfacing/integration with other systems; 1. Electrical; 2. Nav Processor/CCS; and 3. SHINNADS.			
1.7	System trials/alignments; and 1. Calibration/Leadline check.			
1.8	Emerging Technologies			
2	**Describe In-service Navigation Data Distribution Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Operating policy and procedures; 1. Harbour entry/exit; and 2. Navigating narrow passages.			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.07 DESCRIBE THE STABILIZED HORIZON REFERENCE SYSTEM (SHRS)

1. Performance: Describe the Stabilized Horizon Reference System (SHRS)

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe SHRS:

   a. IAW Refs, by completing the following:

      (1) Describe the Generic SHRS; and

      (2) Describe the In-service SHRS.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Stabilized Horizon Reference Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
1.1	Historical background			
1.2	Theory and design principles;			
1.3	Types (advantages/disadvantages);			
1.4	Performance characteristics;			
1.5	System Components; 1. Pitch/roll bars; 2. Control circuitry and 3. Remote indicators.			
1.6	Interfacing/integration with other systems; 1. Nav Data Dist. System (NDDS): 2. Electrical; and 3. Environmental.			

Teaching Points		Instructional Strategies	Time (min)	References
1.7	System trial/alignment			
1.8	Emerging Technologies			
2	**Describe In-service Stabilized Horizon Reference System (SHRS)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow; 1. System Block Diagram.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Operating policy and procedures;			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 001.01 – Describe Control Theory; and

      2) EO 001.02 – Describe Drive Systems.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.08 DESCRIBE AUTOMATIC IDENTIFICATION SYSTEM

1. Performance: Describe Automatic Identification System (AIS)

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the AIS:

   a. IAW Refs, by completing the following:

      (1) Describe the Generic AIS; and

      (2) Describe the In-service AIS.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic AIS System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**85**	
1.1	Historical background			
1.2	Theory and design principles;			
1.3	Types (advantages/disadvantages); 1. Type ‘A’ Units; 2. Type ‘B’ Units; 3. Base Stations; and 4. Aids to Navigation (AtoN).			
1.4	Performance characteristics; 1. Message types and protocol; 2. Broadcast information; 3. Multiplexing methods; and			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Modulation.			
1.5	System Components; 1. Transponder; 2. Antenna Group; and 3. Control and Display Unit.			
1.6	Interfacing/integration with other systems; 1. Nav Data Dist. System; 2. Navigation Radar; 3. Global Positioning System; 4. Electrical; and 5. Environmental.			
1.7	System/integration with other system; 1. Nav Data Dist. System; 2. Electrical; and 3. Environmental.			
2	**Describe In-service AIS System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow; 1. System Block Diagram.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Operating policy and procedures;			

5. Total EO Time: Total: 2.6 x 50 minute periods as follows:

a. 2.6 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil

7. Remarks:

a. Prerequisites:

1) EO 004.01 – Describe Electromagnetic Theory; and

2) EO 004.02 – Describe Antenna Theory.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.09 DESCRIBE METEOROLOGICAL SYSTEMS

1. Performance: Describe the Meteorological System

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Meteorological System:

   a. IAW Refs, by completing the following:

      (1) Describe the Generic Meteorological System; and

      (2) Describe the In-service Meteorological System.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Meteorological System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**85**	
1.1	Historical background			
1.2	Theory and design principles; 4. Anemometer; 5. Barometers; 6. Thermometer; 7. Hydrometer; and 8. Data processors; i. Normalizing; ii. Correlation; and iii. Prediction.			
1.3	Types (advantages/disadvantages);			
1.4	Performance characteristics;			
1.5	System Components; 1. Anemometer;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Barometers; 3. Thermometer; 4. Hydrometer; 5. Processors; and 6. Distributions System.			
1.6	Interfacing/integration with other systems; 1. Fire Control System; 2. Command and Control System; 3. Landing Safety Officer / Flight Co-Ordinator Position; 4. External Communications System; 5. Electrical; and 6. Environmental.			
1.7	System/integration with other system; 1. Nav Data Dist. System; 2. Electrical; and 3. Environmental.			
2	**Describe In-service Meteorological System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow; 1. System Block Diagram.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			

Teaching Points		Instructional Strategies	Time (min)	References
2.9	Operating policy and procedures;			

5. Total EO Time: Total: 2.2 x 50 minute periods as follows:

   a. 2.2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE NAVIGATION SYSTEMS

EO 002.10 SUPPORT NAVIGATION SYSTEMS ONBOARD HMC SHIPS

1. Performance: Support Navigation Systems onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall Support Navigation Systems onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard navigation systems to the satisfaction of the CSEO;

      (2) Conduct a walk-through of the navigation systems with the senior maintainer;

      (3) Deliver a brief on the navigation system suite to the satisfaction of the CSEO;

      (4) Stand four (4) bridge watches with the OOW; and

      (5) Submit a Technical Defect Report on a major problem with one of the fitted Navigation Systems selected by the CSEO.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The NCS ENG shall, to the satisfaction of the CSEO, complete the following for**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
	the Navigation Data Distribution System:			
1.1	Observe and participate in the following; 1. Operations; 2. Tests and Trials 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
1.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
1.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations/signal flow/software; 2. System components; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools;			

Teaching Points		Instructional Strategies	Time (min)	References
	6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Operating policy and procedures; and 11. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
2	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for the Inertial Navigation System (INS):**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Observe and participate in the following: 1. Operations; i. Weapon firings 2. Tests and Trials; 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
2.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
2.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
3	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
	the Global Positioning Systems (GPS):			
3.1	Observe and participate in the following: 1. Operations: 2. Tests and Trials: 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
3.2	Conduct a walk-through of the systems with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss all related considerations for CSE rounds; and 3. Identify MSE systems considerations.			
3.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools;			

Teaching Points		Instructional Strategies	Time (min)	References
	6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Operating policy and procedures; and 11. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
4	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for the Doppler Speed Log System:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Observe and participate in the following: 1. Operations: 2. Tests and Trials: 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
4.2	Conduct a walk-through of the systems with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Identify MSE systems considerations.			
4.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Operating policy and procedures; and 11. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
5	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for the Electronic Charting System (SHINNADS):**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5.1	Observe and participate in the following:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Operations: i. Blind pilotage; ii. Flying Ops; and iii. Entering and Exiting Harbour. 2. Tests and Trials: 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
5.2	Conduct a walk-through of the systems with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
5.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools;			

Teaching Points		Instructional Strategies	Time (min)	References
	6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
6	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for the Echo Sounder:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6.1	Observe and participate in the following: 1. Operations: i. Blind pilotage; ii. Entering and Exiting Harbour. 2. Tests and Trials: 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
6.2	Conduct a walk-through of the systems with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
6.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
7	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
	the Stabilized Horizon Reference System:			
7.1	Observe and participate in the following: 1. Operations: i. Flying Ops; 2. Tests and Trials: 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
7.2	Conduct a walk-through of the systems with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
7.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools;			

Teaching Points		Instructional Strategies	Time (min)	References
	6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Operating policy and procedures; and 11. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
8	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for the Automatic Identification System (AIS):**	**Method:** OJT **Media:** N/A **Environment:** Ship		
8.1	Observe and participate in the following: 1. Operations: i. Flying Ops; 2. Tests and Trials: 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
8.2	Conduct a walk-through of the systems with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
8.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Operating policy and procedures; and 11. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
9	**The NCS ENG shall, to the satisfaction of the CSEO complete the following for**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
	the Meteorological System (MET):			
9.1	Observe and participate in the following: 1. Operations: i. Flying Ops; and ii. Weapons Firings. 2. Tests and Trials: 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
9.2	Conduct a walk-through of the systems with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
9.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies;			

Teaching Points		Instructional Strategies	Time (min)	References
	5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Operating policy and procedures; and 11. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
10	**Present a system briefing to designated ship's staff on the Navigation Suite. This brief shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
10.1	System description			
10.2	System overview			
10.3	Major system components			
10.4	Operating modes			
10.5	Capabilities and limitations			
10.6	Interfaces			
10.7	Redundancies/survivability			
10.8	Combat/CSE liaison considerations			
11	**Stand Department Watches to include the following:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
11.1	Stand eight (8) watches with the W.ENG TECH (Sonar)			
12	**Submit a technical report on a major problem with one of the fitted Navigation**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
	Systems selected by the CSEO selected by the CSEO. (Note: Only one submission is required for PO 002). As a minimum, the report shall cover the following points:			
12.1	Identification/discussion of the problem			
12.2	Operational impact of the problem			
12.3	Options to resolve the problem			
12.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

EO 003.01 DESCRIBE NAVAL COMMUNICATIONS SYSTEMS

1. Performance: Describe Naval Communication Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Naval Communication Systems:

   a. IAW Refs, by completing the following:

      (1) Describe General Communications Systems;

      (2) Describe Data Transmission Methods;

      (3) Describe Modulation Techniques;

      (4) Describe Error Control Techniques;

      (5) Describe Cryptography;

      (6) Describe Spread Spectrum Communications;

      (7) Describe Multiplexing and Multiple Access; and

      (8) Describe Fibre Optic Communication Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe General Communication Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
2	**Describe Data Transmission Methods**	**Method:** Lecture **Media:** PowerPoint	**500**	

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	Analog and Digital Transmission;			
2.2	Data and Signals; 1. Analog Signal; 2. Digital Signal; 3. Baseband Signal; 4. Band Limiting; 5. Data and Signals; and 6. Analog vs. Digital Transmission.			
2.3	Transmission Impairments; 1. Attenuation; 2. Delay Distortion; 3. Noise; 4. Channel Capacity (BW. Baud rate, Bit rate); 5. Standing Wave; 6. Voltage Standing Wave Ratio; and 7. Time Domain Reflectometry.			
2.4	Transmission Media: 1. Twisted Pair; 2. Coaxial Cable; 3. Terrestrial Microwave; and 4. Radio			
2.5	Electromagnetic Wave Propagation; 1. Troposphere Propagation; and 2. Ionosphere Propagation.			
3	**Describe Modulation Techniques**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**550**	
3.1	Analog Data, Analog Signals; 1. Amplitude Modulation;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Frequency and Phase Modulation; 3. Basic Transmitters; and 4. Basic Receivers.			
3.2	Analog Data, Digital Signals; 1. Pulse Coded Modulation; 2. Delta Modulation; 3. Reducing Quantization Noise in PCM; 4. Analog to Digital Conversion; 5. Digital to Analog Conversion; and 6. Modulator Demodulator (MODEM)			
3.3	Digital Data, Analog Signals; 1. Amplitude Shift Keying; 2. Phase Shift Keying; 3. Differential Phase Shift Keying; 4. Frequency Shift Keying; 5. Continuous Phase Shift Keying; 6. M-Array Signalling; and 7. Quadrature Phase Shift Keying.			
3.4	Digital Data, Digital Signals; 1. Unipolar Signal; 2. Polar Signal; 3. Bipolar Signal; 4. Manchester or Biphase Signal.			
3.5	Transmission Efficiency			
4	**Describe Error Control Techniques**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
4.1	Error Detection Techniques			
4.2	Error Correction Techniques			

Teaching Points		Instructional Strategies	Time (min)	References
5	**Describe Cryptography**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
5.1	Introduction			
5.2	Cryptographic Methods			
5.3	Cryptographic Algorithms			
5.4	Security of Cryptographic Systems			
5.5	Protecting Secure Communications			
5.6	Overview of Currently Fielded (Commercial/Military) Cryptographic Algorithms			
6	**Describe Spread Spectrum Communications**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
6.1	Criteria for Spread Spectrum			
6.2	Advantages of Spread Spectrum			
6.3	Processing Gain			
6.4	Direct Sequence Modulation			
6.5	Frequency Hopping Modulation			
7	**Describe Multiplexing and Multiple Access**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
7.1	Frequency Division Multiplexing			
7.2	Time Division Multiplexing			
7.3	Frequency Division Multiplexing			
7.4	Time Division Multiple Access			
7.5	Demand Assigned Multiple Access			
7.6	Code Division Multiple Access			
8	**Describe Fibre Optic Communication Systems**	**Method:** Lecture	**200**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		

5. Total EO Time: Total: 59.5 x 50 minute periods as follows:

    a. 59.5 x 50 min period; and

    b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

    a. Prerequisites:

        1) EO 004.01 – Describe Electromagnetic Theory.

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

EO 003.02 DESCRIBE INTERNAL COMMUNICATION SYSTEMS

1. Performance: Describe Internal Communication Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Internal Communication Systems:

   a. IAW Refs, by completing the following:

      (1) Describe the Generic Telephone Exchange; and

      (2) Describe the In-service Internal Communication System.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Telephone Exchange**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
1.1	Historical Background (minimal)			
1.2	Theory and design principles; and 1. Power line MODEM.			
1.3	Types (advantages/disadvantages); 1. Analog vs Digital; and 2. General topography.			
1.4	Performance characteristics; 1. Capacity/Bandwidth; and 2. Quality of Service.			
1.5	System components; 1. Processors/switches; 2. Terminals/Handsets/Headsets; 3. Amplifiers; 4. Distribution System; and 5. Test and Trials.			

Teaching Points		Instructional Strategies	Time (min)	References
1.6	Interfacing/integration with other systems; 1. Electrical; 2. Environmental; and 3. External Communications.			
1.7	Emerging Technologies; and 1. Fibre/Optical; 2. Wireless; and 3. Voice over Internet Protocol (VoIP) Telephony.			
1.8	Future in-service technologies			
2	**Describe In-service Internal Communication Systems (SHINCOM, Main Broadcast, Emergency Broadcast, TRS, etc)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow; 1. System block diagram.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems (e.g. Electrical, HVAC, etc.)			
2.7	Maintenance requirements			
2.8	Safety considerations (batteries)			
2.9	Security Concerns			
2.10	Operating policy and procedures (Meet-Me-Net, P-Net)			

5. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

1) Describe Naval Communications Systems

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

EO 003.03 DESCRIBE EXTERNAL COMMUNICATIONS SYSTEMS

1. Performance: Describe External Communication Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe External Communication Systems:

   a. IAW Refs, by completing the following:

      (1) Describe the Generic Radio System; and

      (2) Describe In-service External Radio Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Radio System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
1.1	Historical background			
1.2	Theory and design principles			
1.3	Types (advantages/disadvantages); 1. Data/Voice; 2. Plain/secure; 3. Modulation; 4. Integrated/stand-alone; and 5. Broadcast vs. Point-to-Point.			
1.4	Performance characteristics; 1. Frequency; 2. Output power; 3. Sensitivity;			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Bandwidth; 5. Modulation Techniques; 6. Selectivity; 7. EPM; and 8. Frequency Diversity; 9. Antenna Diversity; 10. Polarization; and 11. Directivity / Shaped Antennas.			
1.5	System components; 1. Receiver; 2. Transmitter; 3. Transmission Channel (medium); 4. Antenna and coupler; 5. Input/output devices; and 6. Cryptographic devices.			
1.6	Interfacing/integration with other systems; 1. Electrical; 2. Environmental; 3. Message processing and handling system; 4. Communication control monitoring and distribution system; 5. Emission Control (EMCON); 6. Input/Output Devices; 7. Cryptographic devices; and 8. MODEMs.			
1.7	Emerging Technologies; and			
1.8	Future in-service technologies			
2	**Describe In-service External Radio Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
2.1	Performance characteristics, capabilities and limitations			

Teaching Points		Instructional Strategies	Time (min)	References
2.2	System components/signal flow; 1. System block diagram.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments; 1. Antenna analysis (NESTR); 2. VSWR; 3. Power Out; and 4. Spectral Analysis.			
2.9	Safety considerations; 1. (RF SAFETY)			
2.10	Operating policy and procedures			

5. Total EO Time: Total: 8 x 50 minute periods as follows:

   a. 8 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 003.01 – Describe Communication Theory;

      2) EO 004.01 – Describe Electromagnetic Theory; and

      3) EO 004.02 – Describe Antenna Theory.

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

EO 003.04 DESCRIBE SATELLITE COMMUNICATIONS SYSTEMS

1. Performance: Describe Satellite Communication Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Satellite Communication Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Satellite Communication Systems; and

      (2) Describe In-service Satellite Radio Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Satellite Communication Systems;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
1.1	Historical background			
1.2	Theory and design principles; 1. Orbits; 2. Footprints / System Coverage; 3. Limitations; i. Frequency; ii. Power Out; iii. Bandwidth; and iv. Capacity. 4. Commercial/Military; 5. Access and Approval; and 6. Protocol Selection.			
1.3	Types (advantages/disadvantages/comparison); 1. UHF, SHF, EHF;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Data/Voice; 3. Broadcast/Point to Point; 4. Integrated/stand-alone.			
1.4	Performance characteristics; 1. Frequency; 2. Output power; 3. Sensitivity; 4. Bandwidth; 5. Modulation techniques; 6. Multiplexing; 7. Selectivity; and 8. EPM.			
1.5	System components; 1. Receiver; 2. Transmitter; 3. Transmission Channel (medium); 4. Antenna, control unit and coupler; 5. Input/Output devices; 6. Satellites; and 7. Ground control station.			
1.6	Interfacing/integration with other systems; 1. Electrical; 2. Environmental; 3. Message processing and handling system; 4. Navigation Distribution System; 5. Communication control, monitoring, and distribution system; 6. Emission Control (EMCON); 7. Input/Output Devices; 8. Cryptographic devices; and 9. MODEMs.			
1.7	Emerging Technologies			
2	**Describe In-service Satellite Radio Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Performance characteristics, capabilities and limitations			

Teaching Points		Instructional Strategies	Time (min)	References
2.2	System components/signal flow/software; 1. System block diagram.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems (e.g. CCS, electrical, HVAC, etc.)			
2.7	Maintenance Requirements			
2.8	System trials/alignments; 1. VSWR; 2. Power out; and 3. Spectral analysis.			
2.9	Safety considerations; 1. RF SAFETY.			
2.10	Operating policy and procedures			

5. Total EO Time: Total: 6 x 50 minute periods as follows:

   a. 6 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 003.01 – Describe Communication Theory;

      2) EO 004.01 – Describe Electromagnetic Theory; and

      3) EO 004.02 – Describe Antenna Theory

   b. The trainees should participate in Equipment practical and tour

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

EO 003.05 DESCRIBE THE COMMUNICATION CONTROL AND MANAGEMENT SYSTEM

1. Performance: Describe the Communication Control and Management System

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Communication Control and Management System:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Communication Control and Management Systems; and

      (2) Describe In-service Communication Control and Management Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Communication Control and Management Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Historical background			
1.2	Theory and design principles			
1.3	Types (advantages/disadvantages/comparison); 1. Manual; 2. Automatic; and 3. Hybrid.			
1.4	Performance characteristics;			
1.5	System components; 1. Processor;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Switch/Patch panel; and 3. Test and monitoring system.			
1.6	Interfacing/integration with other systems; 1. Electrical; 2. Environmental; 3. Interior/Exterior communications; 4. Message processing and handling system; 5. Emission Control (EMCON); 6. Input/output devices; 7. Cryptographic devices; and 8. MODEMs.			
1.7	Emerging Technologies			
1.8	Future In-Service technologies			
2	**Describe In-service Communication Control and Management Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow/software; 1. System block diagram.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance Requirements			
2.8	System trials			
2.9	Safety Considerations			
2.10	Operating policy and procedures; 1. COMPLAN; and 2. EMCON.			

5. Total EO Time: Total: 2.5 x 50 minute periods as follows:

   a. 2.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

EO 003.06 DESCRIBE THE MESSAGE HANDLING DISTRIBUTION SYSTEM

1. Performance: Describe the Message Handling Distribution System

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Message Handling Distribution System:

   a. IAW Refs, by completing the following:

      (1) Describe General message processing theory/concepts; and

      (2) Describe In-service Message Processing Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe General message processing theory/concepts**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**15**	
2	**Describe In-service Message Processing Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**20**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow/software;			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			

2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety considerations			
2.10	Operating policy and procedures			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 - MANAGE COMMUNICATIONS SYSTEMS

EO 003.07 SUPPORT COMMUNICATIONS SYSTEMS ONBOARD HMC SHIPS

1. Performance: Support Communication Systems onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall support Communication Systems as fitted on board:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard Communication Systems to the satisfaction of the CSEO;

      (2) Conduct a walk-through of the communication systems with the senior maintainer;

      (3) Deliver a brief on the External Communication Suite to the satisfaction of the CSEO;

      (4) Stand six (6) watches with W ENG TECH (COMM);

      (5) Stand two (2) watches in CCR with NAVCOMM; and

      (6) Submit a Technical Defect Report on a major problem with one of the fitted communication systems selected by the CSEO.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**For all fitted Communication systems, the NCS ENG shall, to the satisfaction of the**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
	CSEO, complete the following:			
1.1	Observe and participate in the following; 1. Operations; i. Set up and monitor/change Comm. plans; ii. Set up a radio circuit manually; iii. Set up and monitor/change SHINCOM programmable nets; iv. Input and transmit/receive messages; and v. Crypto change. 2. Tests and Trials 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
1.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
1.3	Draw the internal and external system diagrams and thoroughly explain the system to the CSEO (without			

Teaching Points		Instructional Strategies	Time (min)	References
	references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
2	**Present a system briefing to designated ship's staff on the internal or external communication (as selected by the CSEO). This brief shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	System description			
2.2	System overview			
2.3	Major system components			
2.4	Operating modes			
2.5	Capabilities and limitations			
2.6	Interfaces			
2.7	Redundancies/survivability			

Teaching Points		Instructional Strategies	Time (min)	References
2.8	Combat/CSE liaison considerations			
3	**CSE/CBT Department Watches;**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	Stand six (6) watches (consecutively) with the W.ENG TECH (COMM)			
3.2	Stand two (2) watches with Nav Comms in the CCR			
4	**Submit a technical report on the internal or external Communication systems (as selected by the CSEO) on a major problem with the system. (Note: Only one submission is required for PO 003). As a minimum, the report shall cover the following points:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Identification/discussion of the problem			
4.2	Operational impact of the problem			
4.3	Options to resolve the problem			
4.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

    1) OJT

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.01 DESCRIBE ELECTROMAGNETIC THEORY

1. Performance: Describe Electromagnetic Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Electromagnetic Theory consistent with the NCS environment:

   a. IAW Refs, by completing the following:

      (1) Describe Electrostatic Fields;

      (2) Describe Magnetostatic Fields;

      (3) Describe Maxwell's Equations;

      (4) Describe Electromagnetic Waves;

      (5) Describe Propagation Effects; and

      (6) Describe Electromagnetic Interference.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Electrostatic Fields**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
1.1	Charge			
1.2	Coulomb's Law			
1.3	Gauss' Law			
1.4	Electric Potential			
1.5	Permittivity			
1.6	Capacitance			

Teaching Points		Instructional Strategies	Time (min)	References
2	**Describe Magnetostatic Fields**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
2.1	Current			
2.2	Magnetic Field			
2.3	Biot-Savart's Law			
2.4	Ampere's Law			
2.5	Permeability			
2.6	Inductance			
3	**Describe Maxwell's Equations;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
3.1	Faraday's Law			
3.2	Gauss' Law (Electric)			
3.3	Gauss' Law (Magnetic)			
3.4	Ampere-Maxwell Law			
3.5	The propagating electromagnetic wave			
4	**Describe Electromagnetic Waves**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
4.1	Electromagnetic Spectrum (DC-Light)			
4.2	Frequency and Wavelength			
4.3	Impedance			
4.4	Polarization			
4.5	Ground Wave			
4.6	Sky Wave			
5	**Describe Propagation Effects**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
5.1	Reflection			
5.2	Refraction			

Teaching Points		Instructional Strategies	Time (min)	References
5.3	Diffraction			
5.4	Attenuation			
5.5	Multipath			
6	**Describe Electromagnetic Interference**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
6.1	Problems			
6.2	Definitions (EME, EMI, EMC)			
6.3	Sources			
6.4	Shielding			
6.5	Shipboard Cabling			
6.6	Grounds/Ground loops.			

5. Total EO Time: Total: 34 x 50 minute periods as follows:

   a. 34 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.02 DESCRIBE ANTENNA SYSTEMS THEORY

1. Performance: Describe Antenna Systems Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Antenna Systems Theory:

   a. IAW Refs, by completing the following:

      (1) Describe Transmission Line Theory;

      (2) Describe Waveguide Theory;

      (3) Describe Antenna Theory;

      (4) Describe the Impact of the Environment; and

      (5) Describe Antenna Design.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Transmission Line Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
1.1	Basics			
1.2	Transmission Line Parameters			
1.3	Transmission Line Equations			
1.4	Input Impedance, Standing Wave Ration (SWR) and Power			
1.5	The Smith Chart			
1.6	Practical Applications			
2	**Describe Waveguide Theory**	**Method:** Lecture	**200**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
2.1	Basics			
2.2	Rectangular Waveguides			
2.3	Transverse Magnetic (TM) Modes			
2.4	Transverse Electric (TE) Modes			
2.5	Wave Propagation in the Guide			
2.6	Power Transmission and Attenuation			
2.7	Waveguide Current and Mode Excitation			
2.8	Waveguide Resonators			
2.9	Practical Applications			
3	**Describe Antenna Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**450**	
3.1	Basics			
3.2	The Hertzian Dipole Antenna			
3.3	The Concept of Far Field			
3.4	The Concept of Near Field			
3.5	Power Flow and Radiation			
3.6	The Half-Wave Dipole and Quarter-Wave Monopole Antennas			
3.7	Antenna Arrays			
3.8	Antenna Properties; 1. Directivity; 2. Gain; 3. Radiation Resistance; 4. Bandwidth; 5. Effective Aperture; 6. The Friis Transmission Equation; and 7. Beam forming.			
4	**Describe the Impact of the Environment**	**Method:** Lecture	**200**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
4.1	The Ionosphere, Troposphere			
4.2	Critical Frequency			
4.3	Critical Angle			
4.4	Maximum Usable Frequency			
4.5	Ground			
5	**Describe Antenna Design**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
5.1	Aperture Antennas			
5.2	Broadband Dipoles and Matching Techniques			
5.3	Antenna Arrays			
5.4	Frequency Independent Antennas			
5.5	Horn Antennas			
5.6	Traveling Wave Antennas			
5.7	Reflector Antennas			

5. Total EO Time: Total: 36 x 50 minute periods as follows:

   a. 36 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   c. Prerequisites:

      1) EO 004.01 – Describe Electromagnetic Theory.

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.03 DESCRIBE RADAR SYSTEM THEORY

1. Performance: Describe RADAR System Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe RADAR System Theory:

   a. IAW Refs, by completing the following:

      (1) Describe RADAR Fundamentals;

      (2) Describe the RADAR Equation;

      (3) Describe Detection Limitations;

      (4) Describe Signal Processing Techniques;

      (5) Describe High Power Transmitters;

      (6) Describe RADAR Tracking Principles; and

      (7) Describe CW RADAR.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe RADAR Fundamentals**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
1.1	Basic Principles			
1.2	Pulse Train Parameters; 1. PW and PRF.			
1.3	Time and Frequency Domain Representations			

Teaching Points		Instructional Strategies	Time (min)	References
1.4	RADAR Types			
1.5	Basic Pulse-RADAR Block Diagram			
2	**Describe the RADAR Equation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**800**	
2.1	RADAR equation and Max Detection Range; 1. RADAR Equation; i. Derivation; and ii. Overview of Parameters.			
2.2	Detection of Signals in Noise; 1) Noise Probability Density Function; 2) Probability of False Alarm; and 3) Probability of Detection.			
2.3	Atmospheric Propagation of RADAR Waves; 1. Electromagnetic Waves; 2. Reflection, and Multipath Effect; i. Software Simulation (AREPS) 3. Refraction and RADAR Horizon; 4. Environmental Noise; and 5. Atmospheric Attenuation.			
3	**Describe Detection Limitations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
3.1	Extraction of Contact Information from RADAR Echoes			

Teaching Points		Instructional Strategies	Time (min)	References
3.2	Contact Information			
3.3	Range Ambiguity			
3.4	Range Resolution			
3.5	Minimum Detection Range			
3.6	Contact Bearing and Elevation			
3.7	Doppler Frequency			
3.8	Velocity Ambiguity			
3.9	RADAR Resolution Cell; 1. Range Resolution; 2. Velocity Resolution; and 3. Angular Resolution.			
3.10	Matching Parameters to Requirements			
4	**Describe Signal Processing Techniques;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
4.1	Integration; 1. Coherent Integration; and 2. Non-Coherent Integration.			
4.2	Sensitivity Time Control			
4.3	Instantaneous Automatic Gain Control (IAGC)			
4.4	Logarithmic Receiver			
4.5	Constant False Alarm Rate (CFAR)			
4.6	Variable Compression			
4.7	Pulse Compression 1. Pulse Compression Techniques 2. Phase Coding vs. Linear FM			
4.8	MTI			
4.9	MTD			
5	**Describe High Power Transmitters**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	

Teaching Points		Instructional Strategies	Time (min)	References
5.1	Magnetron – Basic Features; 1. Coaxial Magnetron.			
5.2	Klystron – Basic Features			
5.3	TWT – Basic Features			
5.4	Solid-State Transmitters			
6	**Describe RADAR Tracking Principles**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
6.1	Angle Tracking			
6.2	Sequential Lobing			
6.3	Monopulse (Simultaneous Lobing)			
6.4	Monopulse vs. Conical Scan			
6.5	Range Tracking			
6.6	Doppler Tracking			
6.7	Track-While-Scan RADARs			
6.8	TWS Process			
7	**Describe CW RADAR**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
7.1	Receiver Bandwidth			
7.2	Application of CW RADAR			
7.3	FM – CW RADAR			
7.4	Modulation Waveform			

5. Total EO Time: Total: 62 x 50 minute periods as follows:

   a. 53 x 50 min period;

   b. 9 x 50 min period - Labs

   c. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. RADAR Labs (Basic RADAR, MTI and MTD)

7. Remarks:

a. Prerequisites:

1) EO 004.02 – Describe Antenna Theory; and

2) EO 005.01 – Describe DSP Theory.

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.04 DESCRIBE APPLICATIONS OF RADAR SYSTEMS

1. Performance: Describe Applications of RADAR Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Applications of RADAR Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic RADAR Systems; and

      (2) Describe In-service RADARs.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic RADAR System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
1.1	Theory and design principles; 1. Layered detection; and 2. Track Management.			
1.2	Types (advantages/disadvantages); 1. Navigation/Air Search/ Surface Search; 2. Track-While-Scan/Track; 3. Short, Medium and Long Range; and 4. Mechanical/Electronic beam steering.			
1.3	Performance characteristics;			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Matching parameters to requirements/type; i. Peak Power, average Power; ii. Frequency; iii. PRF, PRI; iv. Pulse Width (PW) v. Bandwidth (BW) vi. Antenna Rotation Rate (ARR); vii. Antenna Pattern (Beam width, sidelobes) viii. Range ix. Sensitivity; x. EPM; and xi. Range gates.			
1.4	System components; 1. Transmitter; 2. Receiver; 3. Antenna; 4. Processor; 5. Waveguide components; 6. Track Management Processor; and 7. Display			
1.5	Interfacing/integration with other systems; 1. RADAR Video Distribution System; 2. Radiation Control; 3. Electrical; 4. Environmental; 5. Navigation Distribution; 6. Fire Control System (FCS); 7. Identification Friend or Foe (IFF); 8. CCS; 9. Electronic Charting Systems; and			

Teaching Points		Instructional Strategies	Time (min)	References
	10. RADAR Blanker.			
1.6	System trials / alignments; 1. CSAT; 2. Antenna Analysis (NESTR); 3. Min/Max Range; and 4. Datum Alignment.			
1.7	Emerging Technologies: 1. Surface Wave (OTHR)			
1.8	Future In-Service systems			
2	**Describe In-service RADARs**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
2.1	Performance characteristics, capabilities, limitations			
2.2	System components/signal flow/software			
2.3	System Block diagrams			
2.4	Modes of operation			
2.5	System redundancies			
2.6	System diagnostic capabilities/tools			
2.7	Interfacing/integration with other systems			
2.8	Maintenance requirements			
2.9	System trials/alignments			
2.10	Safety considerations; 1. RF SAFETY; and 2. Ionizing Radiation; and 3. High Voltage.			
2.11	Operating policy and procedures; 1. EMCON; 2. Spectrum Management (licensing) 3. Interference with civilian systems; and 4. AREPS (Ranges/Coverage prediction).			

5. Total EO Time: Total: 10 x 50 minute periods as follows:

   a. 10 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 004.03 – Describe RADAR Theory.

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.05 DESCRIBE IDENTIFICATION FRIEND OR FOE SYSTEMS

1. Performance: Describe Identification Friend or Foe (IFF) Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe IFF Systems:

   a. IAW Refs, by completing the following:

      (1) Describe IFF Theory; and

      (2) Describe In-service IFF Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe IFF Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**75**	
1.1	Theory and Design principles; 1. IFF application; i. Modes; ii. Interrogation/Reply signal; iii. Black IFF; iv. Side lobe interrogation suppression; and v. Decoding.			
1.2	Types (advantages/disadvantages/comparison); 1. Modes; 2. Commercial/Military; and 3. Antennas.			
1.3	Performance characteristics;			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Output Power; 2. Frequency; and 3. Range/Counter-detection.			
1.4	System components; 1. Transmitter; 2. Receiver; 3. Coder/Decoder; 4. Antenna; 5. Processor; 6. Transmission line components; and 7. Encryption.			
1.5	Interfacing/integration with other systems; 1. RADAR Video Distribution System; 2. Radiation Control; 3. Electrical; 4. Environmental; 5. CCS; and 6. RADAR Blanker.			
1.6	System trials / alignments			
1.7	Emerging Technologies			
1.8	Future In-Service technologies; 1. Mode 5/S.			
2	**Describe In-service IFF Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Performance characteristics, capabilities, limitations			
2.2	System components/signal flow/software			
2.3	System Block diagrams 1. Modes of operation;			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety considerations; 1. RF SAFETY; and			

Teaching Points		Instructional Strategies	Time (min)	References
	2. High Voltage.			
2.10	Operating policy and procedures; 1. Crypto Management.			

5. Total EO Time: Total: 3.5 x 50 minute periods as follows:

   a. 3.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 004.03 – Describe RADAR Theory.

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.06 DESCRIBE ELECTRO-OPTICS/INFRA-RED SYSTEMS

1. Performance: Describe Electro-Optics/Infra-Red Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Electro-Optics/Infra-Red Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic EO/IR Systems; and

      (2) Describe In-service EO/IR Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic EO/IR System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
1.1	Theory and design principles; 1. Electro-Optics; i. Spectrum; ii. Detectors; iii. Lenses; iv. Atmospheric Effects; v. Attenuation; vi. Suppression; and vii. Basic LASER theory and classification. 2. Infra-red; i. Spectrum; ii. Detectors; iii. Lenses;			

Teaching Points		Instructional Strategies	Time (min)	References
	iv. Atmospheric Effects; v. Emissivity (Grey body/Black body); and vi. Suppression.			
1.2	Types (advantages/disadvantages); 1. Active/Passive; 2. EO/IR (light amplification vs. detection); and 3. IR bands.			
1.3	Performance characteristics; 1. Power; 2. Range; 3. Sensitivity; 4. Band/frequency/wavelength; 5. Resolution; and 6. Coverage.			
1.4	System components; 1. Detector; 2. Lens; 3. Emitter; 4. Signal Processing; 5. Stabilization system; and 6. Display.			
1.5	Interfacing/integration with other systems; 1. Electrical; 2. Environmental; 3. Navigation Distribution System; 4. CCS; and 5. FCS.			
1.6	System trials / alignments			
1.7	Emerging Technologies			
1.8	Future In-Service systems			
2	**Describe In-service EO/IR Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

Teaching Points		Instructional Strategies	Time (min)	References
2.1	Performance characteristics, capabilities, limitations			
2.2	System components/signal flow			
2.3	System Block diagrams; 1. Modes of operation.			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety considerations; and 1. Laser Safety.			
2.10	Operating policy and procedures;			

5. Total EO Time: Total: 7.5 x 50 minute periods as follows:

   a. 7.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 004.01 – Describe Electromagnetic Theory.

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.07 DESCRIBE NAVAL FIRE CONTROL SYSTEMS

1. Performance: Describe Naval Fire Control Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Naval Fire Control Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Naval Fire Control System;

      (2) Describe In service fire control systems

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Naval Fire Control System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**450**	
1.1	Historical background;			
1.2	Theory and Design Principles; 1. Fire Control Solutions; i. Ballistic Equation.			
1.3	Interfacing; 1. Navigation Distribution Systems; 2. CCS/Link; 3. Sensors; 4. Weapons; 5. Radiation Control; 6. Electrical; and 7. Environmental.			
1.4	System trials / alignments;			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Weapon to FCS alignment; 2. Weapon Certification; 3. CSAT; and 4. FCS/INS Alignment.			
1.5	Emerging Technologies			
1.6	Future In-Service Technologies			
2	**Describe In service Fire Control Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
2.1	Performance characteristics, capabilities, limitations			
2.2	System components/signal flow/software; 1. Track designation to kill assessment sequence.			
2.3	Modes of operation.			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety considerations; and 1. RF SAFETY; 2. Laser; and 3. Ionizing Radiation Safety.			
2.10	Manning Requirements			
2.11	Operating policy and procedures			
4.11	Operating policy and procedures			

5. Total EO Time: Total: 16 x 50 minute periods as follows:

   a. 16 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites:

1) EO 004.02 - Describe Antenna Systems Theory; and

2) EO 004.03 – Describe Radar System Theory.

PO 004 – MANAGE ABOVE WATER SURVEILLANCE SYSTEM

EO 004.08 SUPPORT NAVAL ABOVE WATER SURVEILLANCE SYSTEMS ONBOARD HMC SHIPS

1. Performance: Support Naval Above Water Surveillance Systems onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall describe Satellite Communication Systems:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard Above Water Surveillance Systems to the satisfaction of the CSEO Describe In-service Satellite Radio Systems;

      (2) Conduct a walk-through of the Above Water Surveillance Systems with the senior maintainer;

      (3) Deliver a brief on the Radar Suite, and the Fire Control System to the satisfaction of the CSEO;

      (4) Present Above Water Surveillance Systems to designated ship's staff;

      (5) Stand eight (8) watches with the W ENG TECH (RADAR);

      (6) Stand two (2) watches with SAC;

      (7) Stand two (2) watches with the SWC / FCS; and

      (8) Submit a Technical Defect Report on a major problem with one of the fitted Above Water Surveillance Systems selected by the CSEO.

4. Teaching Points:

	Teaching Points	Instructional Strategies	Time (min)	References
1	**The NCS ENG shall, to the satisfaction of the CSEO, complete the following for all fitted Naval Above Water Surveillance Systems;**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Observe and participate in the following; 1. Operations; i. TRACKEX, ADEX, Encounter Ex; ii. Blind pilotage (Ops Room, Bridge); iii. Manual Operation of the RDU; iv. Manual Interrogation of IFF; v. Manipulate RADAR background (AIAT, MIAT, AICAT); vi. Manipulate RADAR RCP's; and vii. SEOSS/OPS. 2. Tests and Trials; i. Naval Electronics Sensor Test Range (NESTRA, NESTRP); ii. RADAR Min/Max Ranges; and iii. Daily/Periodic System Operational (DSOT/PSOT). iv. Weapon Certification 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble Shooting.			

Teaching Points		Instructional Strategies	Time (min)	References
1.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
1.3	Draw the internal and external system diagrams and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/Alignments; 9. Safety considerations; 10. Manning requirements; 11. Application of the RADAR equation to particular RADARs; 12. Operating policy and procedures; and 13. CSEO should confirm knowledge by requesting impact of component failure (components selected by CSEO).			
2	**The NCS ENG shall, to the satisfaction of the CSEO complete**	**Method:** OJT **Media:** N/A		

	Teaching Points	Instructional Strategies	Time (min)	References
	the following for the Fire Control System	**Environment:** Ship		
2.1	Observe and participate in the following; 1. Operations; i. Pre and post firing checks; ii. Pre-fire brief; iii. Loading and unloading ammunition; and iv. Observe all Weapons Systems firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire cew). 2. Tests and Trials; i. Weapons Systems Certification; ii. Daily/Periodic System Operational Tests (DSOT/PSOT); iii. Missile Readiness Checks; and iv. Combat System Alignment Tests (CSAT). 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble Shooting.			
2.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
2.3	Draw the internal and external system diagrams and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials/alignments; 9. Safety considerations; 10. Manning requirements; 11. Application of the RADAR equation to particular RADARs; 12. Operating policy and procedures; and 13. CSEO should confirm knowledge by requesting impact of component failure (components selected by CSEO).			
3	**Present a system briefing to designated ship's officer for one of the fitted Naval Above Water Surveillance Systems selected by the CSEO. This brief shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	System description			
3.2	System overview			
3.3	Major system components			

Teaching Points		Instructional Strategies	Time (min)	References
3.4	Operating modes			
3.5	Capabilities and limitations			
3.6	Interfaces			
3.7	Redundancies/survivability; and			
3.8	Combat/CSE liaison considerations			
4	**CSE/CBT Department Watches;**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Stand eight (8) watches (consecutively) with W.ENG TECH (Radar)			
4.2	Stand two (2) watches with the SAC and RADAR Trackers (RT)			
4.3	Stand two (2) watches with the AAW Operators (SWC/FCS)			
5	**Submit a technical report on a major problem with of the fitted Naval Above Water Surveillance Systems selected by the CSEO. (Note: Only one submission is required for PO 004).As a minimum, the report shall cover the following points;**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5.1	Identification/discussion of the problem			
5.2	Operational impact of the problem			
5.3	Options to resolve the problem			
5.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT

PO 005 – MANAGE UNDER WATER SENSOR SYSTEMS

EO 005.01 DESCRIBE DIGITAL SIGNAL PROCESSING

1. Performance: Describe Digital Signal Processing

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Digital Signal Processing:

   a. IAW Refs, by completing the following:

      (1) Describe concepts of Noise and Signals;

      (2) Describe sampling;

      (3) Describe DFT/FFT;

      (4) Describe Applications of DFT/FFT;

      (5) Describe Digital Filters;

      (6) Describe Windowing; and

      (7) Describe In-Service Systems Applications.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe concepts of Noise and Signals**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
1.1	Noise Characteristics			
1.2	Signal Characteristics			
1.3	Detection of Signals in Noise (Threshold, Probability of			

Teaching Points		Instructional Strategies	Time (min)	References
	Detection, Probability of False Alarm, SNR)			
2	**Describing sampling**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
2.1	Sampling Theorem			
2.2	Nyquist Criteria			
2.3	Aliasing			
2.4	Analog to Digital Converters (ADC) and Digital to Analog Converters (DAC)			
2.5	ADC and DAC (applications)			
2.6	Lab			
3	**Describe DFT/FFT**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
3.1	Definition and Concept of Digital Frequency			
3.2	Properties and Functions			
3.3	Delta Functions and Impulse Response			
3.4	Convolution			
3.5	Correlation and Autocorrelation			
4	**Describe Applications of DFT/FFT**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
4.1	Spectral Analysis			
4.2	Spectral Density Function			
4.3	Frequency Response of System			
4.4	Convolution in the Frequency Domain			
4.5	Lab			
5	**Describe Digital Filters**	**Method:** Lecture	**200**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
5.1	IIR Filters			
5.2	FIR Filters			
5.3	Lab			
6	**Describe Windowing**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
6.1	Definitions of Concept			
6.2	Types (Rectangular, Hanning, Hamming, Kaiser);			
6.3	Gibbs Phenomena			
6.4	Lab			
7	**Describe In-Service Systems Applications**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
7.1	Communications; 1. Heterodyning (Time Domain vs. Frequency Domain); 2. Frequency Domain Analysis 3. Encoding/Decoding Techniques; and 4. Modulation Techniques.			
7.2	SONAR Systems; 1. Beamforming; 2. Match Filtering; 3. Digital Heterodyning; 4. Spectral Analysis; 5. Windowing; and 6. Tracking Filters (Kalmaan and A/B Filters).			
7.3	RADAR Systems; 1. Coherent Detection;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Pulse Compression and Match Filtering; i. Coded Waveforms (Barker Codes); and ii. SAW Devices; 3. MTI FIR Filters; 4. CFAR Adaptive Filter; 5. Clutter Mapping; 6. Sidelobe Cancellation (SLC) Filters; and 7. Tracking Filters (Kalmaan and A/B Filters).			
7.4	Navigation Systems; 1. INS			
7.5	Simulation Project			

5. Total EO Time: Total: 53 x 50 minute periods as follows:

   a. 40 x 50 min period;

   b. 13 x 50 min periods – Labs; and

   c. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. DSP Simulation Software.

7. Remarks:

   a. Nil

PO 005 – MANAGE UNDER WATER SENSOR SYSTEMS

EO 005.02 DESCRIBE SONAR THEORY

1. Performance: Describe SONAR Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe SONAR theory:

   a. IAW Refs, by completing the following:

      (1) Describe Basic Principles of Sonar;

      (2) Describe Oceanographic Principles;

      (3) Describe SONAR Equations;

      (4) Describe Signal Processing Techniques;

      (5) Describe SONAR Equipment;

      (6) Describe Detection Methods;

      (7) Describe Tracking Methods; and

      (8) Describe SONAR Monitoring, Test and Trials.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Basic Principles of SONAR;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
1.1	1. Introduction; 2. Requirements for Underwater detection;			

Teaching Points		Instructional Strategies	Time (min)	References
	i. Submarine threats; ii. Route Survey; and iii. Mines. 3. Historical Background; 4. Definition of SONAR; 5. Objectives of SONAR (Detect to engage sequence); i. Detect; ii. Localize; iii. Classify iv. Track; and v. Prosecute. 6. Active vs. Passive.			
1.2	1. PW; 2. Frequency; and 3. PRI.			
1.3	Time and Frequency Domain Representation			
1.4	SONAR Types; 1. Hull mounted SONARs; 2. Towed Arrays; 3. Towed Bodies; 4. Dipping SONARs; 5. Sonobuoys; and 6. Underwater fixed arrays (SOSUS).			
1.5	Basic SONAR block diagram			
2	**Describe Oceanographic principles;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
2.1	Factors affecting sound velocity in water; 1. Bulk modulus of elasticity; 2. Temperature; 3. Salinity; 4. Depth (pressure) and 5. Wilson's Equation.			
2.2	Ocean thermal structures; 1. Sound Velocity Profile (SVP); i. Gradients; 2. Layers;			

Teaching Points		Instructional Strategies	Time (min)	References
	i. Surface; ii. Seasonal Thermocline; iii. Permanent Thermocline; and iv. Deep Isotherm.			
2.3	Propagation Paths; 1. Surface/Subsurface ducts; 2. Convergence zones; 3. Deep sound channel; 4. Bottom Bounce; 5. Shadow Zones.			
3	**Describe SONAR Equations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
3.1	Objectives			
3.2	SONAR Equation and Max Detection Range; 1. Active SONAR Equations; i. Derivation; and ii. Detailed analysis of parameters. 2. Passive SONAR Equation; i. Derivation; and ii. Detailed analysis of parameters.			
3.3	Cavitation			
3.4	Detection of Signals in Noise: 1. Noise Probability Density Function; 2. Probability of False Alarms; and 3. Probability of Detection.			
3.5	Matching Parameters to Requirements			
3.6	Software Simulation Systems (AESS)			
4	**Describe Signal Processing Techniques**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
4.1	Integration			
4.2	Automatic Gain Control (AGC)			

Teaching Points		Instructional Strategies	Time (min)	References
4.3	Frequency Modulation			
4.4	Beamforming			
4.5	Fast Fourier Transforms (FFT)			
4.6	Correlation			
4.7	Filtering			
5	**Describe SONAR Equipment**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
5.1	Generic Receiver; 1. Sensor (transducer); i. Theory; ii. Types/Construction/Shape; and iii. Frequency/Dimension Relationship. 2. Signal Conditioner; 3. Signal Processor; and 4. Data manager.			
5.2	Generic Transmitter; 1. Trigger; 2. Pulse width generator; 3. Modulator; 4. Frequency generator; 5. Amplifier; 6. Beam Former; and 7. Sensor.			
5.3	Display; 1. Information Presentation (Range vs. Bearing vs. Frequency vs. Time); i. Waterfall; ii. A-Scan; iii. B-Scan; iv. C-Scan; and v. Others.			
6	**Describe Detection Methods**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	

Teaching Points		Instructional Strategies	Time (min)	References
6.1	Extraction of Contact Information; 1. Broadband; 2. Narrowband; and 3. DEMON.			
6.2	Bearing Ambiguity			
6.3	Doppler Frequency			
6.4	Target Motion Analysis (TMA); 1. Purpose; 2. Requirements; and 3. Methodology.			
7	**Describe SONAR monitoring, test and trials;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
7.1	SONAR Performance Figure (SPF) (guest lecturer from FMFCS)			
7.2	Ranging			
7.3	Dynamic			
7.4	Static			
7.5	Tactical			
7.6	Self			

5. Total EO Time: Total: 36 x 50 minute periods as follows:

   a. 36 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Sonar Trainer.

7. Remarks:

   a. Prerequisites:

      1) EO 005.01 – Describe Digital Signal Processing Theory

PO 005 – MANAGE UNDER WATER SENSOR SYSTEMS

EO 005.03 DESCRIBE THE APPLICATIONS OF SONAR SYSTEMS

1. Performance: Describe the Applications of SONAR Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Applications of SONAR Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic SONAR Systems; and

      (2) Describe In-service SONAR Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Basic Principles of SONAR;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
1.1	Historical background			
1.2	Theory and design principles			
1.3	Interfacing/integration with other systems 1. CCS; 2. Navigation Data Distribution System; 3. Electrical; 4. Environment; and 5. Video Switching Unit (VSU).			
1.4	Future In-Service Technologies			
2	**Describe In-Service SONAR Systems**	**Method:** Lecture	**250**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
2.1	Performance characteristics, capabilities and limitations; 1. Output Power; 2. Frequency; 3. Bandwidth; 4. PW; 5. PRI; 6. Bandwidth/Bearing resolution; 7. Aperture; 8. Range; and 9. Depth.			
2.2	System components/signal flow/software; 1. System Block Diagrams.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments; and 1. SPFs. 2. Quiet Running Bill			
2.9	Safety Considerations			
2.10	Manning Requirements; 1. Minimum manning; and 2. Qualifications.			
2.11	Operating policy and procedures; 1. CANTASS Operations; and 2. Operating Logs			

5. Total EO Time: Total: 7 x 50 minute periods as follows:

   a. 7 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites:

1) EO 005.02 – Describe SONAR Theory.

PO 005 – MANAGE UNDER WATER SENSOR SYSTEMS

EO 005.04 DESCRIBE ACOUSTIC COUNTERMEASURE SYSTEMS

1. Performance: Describe Acoustic Countermeasure Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Acoustic Countermeasure Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic Acoustic Countermeasure Systems; and

      (2) Describe In-service Acoustic Countermeasure Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Acoustic Countermeasure System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
1.1	Historical background			
1.2	Theory and design principles; 1. Basic acoustic countermeasure principles.			
1.3	Types (advantages/disadvantages/comparison); 1. Towed; and 2. Jettisoned (NAEB).			
1.4	Performance characteristics; 1. Frequency band; 2. Power out; and 3. Deception/seduction techniques.			
1.5	System components; 1. Launching/towing mechanism;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Body (with transmitter); and 3. Processor/electronics.			
1.6	Interfacing/integration with other systems 1. CCS; 2. Navigation Data Distribution System; 3. Electrical; and 4. Environmental.			
1.7	System trials/alignments			
1.8	Emerging Technologies			
1.9	Future In-Service technologies			
2	**Describe In-Service Acoustic Countermeasure Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**35**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow/software; 1. System Block Diagrams.			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostic capabilities/tools			
2.6	Interfacing/integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety Considerations			
2.10	Manning Requirements;			
2.11	Operating policy and procedures; 1. Towing operations; i. NIXIE; and ii. NIXIE and CANTASS combined.			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 3 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites:

1) EO 005.02 – Describe SONAR Theory.

PO 005 – MANAGE UNDER WATER SENSOR SYSTEMS

EO 005.05 SUPPORT UNDERWATER SENSOR SYSTEMS ONBOARD HMC SHIPS

1. Performance: Support Underwater Sensor Systems onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall describe Satellite Communication Systems:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard Underwater Sensor Systems to the satisfaction of the CSEO;

      (2) Conduct a walk-through of the Underwater Sensor Systems with the senior maintainer;

      (3) Deliver a brief on the CANTASS and HMS-510 systems to the satisfaction of the CSEO;

      (4) Stand six (6) watches with the W.ENG TECH (SONAR); and

      (5) Submit a Technical Defect Report on a major problem with one of the fitted Underwater Sensor Systems selected by the CSEO.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The NCS ENG shall complete the following for the CANTASS Systems to the satisfaction of the CSEO;**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Observe and participate in the following:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Operations; i. Launches and recoveries of CANTASS; ii. CASEX; iii. Use of countermeasures (configuration/set up/use); iv. Launch of Bathythermograph and create SVP on AESS; and v. Sonobuoy Launch. 2. Tests and Trials; i. Winch Certification; ii. Wet end; and iii. Dry end. 3. Maintenance; i. Preventive; a) OCCAP for CANTASS; b) Vibration Analysis (VA): and c) ISOPAR-M Replacement. ii. Corrective. a) Replace Towed Array Module 4. Flash-up/Shutdown; 5. Pre-sail checks; 6. Trouble shooting; and 7. Emergency operation.			
1.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
1.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Performance characteristics, capabilities and limitations; 2. System components/signal flow/software; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (components selected by CSEO).			
2	**The NCS ENG shall complete the following for the Hull Mounted SONAR (HMS) Systems to the satisfaction of the CSEO.**			
2.1	Observe and participate in the following; 1. Operations; i. Raising/lowering of transducer (flooding of dome); ii. CASEX; and iii. Under Water Telephone 2. Tests and Trials; i. SONAR Performance Figure; ii. C5 Dome Inspection; and iii. Ranging. 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			

Teaching Points		Instructional Strategies	Time (min)	References
2.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
2.3	Draw system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components/signal flow/software; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (components selected by CSEO).			
3	**The NCS ENG shall complete the following for the remaining Underwater Sensor Systems to the satisfaction of the CSEO.**			
3.1	Observe and participate in the following; 1. Operations; i. Launches and/or recoveries of NIXIE/Sonobuoy/ Bathythermograph/Probes;			

Teaching Points		Instructional Strategies	Time (min)	References
	ii. CASEX; iii. Use of countermeasures (configuration/set up/use); iv. Creation of Sound Velocity Profile (SVP) on AESS; v. Crane Operations; and vi. Lifting Appliance Certifications. 2. Tests and Trials; 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
3.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
3.3	Draw system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components/signal flow; 3. Modes of operation; 4. System redundancies; 5. System diagnostic capabilities/tools; 6. Interfacing/integration with other systems; 7. Maintenance requirements; 8. System trials; 9. Safety considerations; 10. Manning requirements; 11. Operating policy and procedures; and			

Teaching Points		Instructional Strategies	Time (min)	References
	12. CSEO should confirm knowledge by requesting impact of component failure (components selected by CSEO).			
4	**Present a system briefing to designated ship's staff for the fitted Underwater Sensor selected by the CSEO. This brief shall include:**			
4.1	System description			
4.2	System overview			
4.3	Major system components			
4.4	Operating modes			
4.5	Capabilities and limitations			
4.6	Interfaces			
4.7	Redundancies/survivability; and			
4.8	Combat/CSE liaison considerations			
5	**Participate in CSE/CBT Department Watches;**			
5.1	Stand six (6) watches with W.ENG TECH (SONAR)			
5.2	Stand two (2) watches with SONAROPS			
6	**Submit a technical report on a major problem with of the fitted Underwater Sensor (Note: Only one submission is required for PO 005). As a minimum, the report shall cover the following points;**			
6.1	Identification/discussion of the problem			
6.2	Operational impact of the problem			
6.3	Options to resolve the problem			
6.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

PO 006 – MANAGE COMMAND AND CONTROL SYSTEMS

EO 006.01 DESCRIBE COMMAND AND CONTROL SYSTEM THEORY

1. Performance: Describe Command and Control System Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Command and Control System Theory:

   a. IAW Refs, by completing the following:

      (1) Describe the Theory of Command and Control Systems;

      (2) Describe System Architecture Theory;

      (3) Describe Hardware Design; and

      (4) Describe Emerging Technologies.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the theory of Command and Control Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
1.1	Definitions; 1. Command and Control (C2); 2. Command, Control, Communication (C3); 3. Command, Control, Communication, Computers, Intelligence (C4I); 4. Tactical, strategic, operational; and			

Teaching Points		Instructional Strategies	Time (min)	References
	5. Information Warfare.			
1.2	Command and Control Concepts; 1. Historical background (purpose); 2. Role of CCS; 3. Command Structure; 4. Shipboard; 5. Task group; 6. Threat evaluation and Weapon Assignment (TEWA); 7. Algorithms; 8. Parameters (Threat vs. Engagement); and 9. Cooperative Engagement Capability (CEC).			
1.3	Data types; 1. Inorganic; and 2. Organic.			
1.4	Near real time vs. real time			
1.5	Intelligence			
1.6	Correlation			
1.7	Radar resolution cells (search vs. Fire Control)			
1.8	Fusion			
2	**Describe System Architecture Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Topologies; 1. Centralized; 2. Federated; and 3. Distributed.			
2.2	Pros and Cons			
2.3	Integration			
2.4	Interfacing			
2.5	Protocols			
2.6	Hardware			
2.7	Software:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Software design; 2. Operating systems; and 3. Applications.			
2.8	COTS vs. Proprietary			
2.9	Human computer interface (HCI)			
3	**Describe Hardware Design**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
3.1	COTS vs. Proprietary (MILSPEC)			
3.2	Display Concepts			
3.3	Symbology (e.g. NTDS)			
3.4	Human Computer Interface (HCI)			
4	**Describe Emerging Technologies**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

5. Total EO Time: Total: 10 x 50 minute periods as follows:

   a. 10 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 006 – MANAGE COMMAND AND CONTROL SYSTEMS

EO 006.02 DESCRIBE IN-SERVICE COMMAND AND CONTROL SYSTEM HARDWARE

1. Performance: Describe In-service Command and Control System Hardware

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe In-service Command and Control System Hardware:

   a. IAW Refs, by completing the following:

      (1) Describe the CCS backbone and interfaces; and

      (2) Describe Command and Control System Hardware.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the CCS backbone and interfaces**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Historical background			
1.2	Theory and Design Principles; 1. Equipment Survivability; i. MILSPEC; ii. MIL Hardened; iii. Ruggedized; and iv. COTS.			
2	**Describe Command and Control System Hardware**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**500**	

Teaching Points		Instructional Strategies	Time (min)	References
2.1	Performance characteristics, capabilities and limitations: 1. Number of tracks; 2. Refresh Rate (Track Update Display and sensor); and 3. Others.			
2.2	System Components/signal flow (with Caps and Lims) 1. Physical layout of system components; 2. Processors; 3. Displays; 4. Data Bus Components; i. Generic; and ii. In-Service 5. Data Storage Devices; i. Generic; and ii. In Service 6. Auxiliary equipment; i. Video Switching Unit (VSU): and ii. RADAR Distribution Unit (RDU).			
2.3	Modes of Operation; 1. Software Load configuration; and 2. Manual vs. auto-reconfiguration.			
2.4	System Redundancies; 1. Bus cables and cycling; 2. Dual Interfaces; i. Processor changeover procedures (auto/manual). 3. BAMS; 4. Processors; 5. Displays; 6. Memory Storage ;			

Teaching Points		Instructional Strategies	Time (min)	References
	7. Standard Digital Equipment; and 8. Software reconfiguration.			
2.5	Single Points of Failure			
2.6	System Diagnostic capabilities and tools			
2.7	Interfacing/Integration with other systems; 1. Electrical; 2. Environmental; 3. Navigation Data Distribution System; 4. Weapon system; 5. Sensor system; 6. Weapon Status Panel; 7. Weapon Veto Panel; 8. Data Acquisition System; and 9. Interface Specifications (serial parallel protocols, NTDS Slow/Fast etc.)			
2.8	Maintenance requirements			
2.9	System Trials/Alignments; 1. Combat Systems Interface tests (CSITs); 2. Weapons Certification 3. Combat Systems Operability Tests (CSOTs); 4. Combat System Alignment Tests (CSATs); and 5. POFA.			
2.10	Safety Considerations			
2.11	Operating policies and procedures; 1. Fault reporting.			

5. Total EO Time: Total: 12 x 50 minute periods as follows:

a. 12 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites for this EO are:

1) EO 006.01 – Describe Command and Control System Theory; and

2) EO 007.03 – Describe Network Theory.

PO 006 – MANAGE COMMAND AND CONTROL SYSTEMS

EO 006.03 DESCRIBE CCS SOFTWARE

1. Performance: Describe CCS Software

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe CCS Software:

   a. IAW Refs, by completing the following:

      (1) Describe Generic CCS Software; and

      (2) Describe In-service CCS Software.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic Command and Control System Software**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Historical background			
1.2	Theory and Design Principles;			
2	**Describe In-Service Command and Control System Software**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**500**	
2.1	Performance characteristics, capabilities and limitations: 1. Number of tracks; 2. Programming language; and 3. Others			
2.2	System Components/signal flow; 1. Operating system;			

Teaching Points		Instructional Strategies	Time (min)	References
	i. Message traffic management/prioritization; 2. Modular construction; i. Module types and descriptions a) Base software; b) Interface modules; c) Reconfigurable modules; d) Maintenance modules; and e) Others. 3. In Service CCS Functions; i. Interface; ii. Detect; iii. Track; iv. Classify; v. Engage; vi. Display; vii. Control; viii. Navigation; and ix. Others.			
2.3	Memory Management; 1. MMU vs. Processor memory; i. Module and data locations; and ii. Fixed vs. transient locations. 2. Global database (information stored, where stored, config. etc.); 3. Initialization; and 4. Bus arbitration.			
2.4	Modes of operation; 1. Software load configuration; and 2. Manual vs. auto-reconfiguration.			
2.5	System Redundancies; 1. Reconfigurable modules; 2. Dual interface modules; and 3. Others.			
2.6	System Diagnostic capabilities and tools			

Teaching Points		Instructional Strategies	Time (min)	References
	1. History recording (HR), Debug, History Analysis (HA); 2. Bus sniffer diagnostic tool; and 3. System Monitoring and Control (SMC) pages.			
2.7	Interface/Integration with other systems;			
2.8	Maintenance of Naval software; 1. Change process; i. Release process; ii. Certification; iii. Version Planning Conference; iv. Combat System User Group meeting; and v. Adaptive vs. Corrective vs. Perfective change. 2. Organizations; i. Naval Software Support Infrastructure (ORAC, FMF, CSSC); ii. CFMWC, NETE support; and iii. Combat Systems Engineering Support contract; a) Lockheed Martin Canada.			
2.9	Documentation; 1. Combat Systems User Manual (CSUM); 2. Version Description Documentation (VDD); 3. Sub-System Design Documentation (SSDD); 4. Interface Design Specification (IDS)/Interface Control Document (ICD) (CCS 280); and 5. Standard Military Publications; i. STANAGS (NTDS); and ii. MILSTD/MILSPEC.			
2.10	Operating policies and procedures			

5. Total EO Time: Total: 15 x 50 minute periods as follows:

   a. 15 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Combat System Support Center (East) – assistance may be required for course content, validation and SME guest lectures.

   b. Prerequisites:

      1) EO 006.01 – Describe Command and Control System Theory; and

      2) EO 007.03 – Describe Computer Networks.

PO 006 – MANAGE COMMAND AND CONTROL SYSTEMS

EO 006.04 DESCRIBE CCS OPERATIONS

1. Performance: Describe CCS Operations

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe CCS Operations:

   a. IAW Refs, by completing the following:

      (1) Describe the Roles and responsibilities of the Ops Room watch in all states of readiness and degrees of preparedness;

      (2) Describe In-Service Command and Control System Operation; and

      (3) Describe TEWA using functional sequence diagrams.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the roles and responsibilities of the Ops Room watch in all states of readiness and degrees of preparedness**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2	**Describe In-Service Command and Control System Software**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	
2.1	Modes of Operation; 1. QR modes; i. Manual; ii. Semi-auto; iii. Auto-engage; and iv. Auto-assign.			

Teaching Points		Instructional Strategies	Time (min)	References
	2. QR parameters; i. Range; ii. Speed; and iii. Closest Point of Approach (CPA). 3. Detect to Engage sequence (Signal Flow). i. Range; ii. Speed; and iii. Closest Point of Approach (CPA), 4. Track-While-Scan capability; and 5. Training (TR) module/simulation.			
2.2	Layered defence (detect to engage sequence); 1. Anti-ship missile defence (ASMD); and i. ZIPPOs; and ii. Resolve procedures. 2. Torpedo counter measures (TCM).			
2.3	Recognized Maritime Picture; 1. Track generation; i. EW; ii. RADAR; a) AIAT; b) MIAT; and c) AICAT. iii. Acoustic; and iv. LINK. 2. Track correlation; 3. Track capacity; 4. Track quality; and 5. Track classification (friendly, unknown, hostile, joker, faker, air, surface, subsurface).			
2.4	Normal CCS configuration; 1. IFF background setup;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. RADAR backgrounds; 3. CCS QR mode and QR parameters; 4. Track allocations (LINK, SPS-49, SG-150…); and			
2.5	Display modes and functional assignments (CO, ORO, SWC, RT, etc.)			
2.6	Multi Functional Workstations (MFW) operations			
2.7	Periodic operational software and weapon system restrictions messages			
3	**Describe TEWA using functional sequence diagrams**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**250**	

5. Total EO Time: Total: 11 x 50 minute periods as follows:

   a. 11 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Para 7 (1), 7(2) above may be instructed by CFNOS;

   b. Para 7 (3) above may be instructed by CFMWC; and

   c. Prerequisites:

      1) EO 006.02 – Describe In-Service Command and Control System Hardware; and

      2) EO 006.03 – Describe CCS Software.

PO 006 – MANAGE COMMAND AND CONTROL SYSTEMS

EO 006.05 SUPPORT CCS ONBOARD HMC SHIPS

1. Performance: Support CCS onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall Support CCS onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard CMS 330 system to the satisfaction of the CSEO;

      (2) Conduct a walk-through of CMS 330 system with the senior maintainer;

      (3) Deliver a brief on the CMS 330 system to the satisfaction of the CSEO;

      (4) Stand one (1) watch with the CSE TOW;

      (5) Stand two (2) watches with the ORO/ORS; and

      (6) Submit a Technical Defect Report on a major problem with the CMS 330 selected by the CSEO.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time	References
1	**The NCS ENG shall complete the following for the CCS to the satisfaction of the CSEO**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time	References
1.1	Observe and participate in the following; 1. Operations; i. All Ops Room personnel at all positions; ii. Watch turnover of Ops Room personnel; iii. Manipulate SMC pages; iv. Detect to engage sequence as SWC; and v. History recording/data collection. 2. Tests and Trials; i. Online/offline diagnostics; ii. Weapons Certification iii. Combat System Alignment Test (CSAT); iv. Combat System Interface Test (CSIT); v. Combat System Operability Test (CSOT). 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
1.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			

Teaching Points		Instructional Strategies	Time	References
1.3	Draw the CSE diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components and signal flow; 3. System software; 4. Modes of operation; 5. System redundancies; 6. System diagnostic capabilities/tools; 7. Interfacing/integration with other systems; 8. Maintenance requirements; 9. System trials/alignments; 10. Safety considerations; 11. Manning requirements; 12. Operating policy and procedures; and 13. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
2	**Present a system briefing to designated ship's staff on the fitted CCS. This brief shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	System description			
2.2	System overview			
2.3	Major system components			
2.4	Operating modes			
2.5	Capabilities and limitations			
2.6	Interfaces			
2.7	Redundancies/survivability			
2.8	Combat/CSE liaison considerations			
3	**Stand CSE and CBT Department Watches;**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	Stand one (1) watch with the CSE TOW			
3.2	Two (2) watches with ORO/ORS			
4	**Submit a technical report on a major problem with the fitted CCS as selected by the CSEO. (Note: Only one submission is required for PO 006). As a minimum, the report shall cover the following points:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Identification/discussion of the problem			
4.2	Operational impact of the problem			
4.3	Options to resolve the problem			
4.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

PO 007 – MANAGE DATA PROCESSING SYSTEMS

EO 007.01 DESCRIBE ANALOG ELECTRONICS THEORY

1. Performance: Describe Analog Electronics Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Analog Electronics Theory:

   a. IAW Refs, by completing the following:

      (1) Describe Components (Types, properties, pros and cons, applications) of Semiconductors;

      (2) Describe Components (Types, properties, pros and cons, applications) of Diodes;

      (3) Describe Components (Types, properties, pros and cons, applications) of Bipolar Junction Transistors;

      (4) Describe Components (Types, properties, pros and cons, applications) of Field Effect Transistors (FET);

      (5) Describe Components (Types, properties, pros and cons, applications) of Operational Amplifiers (Op-amps); and

      (6) Describe Components (Types, properties, pros and cons, applications) of Other Circuits.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time	References
1	**Describe Components (types, properties, pros and cons, applications) of Semiconductors**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**350**	
1.1	Atomic structure			

Teaching Points		Instructional Strategies	Time	References
1.2	Doping			
2	**Describe Components (Types, properties, pros and cons, applications) of Diodes**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
2.1	Functional Description of how a diode works			
2.2	Semiconductor description of how a diode works			
2.3	Basic diode circuits: Clippers, clampers, rectifiers;			
2.4	Diode Parameters: 1. I_o 2. I_{pk} 3. T_{rr}; and 4. V_f.			
2.5	Special Diodes; 1. Schottky; 2. LEDs; 3. Constant current; 4. Tunnel; and 5. Zener.			
3	**Describe Components (Types, properties, pros and cons, applications) of Bipolar Junction Transistors**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
3.1	Functional Description of how a transistor works			
3.2	Semiconductor description of how a transistor works			
3.3	Transistor Operating regions: 1. Active; 2. Saturation; and 3. Cut-off			
3.4	The transistors as a switch			
3.5	Transistor amplifiers; 1. Transistor Biasing; i. Why it's done; and ii. Why it's necessary. 2. Small Signal; i. Common Emitter; ii. Common Base; and iii. Common Collector. 3. Large Signal Linear (Power);			

Teaching Points		Instructional Strategies	Time	References
	i. Class A; ii. Class AB; and iii. Class B. 4. Large Signal non-linear: i. Class C.			
4	**Describe Components (Types, properties, pros and cons, applications) of Field Effect Transistors (FETs)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
4.1	Functional Description of how a FET works			
4.2	Semiconductor description of how a FET works			
4.3	Different types of FETs: 1. JFET; 2. E-MOSFET: and 3. D-MOSFET,			
4.4	The FET as a switch			
4.5	FET amplifiers (briefly)			
5	**Describe Components (Types, properties, pros and cons, applications) of Operational Amplifiers (Op-Amps)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
5.1	Functional Description			
5.2	Characteristics of Op-amps (Z_{in}, Z_{out}, Gain, GBW)			
5.3	The two golden rules of Op-Amps			
5.4	Open-loop Op-amps			
5.5	Op-amps with negative feedback			
5.6	Op-amps with positive feedback			
5.7	Construction details of op-amps; 1. Difference amplifier; 2. Output stage; and 3. Frequency compensation.			
6	**Describe Components (Types, properties, pros and cons, applications) of Other Circuits**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
6.1	Active Filters			
6.2	Oscillators: sine wave, relaxation			
6.3	Mixers			
6.4	Voltages Regulators			
6.5	Reading a schematic			

5. Total EO Time: Total: 33 x 50 minute periods as follows:

a. 33 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 007 – MANAGE DATA PROCESSING SYSTEMS

EO 007.02 DESCRIBE DIGITAL ELECTRONIC THEORY

1. Performance: Describe Digital Electronic Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Digital Electronic Theory:

   a. IAW Refs, by completing the following:

      (1) Describe Combinatorial Logic Circuits;

      (2) Describe Sequential Logic Circuits;

      (3) Describe Integrated Circuits Families; and

      (4) Describe Microprocessors/Microcontroller.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time	References
1	**Describe Combinatorial Logic Circuits**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**450**	
1.1	Basic logic gates: AND, OR, NOT			
1.2	Boolean algebra			
1.3	DeMorgan's Theorems			
1.4	Logic Reduction			
2	**Describe Components (Types, properties, pros and cons, applications) of Diodes**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
2.1	Memory elements: Latch, Flip-Flop, Register			

Teaching Points		Instructional Strategies	Time	References
2.2	Finite State Machines: Controlling a process			
2.3	Properties of real logic: propagation delay, setup time, hold time			
2.4	Interfacing Asynchronous signals with a synchronous circuits: switch debouncing etc.			
3	**Describe Components (Types, properties, pros and cons, applications) of Bipolar Junction Transistors**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
3.1	TTL			
3.2	NMOS/PMOS, CMOS			
3.3	ECL/PECL			
4	**Describe Components (Types, properties, pros and cons, applications) of Field Effect Transistors (FETs)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
4.1	Architecture; 1. CPU; 2. Memory; 3. Buses (address, Data, Control, Memory); and 4. Clock (Timing, Trigger Pulse). 5. Registers; and 6. Number systems; i. Octal; ii. Hexadecimal; iii. Fixed point; iv. Floating point; and v. ASCII.			
4.2	Programming and Interfacing			

5. Total EO Time: Total: 33 x 50 minute periods as follows:

   a. 33 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Digital electronics design and simulation software including Xilinux and MultiSim.

7. Remarks: Nil.

PO 007 – MANAGE DATA PROCESSING SYSTEMS

EO 007.03 DESCRIBE COMPUTER NETWORK SYSTEM THEORY

1. Performance: Describe Computer Network System Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Computer Network System Theory:

   a. IAW Refs, by completing the following:

      (1) Describing the networks system;

      (2) Describing the network types;

      (3) Describing the transmissions;

      (4) Describing the Hardware Components;

      (5) Describing the Operating Systems;

      (6) Describing the Network Application;

      (7) Describing the networks security; and

      (8) Describing the SHIPLAN.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time	References
1	**Describe Computer Network Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**1950**	
1.1	Historical Background			
2	**Network Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	

Teaching Points		Instructional Strategies	Time	References
2.1	ISO/OSI Model			
2.2	Definitions; i. Local Area Network (LAN); ii. Metropolitan Area Network (MAN); iii. Wide Area Network (WAN); iv. Enterprise Network; and v. World Wide Web			
2.3	Structures/Topographies; i. Token ring; ii. Ethernet; iii. Carrier Sense (e.g. CSMA); and iv. Contention/Non-contention			
2.4	Protocols; i. Packet Switching; a) TCP/IP; and b) Point to Point (PPP). ii. Circuit Switching			
2.5	Capacity/Bandwidth.			
3	**Network Types**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Client/Server; i. Clients; and ii. Servers (DNS, Mail, Applications)			
3.2	Peer to Peer; and			
3.3	Satellite			
4	**Transmission**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Media and Connectors; i. Coaxial/Triaxial cable; ii. Twisted Pair (UTP, STP); iii. Fibre; and iv. Wireless.			
4.2	Methods; i. Digital Subscriber Line (DSL); ii. ISDN; and iii. ATM.			
4.3	Data Converters;			

Teaching Points		Instructional Strategies	Time	References
	i. Asynchronous Universal Interface (AUI); and ii. Modems			
5	**Hardware Components**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Network Components; i. Routers; ii. Hubs; iii. Switches; iv. Bridges; and v. Gateways			
6	**Operating Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Purpose			
6.2	Elements/Components			
6.3	Network (e.g. Novell) vs. Stand Alone			
6.4	UNIX; and			
6.5	Windows			
7	**Network Application**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	File Transfer Protocol			
7.2	Remote Terminal Applications (SSH, Telnet, etc.); and			
7.3	Ping			
8	**Security**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Physical			
8.2	Encryption; i. Data Encryption Standard (DES); and ii. Pretty Good Protection (PGP).			
8.3	Firewall			
8.4	Virtual Private Network (VPN)			
8.5	Threats; i. Viruses; ii. Worms; and iii. Trojans.			
9	**Bandwidth Management**	**Method:** Lecture		

Teaching Points		Instructional Strategies	Time	References
		Media: PowerPoint **Environment:** Classroom		
9.1	Hardware; i. Statistical Multiplexer; ii. Server; iii. Segmentation; and iv. TELCO (T1/T3, Fractional).			
9.2	Software; i. Compression; and ii. Domino Servers			
10	**SHIPLAN**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

5. Total EO Time: Total: 39 x 50 minute periods as follows:

   a. 39 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 007.01 – Describe Analog Electronics Theory; and

      2) EO 007.02 – Describe Digital Electronics Theory.

PO 007 – MANAGE DATA PROCESSING SYSTEMS

EO 007.04 DESCRIBE TACTICAL DATA LINK SYSTEMS

1. Performance: Describe Tactical Data Link Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Tactical Data Link Systems:

   a. IAW Refs, by completing the following:

      (1) Describe a Generic Tactical Data Link System; and

      (2) Describe In-service Tactical Data Link Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time	References
1	**Describe a Generic Tactical Data Link System**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**300**	
1.1	Historical Background (minimal)			
1.2	Data Link Theory; 1. Purpose; 2. Real time data transfer; 3. Standards i. STANAGS; and ii. NTDS Symbology. 4. Link-11; i. RF Signal; a) Frequency Spectrum Representation; b) Modulation; and c) HF/UHF/Satellite. ii. Data Transfer modes/method; iii. Message Formats; iv. Network protocols;			

Teaching Points		Instructional Strategies	Time	References
	a) Net Control Station (NCS) vs. pickets. v. Grid Reference System and a) Data Link Reference Pont (DLRP). vi. Operating Modes. 5. Link-16; i. RF Signal; ii. Message Formats; iii. Network Protocols; iv. Grid Reference Systems; and v. Operating Modes. 6. Link 22;			
1.3	Emerging Technologies; and 1. Multi-Link.			
2	**Describe In-Service Tactical Data Link Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Performance Characteristics, capabilities, and limitations			
2.2	System components/signal flow/software; 1. Link-11; i. System Block Diagram. 2. Link-16; and i. System Block Diagram. 3. Link 22			
2.3	Modes of Operation			
2.4	System Redundancies			
2.5	System diagnostic capabilities/tools; 1. Link Monitoring System (LMS); 2. POFA; and 3. Loopback;			
2.6	Interfacing/integration with other systems (e.g. CCS, electrical, HVAC, etc.)			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety considerations			
2.10	Operating policies and procedures			

5. Total EO Time: Total: 8.5 x 50 minute periods as follows:

a. 8.5 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites:

1) EO 003.01 – Describe Communication Theory.

PO 007 – MANAGE DATA PROCESSING SYSTEMS

EO 007.05 SUPPORT DATA PROCESSING SYSTEMS ONBOARD HMC SHIPS

1. Performance: Support Data Processing Systems onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall Support Data Processing Systems onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard Data Processing Systems to the satisfaction of the CSEO;

      (2) Conduct a walk-through of Data Processing Systems with the senior maintainer;

      (3) Deliver a brief on the Naval Information System (NavIS) to the satisfaction of the CSEO; and

      (4) Submit a Technical Defect Report on a major problem with the Data Processing System selected by the CSEO.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The NCS ENG shall complete the following all Computer Network Systems to the satisfaction of the CSEO (including HSCD/GCCS-M):**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Observe and participate in the following; 1. Operations; i. Setup of network; ii. Monitor network; iii. Administration of network; and iv. Usage of network. 2. Tests and Trials; i. POFA; and ii. Ping. 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
1.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			
1.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components/signal flow including Network Operations Centre (NOC); 3. List System Software (O/S, non-admin related software); 4. Modes of operation; 5. System redundancies; 6. System diagnostic capabilities/tools; 7. Interfacing/integration with other systems; 8. Maintenance requirements; 9. System trials; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component			

Teaching Points		Instructional Strategies	Time (min)	References
	failure (Components to be selected by CSEO).			
2	**The NCS ENG shall complete the following all Tactical Data Link Systems to the satisfaction of the CSEO**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Observe and participate in the following; 1. Operations; i. Setup of network; ii. Monitor network; iii. Administration of network; and iv. Usage of network. 2. Tests and Trials; i. POFA; and ii. Ping. 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
2.2	Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to: 1. Identify all system components; 2. Discuss considerations for all related CSE rounds; and 3. Identify MSE systems considerations.			

Teaching Points		Instructional Strategies	Time (min)	References
2.3	Draw the system diagram and thoroughly explain the system to the CSEO (without references). This explanation shall include: 1. Performance characteristics, capabilities and limitations; 2. System components/signal flow including Network Operations Centre (NOC); 3. List System Software (O/S, non-admin related software); 4. Modes of operation; 5. System redundancies; 6. System diagnostic capabilities/tools; 7. Interfacing/integration with other systems; 8. Maintenance requirements; 9. System trials; 10. Manning requirements; 11. Operating policy and procedures; and 12. CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).			
3	**Present a system briefing to designated ship's staff for a fitted Data Processing system selected by the CSEO. This brief shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	System description			
3.2	System overview			
3.3	Major system components			
3.4	Operating modes			
3.5	Capabilities and limitations			
3.6	Interfaces			
3.7	Redundancies/survivability			
3.8	Combat/CSE liaison considerations			
4	**Submit a technical report on a major problem with the Data Processing system as selected by the CSEO. (Note: Only one submission is required for PO 007). As a minimum, the report shall cover the following points:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Identification/discussion of the problem			
4.2	Operational impact of the problem			

Teaching Points		Instructional Strategies	Time (min)	References
4.3	Options to resolve the problem			
4.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

PO 008 – MANAGE ELECTRONIC WARFARE (EW) SYSTEMS

EO 008.01 DESCRIBE EW FUNDAMENTALS

1. Performance: Describe EW Fundamentals

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe EW Fundamentals:

   a. IAW Refs, by completing the following:

      (1) Describe Basic EW Theory;

      (2) Describe Electronic Support Measures (ESM);

      (3) Describe Electronic Attack (EA); and

      (4) Describe Electronic Protective Measures (EPM).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Basic EW Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
1.1	Definitions and Purpose			
1.2	Support infrastructure; 1. Library generation; i. Canadian Forces Electronic Warfare Center (CFEWC); ii. Naval Electronic Warfare Center (NEWC); and iii. TRINITY (Comm. research, SIGNINT). 2. Intercept reporting; and 3. Intelligence.			

Teaching Points		Instructional Strategies	Time (min)	References
2	**Describe Electronic Support Measures**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
2.1	Purpose			
2.2	Probability of Intercept; 1. Angular coverage (Elevation, Azimuth); 2. Bearing accuracy/resolution; 3. Sensitivity; 4. Selectivity; 5. Frequency Coverage; 6. Dynamic Range; and 7. Dwell time.			
2.3	Direction Finding			
2.4	Pulse sorting			
2.5	Processing Time			
2.6	Interfacing/integration with other systems			
2.7	Identification			
2.8	Receiver types; 1. Super Heterodyne Receiver; 2. Direct detection (Crystal); 3. Digital IFM; 4. Bragg Cell; and 5. Others.			
2.9	Communication Intercept			
2.10	Multiple-Emitter Phenomena			
2.11	Matching parameters to requirements; 1. Ideal ESM Receiver.			
3	**Describe Electronic Attack (EA)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**350**	
3.1	Purpose			
3.2	Performance Characteristics; 1. Output Power; 2. Deployment range; 3. Hang time (persistence); 4. Frequency Coverage; 5. Sensitivity; 6. Selectivity; 7. Dynamic Range; 8. Bearing Accuracy; 9. Tracking accuracy;			

Teaching Points		Instructional Strategies	Time (min)	References
	10. Variable polarization; 11. Polarization purity; 12. Processing Time; 13. Pulse Sorting Limitations; 14. Library Matching Techniques; and 15. Others.			
3.3	Burn through range			
3.4	Active vs. Passive; 1. Active; i. Jammer Types; a) Responsive; and b) Repeater. ii. Deceptive; a) Range Gate Pull-Off; b) False targets (sidelobes); c) Inverse gain; d) Cross polarization; e) Cross eyed; f) Smart noise; g) Off board seduction; and h) Others. 2. Passive; i. Seduction (decoys); ii. Distraction (decoys); iii. Confusion; iv. Chaff features/properties/composition; v. Signature reduction a) RADAR absorbent material; i) RAM pads; and ii) Paint. b) Structural design; and c) Emission Control.			
3.5	Onboard vs. Off board; 1. Onboard; i. Power out vs. RCS; ii. Counter targeting; and iii. Emitter Identification. 2. Off board; i. Deployment Timing; ii. Emitter Identification; and iii. Fly out patterns.			
4	**Describe Electronic Protective Measures**	**Method:** Lecture	**350**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
4.1	Purpose;			
4.2	Transmitter; 1. Output power; 2. Frequency agility; 3. Frequency diversity; and 4. PRF agility (stagger, jitter, shift).			
4.3	Receiver; 1. Guard gates; 2. IAGC; 3. Dynamic range; and 4. Others.			
4.4	Antenna; 1. Sidelobe blanking; 2. Sidelobe reduction (in design); 3. Coherent sidelobe cancellation; 4. Variable polarization; and 5. Others.			

5. Total EO Time: Total: 30 x 50 minute periods as follows:

   a. 30 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 004.01 – Describe EM Theory;

      2) EO 004.02 – Describe Antenna Theory; and

      3) EO 004.03 – Describe RADAR Theory.

PO 008 – MANAGE ELECTRONIC WARFARE (EW) SYSTEMS

EO 008.02 DESCRIBE ESM SYSTEMS

1. Performance: Describe ESM Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe ESM Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic ESM Systems; and

      (2) Describe In-service ESM Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic ESM Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Types (advantages/disadvantages/comparison); 1. Communication Intercept; 2. Threat warning; and 3. Signature analysis (ELINT).			
1.2	System components; 1. Antenna(s); 2. Receiver; 3. Processor; 4. Library; and 5. Display.			
1.3	Interfacing/Integration with other systems; 1. Blanker (Ship's Emitters); 2. CCS; 3. Navigation Data Distribution System; 4. Electrical; and 5. Environmental.			

Teaching Points		Instructional Strategies	Time (min)	References
1.4	System Trials/Alignments; 1. System analysis (NESTR).			
1.5	Future In-Service ESM Systems			
2	**Describe In-Service ESM Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow/ software			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostics capabilities/tools			
2.6	Interfacing/Integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety Considerations			
2.10	Manning requirements			
2.11	Operating Policy and procedures; 1. Organizations; i. CFEWC; ii. NEWC; and iii. Trinity (Comm. research, SIGNINT).			

5. Total EO Time: Total: 3.5 x 50 minute periods as follows:

   a. 3.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 008.01 – Describe Electronic Warfare Theory.

PO 008 – MANAGE ELECTRONIC WARFARE (EW) SYSTEMS

EO 008.03 DESCRIBE EA SYSTEMS

1. Performance: Describe EA Systems

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe EA Systems:

   a. IAW Refs, by completing the following:

      (1) Describe Generic EA Systems; and

      (2) Describe In-service EA Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Generic ESM Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Types (advantages/disadvantages/comparison); 1. Active/Passive; 2. Onboard/offboard; and 3. Integrated vs. Stand-alone			
1.2	System components; 1. Antenna(s); 2. Receiver; 3. Launcher; 4. Processor; and 5. Library.			
1.3	Interfacing/Integration with other systems; 1. ESM Systems; 2. CCS; 3. Navigation Data Distribution System; 4. Electrical; and			

Teaching Points		Instructional Strategies	Time (min)	References
	5. Environmental.			
1.4	System Trials/Alignments; 1. System analysis (NESTR).			
1.5	Future In-Service Technologies			
2	**Describe In-Service EA Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Performance characteristics, capabilities and limitations			
2.2	System components/signal flow/ software			
2.3	Modes of operation			
2.4	System redundancies			
2.5	System diagnostics capabilities/tools			
2.6	Interfacing/Integration with other systems			
2.7	Maintenance requirements			
2.8	System trials/alignments			
2.9	Safety Considerations; 1. RF Safety; and 2. Ammunition.			
2.10	Manning requirements			
2.11	Operating Policy and procedures; 1. Organizations; i. CFEWC; ii. NEWC.			

5. Total EO Time: Total: 3.5 x 50 minute periods as follows:

   a. 3.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 008.01 – Describe Electronic Warfare Theory.

PO 008 – MANAGE ELECTRONIC WARFARE (EW) SYSTEMS

EO 008.04 SUPPORT EW SYSTEMS ONBOARD HMC SHIPS

1. Performance: Support EW Systems onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall Support EW Systems onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Observe operation and maintenance of shipboard EW Suite to the satisfaction of the CSEO;

      (2) Conduct a walk-through of all EW Systems with the senior maintainer;

      (3) Deliver a brief on the EW Suite to the satisfaction of the CSEO;

      (4) Stand two (2) watches with EW System; and

      (5) Submit a Technical Defect Report on a major problem with an EW Suite selected by the CSEO.

4. Teaching Points:

Teaching Points		**Instructional Strategies**	**Time** (min)	**References**
1	**The NCS ENG shall complete the following all Computer Network Systems to the satisfaction of the CSEO (including HSCD/GCCS-M):**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Observe and participate in the following; 1. Operations; i. EA pre and post firing checks; ii. EA Pre-fire brief;			

Teaching Points		Instructional Strategies	Time (min)	References
	iii. Loading and unloading ammunition; iv. Observe EA System firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire crew); v. Jamming/Deception; vi. ADEW, EWEX, Encounter Ex; vii. Direction finding; and viii. Communications Intercept. 2. Tests and Trials; i. NESTR(A/P) Trial; and ii. Daily/Periodic System Operational Tests (DSOT/PSOT). 3. Maintenance; i. Preventive; and ii. Corrective. 4. Flash-up/Shutdown; 5. Pre-sail checks; and 6. Trouble shooting.			
2	**Conduct a walk-through of the system with the senior maintainer, demonstrating a thorough ability to:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Identify all system components			
2.2	Discuss considerations for all related CSE rounds			
2.3	Identify MSE systems considerations			
3	**Draw the EW system diagrams and verbally explain the system to the CSEO (without references). This explanation shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	Performance characteristics, capabilities and limitations			
3.2	System components/signal flow			
3.3	System software and libraries			
3.4	Modes of operation			
3.5	System redundancies			
3.6	System diagnostic capabilities/tools			
3.7	Interfacing/integration with other systems			
3.8	Maintenance requirements			
3.9	System trials/alignments			

Teaching Points		Instructional Strategies	Time (min)	References
3.10	Safety considerations			
3.11	Manning requirements			
3.12	Operating policy and procedures			
3.13	CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO)			
4	**Present a system briefing to designated ship's staff for a fitted Data Processing system selected by the CSEO. This brief shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	System description			
4.2	System overview			
4.3	Major system components			
4.4	Operating modes			
4.5	Capabilities and limitations			
4.6	Interfaces			
4.7	Redundancies/survivability			
4.8	Combat/CSE liaison considerations			
5	**Stand OPs Departmental Watches**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5.1	Stand two (2) watches with the EWS			
6	**Submit a technical report on a major problem with the EW System as selected by the CSEO. (Note: Only one submission is required for PO 008). As a minimum, the report shall cover the following points:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6.1	Identification/discussion of the problem			
6.2	Operational impact of the problem			
6.3	Options to resolve the problem			
6.4	Actual/recommended solution to the problem			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   b. OJT.

PO 009 – MANAGE AMMUNITION

EO 009.01 DESCRIBE EXPLOSIVE THEORY

1. Performance: Describe Explosive Theory

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Explosive Theory:

   a. IAW Refs, by completing the following:

      (1) Describe properties of Propellant;

      (2) Describe properties of Explosives;

      (3) Define Net Explosive Quantity (NEQ);

      (4) Describe Hazard Classification Codes (HCC); and

      (5) Describe Aggregation and Mixing Rules.

4. Teaching Points:

Teaching Points		**Instructional Strategies**	**Time** (min)	**References**
1	**Describe properties of Propellants**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2	**Describe properties of Explosives**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Environmental effects; 1. Temperature; 2. Humidity; 3. Confinement; 4. Radio frequency (HERO); and 5. Abnormal conditions.			

3	**Define Net Explosive Quantity (NEQ)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4	**Describe Hazard Classification Codes (HCC)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4.1	Hazard Classifications			
4.2	Compatibility groups			
4.3	Mixing of compatibility groups			
4.4	Quantity Distance			
5	**Describe Aggregation and Mixing Rules**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

5. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisite:

      1) EO 001.03 – Describe Naval Gun Systems;

      2) EO 001.04 – Describe Naval Missile Systems; and

      3) EO 001.05 – Describe Underwater Weapon Systems.

PO 009 – MANAGE AMMUNITION

EO 009.02 DESCRIBE IN-SERVICE NAVAL AMMUNITION

1. Performance: Describe In-service Naval Ammunition

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe In-service Naval Ammunition:

   a. IAW Refs, by completing the following:

      (1) Describe Ammunition Types;

      (2) Describe Primers and Fuses;

      (3) Describe Lotting and Batching; and

      (4) Describe Ammunition Logistical Documentation and Forms.

4. Teaching Points:

Teaching Points		**Instructional Strategies**	**Time** (min)	**References**
1	**Describe Ammunition Types**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
1.1	Small arms			
1.2	Gun			
1.3	Missiles			
1.4	Rockets			
1.5	Torpedoes			
1.6	Demolitions			
1.7	Pyrotechnics			
2	**Describe Primers and Fuses**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Percussion; Electric; and			
2.2	Percussion – electric.			

3	**Describe Lotting and Batching**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
3.1	Expiry dates (Service, installed, and shelf life)			
3.2	Rotation of stock			
3.3	Colour coding			
4	**Describe Ammunition Logistical Documentation and Forms**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4.1	Logistical Data Guide			
4.2	Ammunition Restriction			
4.3	Ammunition Log			
4.4	Description of forms			
4.5	Data Cards			
5	**Describe Aggregation and Mixing Rules**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

5. Total EO Time: Total: 9 x 50 minute periods as follows:

   a. 9 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 001.03 – Describe Naval Gun Systems;

      2) EO 001.04 – Describe Naval Missile Systems; and

      3) EO 001.05 – Describe Underwater Weapon Systems.

   b. Tour of CFAD.

PO 009 – MANAGE AMMUNITION

EO 009.03 DESCRIBE AMMUNITION ACCOUNTING

1. Performance: Describe Ammunition Accounting

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Ammunition Accounting:

   a. IAW Refs, by completing the following:

      (1) Describe a Supply Customer Account;

      (2) Describe the MAAL and MCAAL;

      (3) Describe Ammunition Demands;

      (4) Describe Ammunition Expenditure Reporting;

      (5) Describe Ammunition Stock Verifications; and

      (6) Describe Ammunition Accounting Deviations.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe a Supply Customer Account**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	MIMS accounting system			
1.2	AIMS accounting system			
1.3	Actions to be taken for accounting discrepancies			
2	**Describe Primers and Fuses**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Differences between readiness levels			

3	**Describe Ammunition Demands**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
4	**Describe Ammunition Expenditure Reporting**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
5	**Describe Aggregation and Mixing Rules**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
5.1	Logistic Compliance Inspections (LCI)			
5.2	Quarter Inspections			
6	**Describe Ammunition Accounting Deviations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
6.1	Request to go below OP Stock			
6.2	Return of Defective Ammunition			

5. Total EO Time: Total: 4.5 x 50 minute periods as follows:

   a. 4.5 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Prerequisites:

      1) EO 001.03 – Describe Naval Gun Systems;

      2) EO 001.04 – Describe Naval Missile Systems; and

      3) EO 001.05 – Describe Underwater Weapon Systems.

PO 009 – MANAGE AMMUNITION

EO 009.04 DESCRIBE THE AMMUNTION ORGANIZATION

1. Performance: Describe the Ammunition Organization

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Ammunition Organization:

   a. IAW Refs, by completing the following:

      (1) Identify the Ammunition Organizations; and

      (2) Describe the Roles and Responsibilities of the Ammunition Organizations.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Identify the Ammunition Organizations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	DAER			
1.2	DMPOR			
1.3	DAPM			
1.4	J4 Ammo			
1.5	N34			
1.6	FATA			
1.7	CFAD			
1.8	Ships			
1.9	Bases			
2	**Describe the Roles and Responsibilities of the Ammunition Organizations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**175**	

5. Total EO Time: Total: 4.5 x 50 minute periods as follows:

a. 4.5 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites:

1) EO 001.03 – Describe Naval Gun Systems;

2) EO 001.04 – Describe Naval Missile Systems; and

3) EO 001.05 – Describe Underwater Weapon Systems.

PO 009 – MANAGE AMMUNITION

EO 009.05 DESCRIBE THE AMMUNITION AND EXPLOSIVE SAFETY PROGRAM

1. Performance: Describe the Ammunition and Explosive Safety Program

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Ammunition and Explosive Safety Program:

   a. IAW Refs, by completing the following:

      (1) Describe the DND Ammunition and Explosive Safety Program;

      (2) Describe the Unit / Formation Ammunition and Explosive Safety Program;

      (3) Describe Ammunition Directives and Orders;

      (4) Describe Magazine Storage and Handling Requirements;

      (5) Describe Ammunition and Explosive Inspections;

      (6) Describe Ammunition Ancillary Equipment and Support;

      (7) Describe Magazine Certification; and

      (8) Describe Ammunition Deviations and Waivers.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the DND Ammunition and Explosive Safety Program**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	

2	**Describe the Unit / Formation Ammunition and Explosive Safety Program**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3	**Describe Ammunition Demands**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3.1	Explosive Safety Manuals			
3.2	DND Explosive Safety Program			
3.3	Ammunition and Explosive Instructions			
4	**Describe Ammunition Expenditure Reporting**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
4.1	Prohibited Articles			
4.2	Segregation			
4.3	Lockers			
4.4	Ready-use			
4.5	Compatibility			
4.6	Fraction Boxes / Tag			
4.7	Maintenance Requirements			
5	**Describe Ammunition and Explosive Inspections**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
5.1	C-03-010-507/NT-001 – Trial Agenda for Shipboard Ammunition Stowage's			
6	**Describe Ammunition Ancillary Equipment and Support**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
6.1	Lifting Appliances			
6.2	Flood and Spray			
7	**Describe Magazine Certification**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
8	**Describe Ammunition Deviations and Wavers**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	

5. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks:

a. Prerequisites:

1) EO 001.03 – Describe Naval Gun Systems;

2) EO 001.04 – Describe Naval Missile Systems; and

3) EO 001.05 – Describe Underwater Weapon Systems.

PO 009 – MANAGE AMMUNITION

EO 009.06 PARTICIPATE IN AMMUNITION OPERATIONS ONBOARD HMC SHIPS

1. Performance: Participate in Ammunition Operation onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall Participate in Ammunition Operation onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Participate in Ammunition Operations;

      (2) Liaise with ammunition stakeholders under supervision;

      (3) Prepare Ammunition Documentation under supervision;

      (4) Participate in Inspections and Rounds;

      (5) Respond to Ammunition Accidents and Incidents under supervision; and

      (6) Manage work in proximity to ammunition/magazines under supervision.

4. Teaching Points:

Teaching Points		**Instructional Strategies**	**Time** (min)	**References**
1	**Participate in Ammunition Operations**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Embarking/Disembarking and internal transfer of Ammunition; 1. Alongside in Home Port i. Operational necessity; and ii. Jetty limits. 2. Alongside in Foreign Port; i. Local Regulations; and ii. Ammunition Depot or buoy. 3. At sea ammo transfers.			
2	**Liaise with ammunition stakeholders under supervision**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Ammunition Depot			
2.2	Formation Ammunition Technical Authority 1. Formation Ammunition Technical Officer (FATO); and 2. Formation Ammunition Inspector.			
2.3	Fleet/Formation Staff			
3	**Prepare Ammunition Documentation under supervision**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	Ammunition Orders; and			
3.2	Ammunition Check-list.			
4	**Participate in Inspections and Rounds**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Annual			
4.2	Adjacent compartment rounds			
4.3	Magazine rounds			
4.4	Free from ammunition certification			
5	**Respond to Ammunition Accidents and Incidents under supervision**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5.1	Definitions			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Accident; 2. Incident; 3. Defect; and 4. Malfunction.			
5.2	Reporting and investigation requirements;			
5.3	Damaged ammunition; 1. Retaining; 2. Jettisoning; and 3. Required reports.			
6	**Manage work in proximity to ammunition / magazines under supervision**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6.1	Docking Requirements 1. Routine; and 2. Emergency.			
6.2	Forces protection ammunition			
6.3	Burning and welding; 1. Definition; and 2. Max working distance.			
6.4	Waivers			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

PO 010 – MANAGE SHIP'S SIGNATURE

EO 010.01 DESCRIBE SHIP'S SIGNATURE MANAGEMENT

1. Performance: Describe Ship's Signature Management

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Ship's Signature Management:

   a. IAW Refs, by completing the following:

      (1) Describe Signature Elements;

      (2) Describe Policy and Procedures;

      (3) Describe Organizations;

      (4) Describe the types of ranging;

      (5) Describe Reduction Methods; and

      (6) Describe Documentation / Reports.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Signature Elements**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Definitions			
1.2	Theory			
1.3	Characteristics			
1.4	Normal/abnormal signature limits (parameters)			
1.5	Potential risk			
2	**Describe Policy And Procedure**	**Method:** Lecture **Media:** PowerPoint	**25**	

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	Overview of references; 1. NAVORD 46-109; and 2. CFCD 109 (Vol 1-4).			
3	**Describe Organizations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3.1	Shipboard (Unit)			
3.2	Formation			
3.3	Maritime Staff HQ			
4	**Describe the types of ranging**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
4.1	In water acoustic			
4.2	In water non-acoustic			
4.3	Infrared			
4.4	Check ranging			
4.5	Mission ranging			
4.6	Requirements			
5	**Describe Reduction Methods**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
5.1	Maintenance requirement			
5.2	RAM pads			
5.3	RADAR Absorbent Paint			
5.4	IR Suppression			
5.5	Quiet Running Bill			
5.6	Quiet Running States			
5.7	Anecoic Tile			
5.8	Degaussing			
5.9	Deperming			
5.10	Active Shaft Grounding			
5.11	Ship's Aspect			
5.12	Others			
6	**Describe Documentation / Reports**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
6.1	Semi Annual Report			

Teaching Points		Instructional Strategies	Time (min)	References
6.2	Range Reports			
6.3	Types of reports 1. Quick Look; and 2. Full.			
6.4	OPDEFs			

5. Total EO Time: Total: 4 x 50 minute periods as follows:

   a. 4 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. NESTR Tour

7. Remarks: Nil.

PO 010 – MANAGE SHIP'S SIGNATURE

EO 010.02 SUPPORT SHIP'S SIGNATURE MANAGEMENT ONBOARD HMC SHIPS

1. Performance: Support Ship's Signature Management onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall support Ship's Signature Management onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Participate in the Signature Management Organization (SMO);

      (2) Review and interpret trial results;

      (3) Observe and participate in the planning and conduct of ship's signature ranging;

      (4) Identify and understand Ship's Signature Logs;

      (5) Advise Command on ship's signature;

      (6) Participate in the conduct of Noise Rounds;

      (7) Participate in completion of ship's signature documentation; and

      (8) Demonstrate a detailed understanding of Signature Management to the satisfaction of the CSEO.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The NCS ENG shall, to the satisfaction of the CSEO and under supervision**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Participate in the Signature Management Organization (SMO)			
1.2	Review and interpret trial results			
1.3	Observe and participate in the planning and conduct of ship's signature ranging			
1.4	Identify and understand Ship's Signature Logs			
1.5	Advise Command on ship's signature			
1.6	Participate in the conduct of Noise Rounds			
1.7	Participate in completion of ship's signature documentation (SOPs/reports			
2	**Demonstrate a detailed understanding of Signature Management by explaining the following to the satisfaction of the CSEO**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Role and responsibility of the Signature Management Organization			
2.2	Types of signatures			
2.3	Signature reduction methods			
2.4	Range resources co-ordination			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

PO 011 – MANAGE SHIPBOARD HAZARD

EO 011.01 DESCRIBE THE SHIPBOARD RISK MANAGEMENT PROGRAMS

1. Performance: Describe the Shipboard Risk Management Programs

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Shipboard Risk Management Programs:

   a. IAW Refs, by completing the following:

      (1) Describe the following Risk Management Programs; and

      (2) Describe Ship's Safety Committee.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The NCS ENG shall, to the satisfaction of the CSEO and under supervision**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
1.1	Shipboard Environmental Management System (SEMS)			
1.2	Hazardous Material Management			
1.3	General Safety			
1.4	Radiation Safety (Ionizing)			
1.5	Non-Ionizing EM Radiation Hazards (RF SAFETY) safety			
1.6	Solid Waste Management			
2	**Describe Ship's Safety Committee**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

5. Total EO Time: Total: 2 x 50 minute periods as follows:

a. 2 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 011 – MANAGE SHIPBOARD HAZARD

EO 011.02 DESCRIBE THE SHIPBOARD NON-IONIZING RADIATION HAZARD SAFETY PROGRAM

1. Performance: Describe the Shipboard Non-Ionizing Radiation Hazard Safety Program

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe the Shipboard Non-Ionizing Radiation Hazard Safety Program:

   a. IAW Refs, by completing the following:

      (1) Describe SEMS;

      (2) Describe Safety Code 6;

      (3) Describe Leak Testing; and

      (4) Describe Formation Directives.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe NAVORD 43-2**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
1.1	Organization			
1.2	HERO			
1.3	HERP			
1.4	HERF			
1.5	Personnel Aloft			
1.6	RF SAFETY Emitter chart			
1.7	Positive Control of Emitters			
1.8	Control Chits			

2	**Describe Safety Code 6**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
3	**Describe Leak Testing**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4	**Describe Formation Directives**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

5. Total EO Time: Total: 7 x 50 minute periods as follows:

   a. 7 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 011 – MANAGE SHIPBOARD HAZARD

EO 011.03 DESCRIBE HAZARDOUS MATERIAL CLEAN-UP PROCEDURES

1. Performance: Describe Hazardous Material Clean-Up Procedures

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe Hazardous Material Clean-Up Procedures:

   a. IAW Refs, by completing the following:

      (1) Describe HAZMAT Fundamentals

      (2) Describe WHMIS;

      (3) Describe Shipboard Spill Clean-up; and

      (4) Participate in a Spill Clean-up Exercise.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe HAZMAT Fundamentals**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
1.1	Legislation and DND Policy			
1.2	Hazard classifications			
1.3	National Institute of Occupational Safety and Health (NIOSH) pocket guide			
2	**Describe WHMIS**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
2.1	Program			
2.2	Material Safety Data Sheets (MSDS)			
2.3	Labelling			

Teaching Points		Instructional Strategies	Time (min)	References
2.4	Education			
3	**Describe Shipboard Spill Clean-up**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
3.1	Organization; 1. Roles; and 2. Responsibilities.			
3.2	Equipment; 1. PPE; 2. Clean-up equipment; and 3. Detectors.			
3.3	Major risks; 1. Oxygen deficiency; 2. Toxic; 3. Explosive; and 4. Caustic.			
3.4	Specific shipboard HAZMAT; 1. Otto Fuel II; 2. JP10; 3. Refrigerants; 4. ISOPAR-M; 5. Cleaning products; 6. Fuel and lubricants; and 7. Others.			
3.5	Procedures; 1. Rapid Response; 2. Containment; 3. Casualties; 4. Clean-up; 5. Disposal; and 6. Investigation and reporting.			
4	**Participate in Spill Clean-up Exercise**	**Method:** Interactive Demo **Media:** N/A **Environment:** Damage Control Training Facility	**200**	

5. Total EO Time: Total: 14 x 50 minute periods as follows:

   a. 14 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids:

a. Damage Control Training Facility.

7. Remarks:

a. All instruction to be provided by TE staff at Damage Control Training Facility (DCTF). The students are not required to complete an EC for the practical demonstration.

PO 011 – MANAGE SHIPBOARD HAZARD

EO 011.04 PARTICIPATE IN RISK MANAGEMENT PROGRAMS ONBOARD HMC SHIPS

1. Performance: Participate in Risk Management Programs onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall Participate in Risk Management Programs onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Understudy the Unit RF SafetyO; and

      (2) Observe the HAZMAT clean-up team.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Understudy the Unit RF SafetyO including:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Present general RF SAFETY brief to ship's company			
1.2	Present a man aloft brief to designated personnel			
1.3	Co-ordinate man aloft/RF SAFETY evolutions during a DMP/SWP/DWP			
1.4	Conduct leak tests			
1.5	Observe RF SAFETY surveys			

2	**Observe the Spill clean-up team from the following positions:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Dressing area			
2.2	Spill area			
2.3	Command area			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

PO 012 – DIRECT EMERGENCY RESPONSE TEAM (ERT) OPERATIONS

EO 012.01 DESCRIBE ERT OPERATIONS

1. Performance: Describe ERT Operations

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment: Classroom

3. Standard: The trainee shall describe ERT Operations:

   a. IAW Refs, by completing the following:

      (1) Describe the roles and responsibilities of the ERT Organization;

      (2) Describe ERT Policy and Procedures;

      (3) Describe ERT Resources; and

      (4) Describe ERT Training Requirements.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the roles and responsibilities of the ERT Organization**	**Method:** Lecture **Media:** PowerPoint **Environment:**Classroom	**50**	
1.1	CO/Command			
1.2	ERT Ops			
1.3	ERT HQ			
1.4	Fwd ERT/Aft ERT/CCR			
1.5	Liaison with HQ1/Snr ET			
1.6	Logistics			
1.7	Action and Emergency Procedures			
2	**Observe the Spill clean-up team from the following positions:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Dressing area			
2.2	Spill area			
2.3	Command area			

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Students should cover voice procedures and plotting with a practical exercise.

PO 012 – DIRECT EMERGENCY RESPONSE TEAM (ERT) OPERATIONS

EO 012.02 PARTICIPATE IN ERT OPERATIONS ONBOARD HMC SHIPS

1. Performance: Participate in ERT Operations onboard HMC Ships

2. Conditions:

   a. Given: Nil;

   b. Denied: N/A; and

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The trainee shall participate in ERT Operations onboard HMC Ships:

   a. IAW Refs, by completing the following:

      (1) Participate in the ERT exercises and drills;

      (2) Participate in Hot Wash-up briefs;

      (3) Draw a functional diagram for the ERT Organization and verbally explain it to the CSEO;

      (4) Draw system level diagrams for both the 60 Hz and 400 Hz Power Generation and Distribution Systems (PG&D) and verbally explain them to the CSEO; and

      (5) Assist with the planning of an ERT exercise.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Participate in the ERT exercises/drills at the following ERT positions (in order of priority)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	ERT Ops			
1.2	ERT HQ Communicator			

Teaching Points		Instructional Strategies	Time (min)	References
1.3	FWD or AFT ERT (communicator and roundsman);			
1.4	HQ1			
1.5	Command (Bridge)			
2	**Participate in all Hot Wash-up briefs on completion of ERT Operations**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3	**Draw a functional diagram for the ERT Organization and verbally explain it to the CSEO. This explanation shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
	CO/Command			
	ERT Ops			
	ERT HQ			
	Fwd ERT/Aft ERT/CCR			
	Liaison with HQ1/Snr ET			
	Supply			
	All lines of communication			
4	**Draw system level diagrams for both the 60hz and 400hz Power Generation and Distribution Systems (PG&D) and verbally explain them to the CSEO. This explanation shall include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Key components			
4.2	Operational description			
4.3	Typical distribution from Main Switchboards to equipment			
4.4	Modes of operation			
4.5	Control and Monitoring			
4.6	System Inter-relationships			
4.7	Redundancy and survivability			
5	**Assist with the planning of an ERT exercise**	**Method:** OJT **Media:** N/A **Environment:** Ship		

5. Total EO Time: Total: N/A

6. Training and Learning Aids: Nil.

7. Remarks:

   a. OJT.

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01	A-AD-121-F01/JX-000	Manual of Abbreviations
A02	A-AE-219-00l/AG-001	Organization , Concept and Policy
A03	A-QA-193-001/AG-001	Manual of DND Quality Assurance Management, Vol. I
A04	A-LM-501-001/AG-003	Code Control Guide
A05	A-LM-502-001 /FP-001	Field Supply Operation Transaction Guide
A06	A-PD-121-C01/FP-000	Staff Procedures and Military Writing
A07	A-PD-131-001/PT-001	Leadership Manual
A08	A-SJ-100-001/AS-001	Security Orders for DND, Vol. I
A09	C-01-000-000/AX-000	NDID
A10	C-01-000-102/AG-000	NDID Coding System
A11	C-O1-007-005/DA-001	NDID Coding Chart
A12	C-02-015-001/AG-000	Instructions for Completing CF 777 Unsatisfactory Condition Report
A13	C-02-040-009/AG-001	General Safety Manual
A14	C-03-005-012/AM-001	Naval Maintenance Management System, Vol. 1
A15	C-03-005-033/AA-001	Naval Engineering Manual, Vol. II
A16	C-03-015-024/AM-000	Maritime Pre-Installation Failure Reporting
A17	A-GG-040-006/AG-002	Accident, Incident, Defect & Malfunctions
A18	C-09-153-001/TS-003	Ammunition Safety Manual, Vol. 3
A19	C-09-215-001/TX-000	Ammunition Restrictions
A20	C-55-040-001/TS-00l	Radiation Hazard
A21	L-LM-000-000/AX-000	Instructions for Completing an ESL Change Notice
A22		Canadian Forces Administrative Orders (CFAOs)
A23	CF 1305	Maintenance Action Form Log
A24	CF 411 - CF 419	Ammunition Log
A25	CFCD 102	
A26	CFCD 109	

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A27		CF Medical Orders (Relevant Safety Extracts)
A28		Current Maintenance Administration Bulletins (MABs)
A29		Engineering and Maintenance Readiness Inspection Checklist for Ships
A30		Engineering Change Proposal (ECP) Completion Guide
A31		Equipment Support List (ESL)
A32		Fleet SOPs
A33		Formation Orders
A34		Guide to Divisional System
A35		Harbour Control Instructions
A36		Maritime Command Ammunition Allocation List (MCAAL)
A37		Maritime Command Orders (NAVORDs) Vol. 1, 2 & 3A
A38		Maritime Command Ships Standing Orders (SSOs)
A39		Material Authorization Change Request (MACR) Guide
A40		Material Authorization Ships Table (MAST) Manual
A41		Missile Condition Log
A42		Naval Technical Bulletins (as required)
A43	B-0G-282-000/FP-000	Shipborne Helicopter Operating Procedures (SHOP)
A45		Ammunition Directives
A46		MARLANT/MARPAC Ammunition Allocation List Part 1 (MAAL/MPAAL)
A47		Identification of Ammunition
A48		Ammunition Logistic Data
A49		Captain’s Ship Book
A50		Torpedo Log book
A51		Harpoon Log Book
A52		DND Explosive Safety Program
A53		QR&Os
A54	A-PM-521-001/AG-K01	Resource Manager Report (REMAR)
A55		Sea Training Guides
A56		Towed Body Log
A57		WHMIS Instruction Guide
A58		General Safety Policy Manuals
A59		Emergency Response Guidebook

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A60	A-LM-187-004/JS-001l	Storage and Handling of Hazardous Materials
A61		NIOSH Pocket Guide to Chemical Hazards
A62		EMS(s) Chapter 10
A63		AL4 - Ships Standing Orders - HAZMAT Spill Response
A73	DAOD 4003-0	Environmental Protection & Stewardship
A74	DAOD 4003-1	Hazardous Material Management
A75	ED 4003-1/99	Spill Reporting
A76	NAVORD 4-12	Policy Directive - Environmental Program
A77	NAVORD 66-05	Hazardous Material Management
A78		Respiratory Protection Policy
A79	C-21-132-000/MS-001	Meteorological System
A80	C-21-134-003/MS-001	AN/WQQ-502(V)3 Bathythermograph Mk8AR-1662
A81	C-21-154-000/MS-001	S/D Converter Junction Box
A82	C-46-495-000/MS-001	AN/UYH-504 Recorder/Reproducer Set, Signal Data (Shipboard Mass Storage System)
A83	C-46-498-000/MS-001	AN/UYK-507(V) Data Processing Set
A84	C-51-406-000/MS-001	AN/SRR-504(V) Receiving Set Radio (Rediffusion Systems) WK 2182 C
A85	C-51-428-000/MS-000	AN/SRR-503(V) Distress Receiver
A86	C-51-501-000/MS-001	Exterior Communications System, Technical Manual
A87	C-51-528-000/MS-001	OE-82C/WSC-1(V), Antenna Group, Operation and Maintenance Manual
A88	C-51-529-001/MS-003	ON-143(V)/USQ Interconnect Group, Operation and Maintenance
A89	C-51-530-000/MS-001	AN/SSR-1A Receiving Set Satellite, Operation and Maintenance
A90	C-51-530-000/MB-001	AN/SSR-1A Receiving Set Satellite, Organizational Maintenance Instructions
A91	C-51-559-000/MF-001	RT-5073(V)1/WSC-3,Communication Set Satellite, Operation and Maintenance, Vol. 1
A92	C-51-561-000/MS-001	R-5113/SRR, UHF Receiver, Operation and Maintenance
A93	C-51-568-000/MS-001	RF-590 Receiver, Operation and Maintenance Manual
A94	C-51-574-000/MS-001	UHF LOS Radio Communications System Manual

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A95	C-51-575-000/MS-001	Communication Control and Monitoring System, Book 1
A96	C-51-576-000/MS-001	Ancillary Radio Equipment System Manual
A97	C-51-584-000/MS-001	HF Emergency Communication System, Technical Manual
A98	C-51-586-000/MS-001	UHF Satellite Radio Communications System Manual
A99	C-51-598-000/MS-001	OA-5180/S, Wideband Audio Switching Group, Diagrams and Parts Lists, Vol. 2
A100	C-51-637-000/MS-001	Distress Monitor Panel LS-5026/SRC
A101	C-51-639-000/MS-001	BZ-5021/U, Operator Alarm Panel, Instruction Manual
A102	C-51-809-000/MB-001	NERA Saturn BM Marine MK2 - Class 2 - Operator's Manual
A103	C-52-289-001/MS-001	AN/SSC-505(V) Shincom, Description and Maintenance Manual, Vol. 1
A104	C-52-289-001/MB-001	AN/SSC-505(V) Shincom, Description and Maintenance Manual, Vol. 4
A105	C-52-370-000/MS-001	Entertainment, Training and Briefing System, Technical Manual
A106	C-53-146-000/MN-001	AN/VRC-512, RT-524/VRC, RT-246/VRC, RT-246A/VRC, RT-524A/VRC Receiver / Transmitter Radio Set, Maintenance Manuals (2 Books)
A107	C-53-168-000/MS-001	RF-601A Antenna Coupler Group, Instruction Manual.
A108	C-53-279-000/MS-002	AN/VRC-49 Receiver/Transmitter Radio Set, Operator Manual
A109	C-53-279-000/MS-003	AN/VRC-512 Receiver/Transmitter Radio Set, Operation Manual
A110	C-51-670-000/MS-001	CD-5021/UR Transmit System Control Unit, Instruction Manual
A111	C-57-716-000/MS-001	AN/URN-25 TACAN Navigation Set, Vol. 1
A112	C-57-716-000/MS-002	AN/URN-502 TACAN Navigation Set, Vol. 2
A113	C-57-716-000/MS-003	AN/URN-25 TACAN Navigation Set, Vol. 3
A114	C-57-546-000/MS-001	AN/URN-25 Tacan Navigation Antenna Group OE-273(V) and OE-273A(V), Vol. 1
A115	C-57-546-000/MS-002	AN/URN-25 Tacan Navigation Antenna Group OE-273(V) and OE-273A(V), Vol. 2
A116	C-57-611-002/MS-001	AN/SQN-502(V) Speed Log Doppler Sonar System, Operation Manual

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A117	C-57-611-002/MS-002	AN/SQN-502 Speed Log Doppler Sonar System, Service Manual
A118	C-57-624-N00/MS-000	AN/SRN-502 Omega II Navigation Set, Operation Manual
A119	C-57-624-000/MS-001	AN/SRN-502 Omega II Navigation Set, Service Manual
A120	C-57-624-000/MS-Z01	Automatic OMEGA II Receiver, CPF Supplement
A121	C-57-625-000/MS-001	AN/SQN-501 Sounding Set Sonar, Vol. 1
A122	C-57-625-000/MS-002	AN/SQN-501 Sounding Set Sonar, Vol. 2
A123	C-57-625-000/MY-001	AN/SQN-501 Sounding Set Sonar, Vol. 3
A124	C-57-633-A00/MS-001	AN/SYN-502 Inertial Navigation Remote Display
A125	C-57-634-C00/MS-001	CS-415 Inertial Navigation System Alarm Unit, Operation and Maintenance
A126	C-57-7O5-000/MS-001	AN/SSN-502 Navigation Switchboard, Operation and Maintenance, Vol. 1
A127	C-57-794-000/MS-001	Operation and Service Manual Sins-HALIFAX Ring Laser Gyro Navigator (RLGN)
A128	C-58-251-000/MB-001	AN/SLQ-501 B CANEWS, ESM Set, System Operation
A129	C-58-251-000/MS-Z01	AN/SLQ-501 B CANEWS,ESM Set, Supplement
A130	C-58 252-000/MS-001	AN/SLA-10B Blanker-Video Mixer, Operation and Maintenance
A131	C-58-258-000/MS-003	AN/SLQ-502 ECM Offboard System (SHIELD II), Vol. 1
A132	C-58-258-000/MS-004	AN/SLQ-502 ECM Offboard System (SHIELD II), Vol. 2
A133	C-58-259-000/MD-001	AN/SLQ-503 RAMSES ESM Set, Theory of Operation, Vol. 1, Books 1 through 5
A134	C-58-260-CC0/MQ-001	AN/SRD-502 Surveillance Receiver, Operation and Programming, Vol. 1
A135	C-58-260-CC0/MR-001	AN/SRD-502 Surveillance Receiver, Test and Adjustments, Vol. 2
A136	C-58-260-CC0/ML-001	AN/SRD-502 Surveillance Receiver, Troubleshooting, Vol. 3

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A137	C-58-260-CC0/ML-002	AN/SRD-502 Surveillance Receiver, Troubleshooting, Vol. 4
A138	C-58-260-CC0/MB-001	AN/SRD 502 Surveillance Receiver, Operator's Handbook, Vol. 5
A139	C-58-260-000/MS-001	AN/SRD-502 Direction Finder Set, Vol. 1 (2 Books)
A140	C-58-260-CD0/MS-001	AN/SRD-502 Plasma Display
A141	C-59-678-000/MS-001	AN/SPX-501(V)3 IFF Mk XII System, Operation and Maintenance
A142	C-59-723-000/MB-001	AN/SPS-49(V) Radar System, Operation, Scheduled Maintenance and General Information, Vol. 1
A143	C-59-723-000/MD-001	AN/SPS-49(V) Radar System, Functional Descriptions, Vol. 2
A144	C-59-723-000/MT-001	AN/SPS-49(V) Radar System, Troubleshooting Procedures, Vol. 3
A145	C-59-723-000/MT-002	AN/SPS-49(V) Radar System, Troubleshooting Logic and Flow Diagrams, Vol. 4
A146	C-59-723-000/MT-003	AN/SPS-49(V) Radar System, Troubleshooting Logic and Flow Diagrams, Vol. 5
A147	C-59-723-000/DS-001	AN/SPS-49(V) Radar System, Troubleshooting Schematics, Vol. 6
A148	C-59-724-000/MS-001	AN/SPS-505 Radar System, Operation, Description and Block Diagrams
A149	C-59-724-D00/MS-001	AN/SPS-505 Radar System, Cooling Unit/Dry Air Unit
A150	C-59-777-000/ML-001	AN/SPS-504 Radar System, Service Manual and Addendum, Vol. 1 (2 Books)
A151	C-59-777-000/MB-001	AN/SPS-504 Radar System, Operation Manual and Addendum, Vol. 2 (2 Books)
A152		Radiation Control Unit
A153	C-59-791-000/MS-001	Interface Unit for Radiation Control Unit
A154	C-59-765-001/MS-001	AN/UYS-503(V) Acoustic Signal Processor, Operation and Maintenance Manual, Vol. 1
A155	C-59-667-000/MS-003	Shipboard Sonobuoy Receiver Assembly, Operation and Maintenance Manual
A156	C-59-766-000/MB-001	AN/USH-33(V) Recorder Reproducer, Operator's Manual, Vol. 1
A157	C-59-844-000/MS-001	Operation, Installation and Servicing Radar Set AN/SPS-506 Type 1007

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A158	C-60-172-A00/MS-001	Radar Distribution Unit, (RDU) Sensor Switchboard, SA-5148/USQ-505 (2 Books)
A159	C-60-172-B00/MS-001	Video Switching Unit (VSU), Sensor Switchboard, SA-5151/USQ-505 (2 Books)
A160	C-160-187-000/MS-001	Weapons Status Panel, Operation and Maintenance
A161	C-60-186-A00/MS-001	AN/SSQ-532 Weapons Veto Panel, Operation an Maintenance
A162	C-60-186-C00/MS-001	AN/SSQ-532 Weapons Veto Panel, Remote Switching Box
A163	C-69-700-000/MS-000	AN/WQC-501(V) Communication Set Sonar, Underwater Telephone
A164	C-69-713-000/MS-001	AN/SLQ-25 Torpedo Countermeasures Transmitting Set, Operation and Maintenance Instructions
A165	C-69-713-M00/MS-001	AN/SLQ-25 Control and Interface
A166	C-69-725-000/MS-001	Torpedo Launcher Control (TLC), Instruction Manual
A167	C-69-745-000/MS-001	Description, Operation, Maintenance, Towed Array Handling and Stowage Group OK-410(V)1/SQR, Vol. 1, Book 1
A168	C-69-748-000/MS-001	Maintenance Manual CANTASS Ship Electronic Subsystem Book 1 of 2
A169	C-700-266-000/MD-001	AN/SPG-503 STIR Fire Control System, Part 1, System Description
A170	C-70-266-000/MD-Z01	AN/SPG-503 STIR Fire Control System, Part 1 Supplement, System Description Supplement
A171	C-70-266-000/MS-001	AN/SPG-503 STIR Fire Control System, Gun Missile Fire Control System
A172	C-70-267-000/MD-001	Guided Missile Vertical Launching System Mk48 MOD 0, Description Manual
A173	C-70-268-000/MD-001	AN/SWG-501 Harpoon Missile System, Description and Operation, Vol. 1
A174	C-70-270-000/MB-001	57mm Gun System, Operation Manual, Vol. 1
A175	C-70-270-000/MD-001	57 mm Gun System, Description Manual, Vol. 2 (2 books)
A176	C-70-270-000/DS-001	57mm Gun System, Diagrams, Vol. 3

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A177	C-70-312-000/MD-001	Transmitter Group Mk73 MOD 1, Description Manual
A178	C-70-321-000/MS-001	CIWS Introduction, Vol. 1
A179	C-70-321-000/MS-002	CIWS Description, Vol. 2
A180	C-70-32l-000/MB-001	CIWS Operations, Vol. 4
A181	C-70-321-000/MS-Z01	CIWS Supplemental Data for Volumes 1 to 10
A182	C-70-321-000/MS-004	CIWS Peculiar Support Equipment, Vol. 1
A183	C-70-321-000/MS-007	CIWS Peculiar Support Equipment, Vol. 4
A184	0967-LP-614-7010	ON-143 (V) Interconnecting Group
A185	110181-0043-01A	Falcon Series HF Communication System
A186	110208-50038	RF-382 Fast Tune Auto-coupler
A187	1125-2388-045	51M-81 UHF 243MHZ Guard Receiver
A188	9853-355-000/MX-000	AN/PRC-515 Radio Set
A189	C-21-134-002/MF002	Bathy THERMOGRAPH MK 8
A190	C-39-101-000/MS-001	Gyro Stabilized Horizon Reference System
A191	C-39-115-B00/MS-000	Instruction Book - Synchro Signal Re-Transmission Unit
A192	C-39-115-A00/MJ-000	Instruction Manual, VG-204F Displacement Jet Gyro
A193	C-41-L00-014	AMD NULKA Maintenance Manual Vol. 1
A194	C-46-000-000/AX-000	Data Processing Equipment Index
A195	C-46-250-000/MS-001	AN/UYC-503 (V) Bootstrap
A196	C-46-254-000/MS-001	AN/USH-504 (V) Q2600 Magnetic Tape Unit Operation and Maintenance Vol. 1
A197	C-46-337-000/MS-001	AN/UYK-505 (V) Data Processing Set Vol. 1
A198	C-46-380-000/DC-001	Bus Interface Set (NODE), AN/UYC-503 (V) Vol. 2
A199	C-46-380-000/DS-001	AN/UYK-502 (V) Data Processing Set Technical Manual
A200	C-46-421-000/MS-001	Operation and Maintenance Manual for SHINPADS Standard Display AN/UYQ-501 (V)
A201	C-46-448-000/DC-001	AN/UYC-504 (V) Data Processing/Bus Interface
A202	C-46-448-000/MS-001	AN/UYC-504 (V) Data Processing/Bus Interface
A203	C-46-449-000/MS-000	AN/UYC-504 (V) Technical Manual BAM
A204	C-46-449-000/MS-001	AN/UYC-504 (V) Technical Manual
A205	C-46-450-000/MS-001	Bus Interface Set NODE Vol. 1 Operation and Maintenance Manual with Parts List

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A206	C-46-450-000/MS-Z01	AN/UYC-503 (V) Technical Manual
A207	C-46-484-002/MS-001	HSP-3609 High Speed Printer Operating and Service Manual M365-90
A208	C-46-497-000/MS-001	Data Processing and Bus Interface
A209	C-46-498-000/MS-001	AN/UYK-507 Operation and Maintenance Manual
A210	C-46-498-000/MS-Z01	AN/UYK-507 Remote Terminal User Manual
A211	C-50-251-A00/MJ-000	618M-3/3A Collins VHF Communications Transceiver
A212	C-50-251-A02/MS-000	Collins 303N Controls (5XX Series)
A213	C-51-000-000/AX-000	Naval External Communication Equipment Index
A214	C-51-007-001/MS-000	Cesium Beam Frequency Standard Hewlett Packard 5061A
A215	AMC 8/12/BNC	Equipment Manual Antenna Distributor
A216	C-46-495-000/MS-001	Recorder/reproducer set, Signal Data AN/UYH-504
A217	C-50-254-000/MS-001	Control, Frequency, Selector C-5556/ARC-164(V)
A218	C-50-254-000/MY-000	Illustrated parts breakdown Radio Set AN/ARC-164(V)
A219	C-51-339-000/NY-001	Radio Antennas (Whip and Wire)
A220	C-51-339-000/NY-002	Communication Antenna Group – UHF, VHF
A221	C-51-339-K00/MS-001	Ultra High Frequency Antenna AS-5042/URC
A222	C-51-050-001/MD-001	Handbook for Communications System
A223	C-51-501-000/MS-001	External Naval Communications System
A224	C-51-528-000/MS-001	Antenna Group OE-82C/WSC-1(V)
A225	C-51-528-000/MS-002	Antenna Group OE-82C/WSC-1(V)
A226	C-51-304-000/MS-000	BZ-5007/U Alarm – monitor
A227	C-51-529-000/MS-003	Interconnecting Group ON-143(V)6/USQ
A228	C-51-407-000/MS-001	Sealand 66 Maintenance Manual
A229	C-51-474-000/MS-000	AS-5104/SRC UHF Polemast Antenna Assembly
A230	C-51-475-000/MS-001	OE-5O12/SRC Serial Control Group Interface Group
A231	C-51-475-G00/MS-001	CU-5121/SC Antenna Control Interface Module

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A232	C-51-521-000/MS-000	CV5144 U Converter, Patch Panel Technical Manual
A233	C-51-529-001/MS-Z01	ON-143 (V) 6/USQ Interconnecting Group Technical Supplement 2
A234	C-51-534-000/MS-001	R-5099/U Radio Receiver Technical Manual
A235	C-51-552-000/MS-001	Technical Manual Model 1280R (MS-5062/UGC)
A236	C-51-658-000/MS-001	TD-1271 B/U Multiplexer
A237	C-51-573-000/MS-001	Operation and Maintenance Instruction with Parts Identification List, Book 1, Radio Set AN/SRC-513
A238	C-51-574-000/MS-001	Technical Manual, Radio Set AN/SRC-514
A239	C-51-655-000/MB-001	MX2400 Integrated SATCOM Terminal Operator's Manual
A240	C-51-657-000/MS-001	Operation and Maintenance Instructions with Parts List - Satellite Communications Set AN/WSC-3(V)17
A241	C-51-658-000/MS-001	TD-1271 B/U Multiplexer
A242	C-51-659-000/MS-OO1	Operation and Maintenance Instructions with Parts List, Organizational Level - Single DAMA Control Monitor
A243	C-51-661-000/MS-001	Operation and Maintenance Instructions UHF RT-1107(V) 11/WSC-3(V)
A244	C-51-665-000/MS-001	Instruction Manual Power Amplifier RF-1110B (AN-5334/UR)
A245	C-51-666-000/MS-001	Operation and Maintenance Instructions, Radio Frequency switching SA-5147/UR
A246	C-51-667-000/MS-001	Operation and Maintenance Instructions with Parts Identification List, Indicator Operational Control ID5138/SR
A247	C-51-668-000/MS-001	Operation and Maintenance Instructions with Parts Identification List, Bandpass Filter Unit, F-5088/SR
A248	C-51-669-000/MS-001	Output Combiner Unit (CV-5165/UR Frequency Mixer Stage)
A249	C-51-670-000/MS-001	Instruction Manual Control Transmitter CB-5021/UR
A250	C-51-678-000/MS-001	Instruction Manual Instruction Manual Filter Bypass, F-5082A/URC
A251	C-51-693-000/MB-001	RF-5000 Falcon HF Communication System

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A252	C-51-711-000/ME-001	Spectra VHF Radio
A253	C-51 -806-000/MS-001	Technical Manual Operation and Maintenance Instructions, Operational Radio frequency Conditioning Unit (commonly known as RFCU) HFDU-6130-2
A254	C-51-808-000/MB-001	Operators Manual for MX2464 Magnaphone B Integrated SatCom Terminal
A255	C-51-81-000/MS-001	Technical Manual Miniloop Tuneable Antenna Model MLA-115, 1.8-15 MHZ
A256	C-51-811-000/MS-001	Intermediate Maintenance Manual CD98/URC, Receive Mute Matrix Unit (RMMU)
A257	C-51-812-000/MS-001	Operators Manual, Control/Monitoring Subsystem
A258	C-52-000-000/AX-000	Naval Alarm, Signaling, Interior Comm Index
A259	C-52-101-000/MS-000	AN/SIH-503 Public Address System Instruction Manual
A260		AN/PIC-2 Public Address Set
A261	C-52-239-000/MS-001	AN/SIA-120B (U) Amplifier Control Group
A262	C-52-257-000/MM-000	Integrated Central Broadcast System
A263	C-52-263-000/MS-000	Integrated Central Broadcast System and Audio Broadcast System
A264	C-52-394-000/MS-001	AN/SSC-512 (V) Trump Shincom
A265	C-52-394-000/MY-001	AN/SSC-512 (V) Trump Shincom
A266	C-52-415-000/MF-00	J-5222/SIC Comm Equipment Interface Unit and PP-S387 SIC
A267	C-52-448-000/MB-001	Operators Handbook, Shincom 2100
A268	C-52-448-000/MS-001	Description and Maintenance Instructions for Shincom 2128 System (IRO)
A269	C-53-279-000/MN-000	AN/VRC Radio Set
A270	C-53-279-F00/MS-000	AS-5041/URC Ultra High Frequency AE
A271	C-54-000-000/AX-000	Fixed Comm Equipment Index
A272	C-55-000-000/AX-000	Microwave and Visual Comm Index
A273	C-57-118-000/MB-000	AN/URN-20A (V)1 Tacan
A274	C-57-118-000/MB-001	AN/URN-20A (V)1 Radio Set Modified
A275	C-57-118-000/ME-001	AN/URN-20A (V)1 Radio Set
A276	C-57-382-000/MS-001	Sonar Sounding Set UQN-4
A277	C-57-540-000/MB-001	Omega MX1105 Navigators Manual
A278	C-57-546-000/MS-001	OE-273 (V) URN Antenna Group for Tacan

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A279	C-57-546-000/MS-002	OE-273 (V) URN Antenna Group for Tacan
A280	C-57-608-000/MS-001	Navigation Set, Loran-C/GPS AN/SRN-501
A281	C-57-610-000/MS-001	AN/WSN-S Technical Manual
A282	C-57-611-001/MS-001	Doppler Speed Log System
A283	C-57-627-000/MS-001	Technical Manual INDB AN/SYN-503
A284	C-57-627-000/MS-002	Technical Manual INDB AN/SYN-503
A285	C-57-686-000/MS-001	Inspection and Operation Instructions Satellite emergency Position Maintenance Beacon (EPIRB) Kannad 506
A286	C-57-689-000/MB-001	Navigational Computer AN/USN-502
A288	C-57-689-000/MS-001	Installation and Maintenance Manual With Parts List, Inertial Navigation Set AN/WSN-502
A289	C-57-689-000/MS-002	Inertial Navigation Set AN/WSN-502
A290	C-57-732-000/MB-001	Inspection and Operation Instructions for Radar Transponder Rescuer SART
A291	C-57-735-000/MS-001	Naval GPS Navigation System (Trimble) Book 1 of 2
A292	C-57-735-000/MS-001	Naval GPS Navigation System (Trimble) Book 2 of 2
A293	C-57-787-000/MS-001	SHINNADS
A294	C-57-799-000/CS-001	Special Maintenance Instructions, SQS-510 Sonar Set
A295	C-58-144-000/MJ-001	AN/ULQ-6A Counter Measures Set
A296	C-58-144-000/MB-000	AN/ULQ-6A Counter Measures Set
A297	C-58-144-000/MS-001	AN/ULQ-6C Counter Measures Set
A298	C-58-145-000/CS-001	Special Information Instructions
A299	C-58-146-000/MJ-000	AN/SLA-15 Antenna Group for ULQ
A300	C-58-146-000/MS-001	Technical Manual Volume 1
A301	C-58-146-000/MS-001	Technical Manual Volume 2
A302	C-58-251-000/MD-001	AN/SLQ-501 RAMSES Description (Confidential)
A303	C-58-252-000/MM-001	AN/SLA-10B Blanker-Video Mixer Group
A304	C-58-258-001/MS-001	Shield II Type 210 Chaff and IR Decoy System
A305	C-58-258-001/MS-002	Shield II Type 210 Chaff and IR Decoy System
A306	C-58-376-000/MS-001	AN/SRD-503 DF Set Operation and Maintenance

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A307	C-59-010-009/MS-000	AN/UPA-59A (V) Operation and Maintenance Manual with parts list
A308	C-59-010-009/MS-001	AN/UPA-59A & 59B Decoder Group
A309	C-59-107-000/MS-000	Control Transponder Set
A310	C-59-107-000/MS-001	C620/APX Operation and Maintenance Manual with parts list
A311	C-59-667-000/MS-001	Receiving Set Radio, AN/ARR-75
A312	C-59-667-000/MS-003	Sonobuoy Processing System (SPS)
A313	C-59-747-000/MD-001	AN/SPS-501 System Description LW-08
A314	C-59-756-000/MD-001	AN/SPS-502 System Description DA-08
A315	C-59-678-000/MS-001	AN/SPX-501 (V)
A316	C-59-678-000/MS-Z01	Supplement for CPF/TRUMP Systems Operation and Maintenance Instructions
A317	C-59-678-000/MS-Z02	AN/SPX-501 (V)3
A318	C-59-680-000/MF-000	AN/TPX-54 (V) RX/TX Group Interrogator
A319	C-59-680-000/MF-001	OR5014/TPX-54 (V) RX/TX Group Interrogator
A320	C-59-680-000/MS-001	TS-5144/SPX-501 (V) and TS-5145 Operation & Maintenance Instructions
A321	C-59-680-000/MF-Z10	Technical Manual Supplement lFF1
A322	C-59-680-000/MS-001	IFF Test Set Loop
A324	C-59-747-000/MD-001	LW-08 Part 1; System Description
A325	C-59-747-000/MD-Z01	LW-08 System Description Supplement
A326	C-59-747-000/MD-002	LW-08 Part 2; Technical Description
A327	C-59-747-000/MB-001	LW-08 Part 3; Operating Instructions
A328	C-59-747-000/AX-001	LW-08 Part 8; Catalogue of General Information
A329	C-59-755-000/MD-001	ATMS Part 1; System Description
A330	C-59-755-000/MD-002	ATMS Part 2; Technical Description
A331	C-59-755-000/MB-001	ATMS Part 3; Operating Instruction
A332	C-59-756-000/MD-001	DA-08 Part 1; System Description

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A333	C-59-756-000/MD-Z01	DA-08 Part 8; System Description Supplement
A334	C-59-756-000/MD-002	DA-08 Part 2; Technical Description
A335	C-59-756-000/MB-001	DA-08 Part 3; Operating Instruction
A336	C-59-756-000/AX-001	DA-08 Part 8; Catalogue of General Information
A337	C-59-756-000/MB-001	DA-08 Part 3; Operating Instruction
A338	C-59-756-000/AX-001	DA-08 Part 8; Catalogue of General Information
A339	C-59-755-000/MD-001	ATMS Part 1; System Description
A340	C-59-755-000/MD-002	ATMS Part 2; Technical Description
A341	C-59-755-000/MB-001	ATMS Part 3; Operating Instruction
A342	C-59-755-000/MQ-001	ATMS Part 6B4; Computer Message Formats
A343	C-59-755-000/AX-001	ATMS Part 8; Catalog of General Information
A344	C-59-756-000/MD-001	LW-08 Part 1; System Description
A345	C-59-756-000/MD-Z01	LW-08 Part 1; System Description Supplement
A346	C-59-756-000/MD-002	LW-08 Part 2; Technical Description
A347	C-59-756-000/MB-001	LW-08 Part 3; Operating Description
A348	C-59-756-000/AX-001	LW-08 Part 8; Catalog of General Information
A349	C-59-770-000/MS-001	AS 177 Antenna
A350	C-59-795-000/MS-001	Sonobuoy Processor Sub-system SPSS3-1
A351	C-59-849-000/MS-001	Pathfinder Instruction Manual Vol. 1
A352	C-59-849-000/MS-002	Pathfinder Instruction Manual Vol. 2
A353	C-51-530-000/MS-001	Receiving Set, Satellite Signal AN/SSR-1 and AN/SSR-1A
A354	C-51-530-000/MB-001	Receiving Set, Satellite Signal AN/SSR-1 and AN/SSR-1A
A355	C-60-162-000/MS-001	Radar Indicator Group AN/SPA-25F
A356	C-60-162-000/MS-002	Radar Indicator Group AN/SPA-25F
A357	C-60-179-000/MS-001	AN/SYQ-503 RDDS Operating and Maintenance Instruction
A358	C-60-179-000/MS-001	AN/SYQ-503 RDDS Technical Manual Vol. I
A359	C-60-179-000/MS-001	RDDS Technical Manual Vol. II

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A360	C-60-211-000/MS-002	Technical Manual Operational and Maintenance Instructions MX-512-PV/RC Remote Control Link Monitor
A361	C-63-000-000/AX-001	Test Equipment Index General Purpose
A362	C-63-278-000/MS-001	Technical Manual for HF/MF Receiving Antenna Multicoupler Model AMC/21
A363	C-64-376-000/MS-001	Fire Veto Panel
A364	C-67-R00-000/MS-002	AN/UPM-504 (V) Radar Test Set
A365	C-67-R00-000/MS-Z01	Test Set Supplement
A366	C-69-125 000/MS-001	Sonar AN/SQS-505 (V)
A367	C-69-462-000/MS-000	AN/SQA 502 Sonar Hoist Mechanism Handbook
A368	C-69-675-B00/MG-000	Test Set Assembly MK432
A369	C-69-675-A00/MG-000	Control Panel MK 264
A370	C-69-676-000/MG-002	SVTT MK 32 MOD 5
A371	C-69-690-000/MS-001	Transmitter Group Heading Reference Data 0T-5002/AN/SQS-505 (V)
A372	C-69-700-000/MS-000	Communication Set Sonar AN/WQC-501 (V) UWT
A373	C-69-704-000/MS-001	Tow Condition Monitor
A374	C-69-713-000/MS-001	Transmitter Set Torpedo Countermeasures AN/SLQ-25 (U)
A375	C-69-770-000/MS-001	Torpedo Handling and Storage System
A376	C-69-766-000/MS-001	CS Hull Outfit Operator and Maintenance Instruction
A377	C-69-771-000/MG-002	SVTT MK 32 MOD 9
A378	C-69-799-000/MS-001	Sonar Set AN/SQS-510 System Maintenance Manual
A379	C-70-110-000/MS-000	Wind Indicating Equipment Type F
A380	C-70-312-000/MD-001	Equipment Description Group MK73 MOD 1
A381	C-70-321-000/MS-001	CIWS MK 15 Mods 11 through 14
A382	C-70-325-000/MD-001	LIROD Part 1; System Description

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A383	C-70-325-000/MD-Z01	LIROD Part 1A; System Description Supplement
A384	C-70-325-000/MD-002	LIROD Part 2A; Technical Description Computer
A385	C-70-325-000/MD-003	LIROD Part 2B1; Technical Description Optronics
A386	C-70-325-000/MD-004	LIROD Part 2B2; Technical Description Display
A387	C-70-325-000/MD-005	LIROD Part 2C; Technical Description Director
A388	C-70-325-000/MD-006	LIROD Part 2D1; Technical Description Radar
A389	C-70-325-000/MD-007	LIROD Part 2D2; Technical Description Radar Timing
A390	C-70-325-000/MD-008	LIROD Part 2E; Technical Description System Support
A391	C-70-325-000/CS-001	LIROD Part 8; Catalogue of General Information
A392	C-70-326-000/MD-001	76/62 SRGM Technical Manual
A393	C-70-326-000/MD-002	76/62 SRGM Technical Manual
A394	C-70-326-000/MD-003	76/62 SRGM Technical Manual
A395	C-70-326-000/ME-001	76/62 SRGM Technical Manual
A396	C-70-326-000/NK-019	76/62 Lower Hoist Test
A397	C-70-327-000/MD-001	STIR Part 1; System Description
A398	C-70-327-000/MD-002	STIR Part 2A; Technical Description Computer Sub-system
A399	C-70-327-000/MD-003	STIR Part 2B; Technical Description Display
A400	C-70-327-000/MD-004	STIR Part 2C; Technical Description Director
A401	C-70-327-000/MD-005	STIR Part 2D1; Technical Description Radar
A402	C-70-327-000/MD-007	STIR Part 2E; Technical Description Weapons
A403	C-70-327-000/MD-008	STIR Part 2F; Technical Description System Support
A404	C-70-327-000/MB-001	STIR Part 3; Operating Instruction
A405	C-70-327-000/CS-001	STIR Part 8; Catalogue of General Information
A406	C-70-328-000/DC-001	Vertical Launching System MK41 MOD C

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A407	C-70-328-000/MB-001	LCCP Computer Program Operator's Manual
A408	C-70-328-000/MD-001	Vertical Launching System MK41 MOD C
A409	C-70-330-000/MM-003	CWI Sub-system
A410	C-70-330-000/MS-001	CWI Operation/Technical Description
A411	C-70-330-000/MS-002	CWI System Description
A412	C-70-336-000/MS-001	Fire Control Interface Unit
A413	C-70-353-000/NG-002	WDS/CCS Integration System
A414	EE13O-BG-OMI-010	AN/USG64 (V) 7, AN/USG-64 (V) 8 Satcom Control System Overview
A415	EE160-GP-OMI-010/W1	AN/USC-43 (V)ANDVT
A416	EE160-WA-OMP-010/W1	OK-454 (V) WSC Single Dama
A417	ES701038 AN/SSC-512	(V) Terminal User's Instruction Supplement
A418	SPAR-ASG-AMN-1013	AN/SSC-505 (V) Interior Communications System Description and Maintenance Instructions
A419		MX2400 Magnavox Integrated Satcom Terminal
A420	C-51-547-000/MS-001	UHF Radio Receiver-Transmitter RT-5085
A421	C-51-547-000/MS-002	UHF Radio Receiver-Transmitter RT-5085
A422	C-51-559-000/MF-001	Receiver Transmitter Radio RT-5073(V)WSC-3 and RT-5073(V)1WSC-3
A423	C-51-568-000/MS-001	RF-590 Radio Receiver (R-5095B/G)
A424	C-51-576-000/MS-001	Ancillary Radio Equipment Group
A425	C-51-633-000/MS-001	Radio Transmitter T-5086A/URT (RF-1310-13)
A427	C-51-699-000/MS-001	RF-5710 HF Modem (MD-5077(V)
A428	C-51-756-000/MS-001	RF-2601 Fast Tune Automatic Antenna Coupler
A429	C-51-762-000/MS-001	RF-591A-01 Pre/Postselecto
A430	C-51-810-A00/MS-001	Miniloop HF Tunable Antenna
A431	C-51-814-000/MS-001	Narrowband Transceiver Subsystem
A432	C-51-866-000/MS-001	Indicator Fault Locating ID-5139/UR
A433	C-51-870-000/MS-001	Transmit/Receive Switch RE-5062A/UR
A434	C-51-872-000/MS-001	Bandpass Filter F-5094/UR
A435	C-51-873-000/MS-001	Amplifier, Radio Frequency AM-5314A/URT
A436	C-53-279-000/MS-002	Radio Sets AN/VRC-49

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A437	C-53-626-000/MB-000	RF-5200 Flacon Series Manpack Tactical Communication Systems Operations Manual
A438	C-54-962-000/MB-001	Modem Communications 2440
A439	C-56-560-000/MB-001	Modem Communications V.3224/V.3225
A440	C-57-118-000/MS-000	AN/URN-20A(V) TACAN Radio set
A441	C-57-118-000/MS-002	AN/URN-20A(V) TACAN Radio set
A442	C-57-118-000/MS-007	AN/URN-20A(V) TACAN Radio set
A443	C-58-244-000/MS-000	Drier, Air-Gas, Desiccant HD-5029/SRC
A444	C-59-803-ZA0/MS-001	CP-2205(P) (V)/USQ-125 Model No. MX-512PV Link-11 Data Terminal P/N 34000-()-() MX-512PV VME Card Set Link-11Data Terminal P/N 33510-()-() MX-512PV VME DTS Module P/N 34168-()-
A445	C-59-803-ZB0/MS-001	Link-11 Monitor System Single-Tone (LMS-11ST)
A446	957031-A	SHINPADS Standard Display AN/UYQ-501(V)
A447	C-46-498-000/MS-001	Technical Manual for AN/UYL-507(V) Data Processing Set
A448		AN/USQ-69(V) Learner's Guide
A449		HALIFAX Class Combat System User's Manual, Rev M, 13 January 2011
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01		Canadian Environmental Protection Act
B02		Canadian Environmental Assessment Act
B03		Fisheries Act
B04		Transportation of Dangerous Goods Act
B05		Hazardous Products Act
B06		Arctic Waters Pollution Prevention Act
B07		Territorial Lands Act
B08		Pest Control Products Act
B09		Canada Shipping Act
C		**Civilian Publications and Manuals that relate to this Qualification**
C01	PMX Doc. No. 101733100	Communication Control and Monitoring System User's Manual.
C02	PMX Doc. No. 101731809	User's Manual Maintenance Test Combat System Operational Test

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
C03	PMX Doc. No. 101731810	User's Manual Maintenance Test Combat System Alignment Test
C04	PMX Doc. No. 101731811	User's Manual Maintenance Test Combat System Interface Test
C05		User's Manual Maintenance Test Link 11 POFA Test.
C06		Underwater Acoustic System Analysis; Burdic, William S.; Prentice-Hall 1984
C07		Principles of Underwater Sound, 3rd ed.; Urick, Robert J.; McGraw-Hill 1983
C08		Principles of Naval Weapon Systems; Payne, Craig M.; Naval Institute Press 2006
C09		Radar Fundamentals; Faulconbridge, R. Ian; Argos Press (Scitech Publishing) 2002
C10		Communication Electronics, 2nd ed.; Frenzel, Louis E.; Glencoe/MacMillan/McGraw-Hill 1995
C11		Elements of Electromagnetics, 3rd ed.; Sadiku, Matthew N.O.; Oxford Press 2001
C12		Electronic Principles, 3rd ed.; Malvino, Albert P.; McGraw-Hill 1984
C13		Network+ Study Guide, 3rd ed.; Groth, David; Sybex 2001
C14		Industrial Hydraulics Manual, 2nd ed.; Vickers 1989
C15		Modern Control Systems, 8th ed.; Dorf, Richard C.; Bishop, Robert H.; Addison Wesley 1998
C16		Digital Systems, 5th ed.; Tocci, Ronald J.; Prentice-Hall 1991

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
AT0040	Promote environmental awareness	Train	011	
CT0010	Plan/conduct/assess shipboard battle problems/damage control drills	Train	012	
CT0015	Plan/conduct/assess shipboard Combat Systems Engineering drills	Train	012	
DT0050	Conduct engineering rounds	Train	001 to 008	
DT0055	Conduct noise-short rounds	Train	010	
DT0065	Assess pre-sail checks	Train	001 to 008	
DT0075	Direct the ship's Emergency Response Team (ERT)	Train	012	
DT0080	Monitor pre-firing and post firing checks	Train	001	
DT0120	Perform as ship's signature Management Officer	Train	010	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
DT0135	Assess the impact of MSE systems on Combat Systems	Train	001 to 008	
ET0010	Assess battle damage to ship	Train	012	
ET0015	Develop ERT policy/procedures	Train	012	
ET0020	Advise Command on the impact of damage, equipment/systems casualties and availability of spares	Train	012	
FT0005	Ensure compliance with policies/procedures governing handling, stowage and disposal of hazardous material	Train	011	
FT0015	Perform as ship's ammunition Officer	Train	009	
FT0035	Perform as Radio Frequency Safety Officer (RF Safety O)	Train	011	
FT0040	Ensure compliance with general safety policies and procedures	Train	011	
FT0045	Ensure compliance with general safety policies and procedures	Train	011	
GT0055	Monitor the start-up, running, and shutdown of equipment and systems	Train	001 to 008	
GT0060	Review logs and system operating records	Train	001 to 008	
GT0075	Isolate system faults	Train	001 to 008	
GT0090	Interpret engineering documentation	Train	001 to 008	
IT0020	Analyze shipboard noise sources	Train	010	
OT0040	Manage weapon systems	Train	001	
OT0045	Manage navigation systems	Train	002	
OT0050	Manage Communication Systems	Train	003	
OT0055	Manage Above water warfare systems	Train	004	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
OT0060	Manage underwater warfare systems	Train	005	
OT0065	Manage command and control systems	Train	006	
OT0070	Manage Data Processing Systems	Train	007	
OT0075	Manage Electronic Warfare Systems	Train	008	

ANNEX C – QS SCALAR

N/A

ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required. The original QSWB was not found.

Title				
Location of Board	CFNES S-37 Rm B-15			
Dates	13 Feb – 09 Mar 2012			
Board Composition				
	Surname	Initials	Rank	Unit
Chair	Tremblett	G	LCdr	CCFL HQ
TM	Prince	M	Capt	CFNES OT Div
SME	Cyr	K	LCdr	MARPAC HQ N42
SME	Rector	I	Lt(N)	CFNES OT Div
SME	Walker	G	Lt(N)	FMFCS/FTA
SME	Castelli	A	Lt(N)	PMO JSS
Secretary	Goodmurphy	J	NCdt	CFNES OT Div

BOARD OBJECTIVE:

1. A QSPWB was conduct in Oct 2010 and it became apparent that a Military Employment Structure Implementation Plan (MESIP) was required for the NTO MOSID in order to define clearly the job requirements. As waiting for the MESIP to be completed (3 to 5 years), a QSPRB was convened to bridge the gap of the current training.

2. Based on the QSPRB convening order, the QSPRB mandate was to:

   a. Separate the MS Eng BOQ and the NCS Eng BOQ QPSs common phase into one QSP called Naval Engineering Common. This will facilitate document management and will not affect training;

   b. Review and amend the existing POs, EOs and Assessment;

   c. Update the impact of the new equipment on the job requirements and the training.

DISCUSSIONS / DECISIONS:

3. Initial basic changes to the CSE QSP are as follows:

   a. Stripped out the NEC subjects that are common to both NCS Eng and MS Eng BOQ courses. The CSE POs are:

i. PO 413 – Manage Operational Readiness;
ii. PO 414 – Manage Naval Maintenance;
iii. PO 415 – Conduct Engineering Analysis
iv. PO 416 – Manage Naval Tests and Trials;
v. PO 417 – Manage Training;
vi. PO 418 – Lead Personnel;
vii. PO 419 – Manage Projects;
viii. PO 420 – Manage the Life Cycle of Materiel; and
ix. PO 421 – Conduct Business Management.

b. Re-numbered remaining POs from 4XX to 0XX to be in line with the version template, re-wrote each standard statement to include an appropriate verb;

c. Re-formatted the new CSE QSP to the new version 5 template;

d. Re-drafted chapters 1 and 3 to meet the requirements of the new template and with consideration of the NEC and MSE QSPs;

e. Redrafted Board Assessment Guide and Instructions to the Candidate (Annex L) so that it incorporates the new guide as well as incorporating wording so that all three QSPs, NCS Eng, MS Eng and NEC are similar;

f. Re-drafted the OJPR (Annex E) to have the same signature style as the NEC and MSE OJPRs;

g. Re-titled PO 404 Above Water Sensor Systems to PO 004 Manage Above Water Surveillance Systems to clearly define the scope of the PO;

h. Removed Fire Control from PO 001 – Manage Naval Weapons and placed it in to PO 004 Manage Above Water Surveillance Systems in order to group all RADAR systems in one PO; and

i. It was decided in paragraph 9 of the OJPR to make the requirement for practice boards by the CSEO implicit as the level of boardsmanship of trainees has been lacking. This will provide a means for the CSEO to be held to account for the training of his trainees as well as provide a management tool for the CSEO during the training year.

4. All remaining POs, 001 to 012, were reviewed for content. In general, the content of most POs was considered adequate until the future QSP rewrite. In all POs, all teaching points titled “Emerging Technologies” were re-titled “Future Technologies” and all teaching points that were titled “Future In-Service Technologies” where that technology was being introduced as part of the mid-life refits of the HFX Class were removed. Four specific POs had significant discussion:

a. PO 001 – Manage Naval Weapons Systems: EO 001 – Describe the Theory of Control and Drive Systems was broken into two separate EOs: EO 001.1 – Describe Control Theory and EO 001.2 – Describe Drive Systems. This EO had 61 periods allocated and it was felt that two separate EOs would provide more granularity with instructing the material.

b. PO 003 – Manage Communications Systems: An OT Div instructor provided a revamped PO on Naval Communications as a suggested replacement. The suggested revisions were compared to the existing QSP. A number of the suggested changes reduced the overall theory and it was determined that this reduction would not provide all of the basics that are required of an NCS Eng. The structures changes, realignment of topics and teaching point, were adopted. One other small change was to reduce the TACAN tour time from 30 minutes to 15 minutes as this system will be removed from service on the HFX class (post HCM) and only remain in service with the IRO and PRO.

c. PO 009 – Manage Ammunition: There was a significant re-write of this PO in response to concerns raised from the Fleet. The next A/HoD QSPRB or QSPWB needs to address an update to the material required for the A/HoD. The MARLANT FATO will develop a MLP for this course to assist OT Div in developing instruction. As part of the review, no time was added and the PO was changed as follows:

   i. Removed:
      1. EO 409.1 – Shipboard Ammunition Operations; and
      2. EO 409.2- Ammunition Operations in HMCS Ships.

   ii. Added:
      1. EO 009.01 – Describe Explosive Theory;
      2. EO 009.2 – Describe In-service Naval Ammunition;
      3. EO 009.3 – Describe Ammunition Accounting;
      4. EO 009.4 – Describe the Ammunition Organization
      5. EO 009.5 – Describe the Ammunition Explosive Safety Program; and
      6. EO 009.6 – Participate in Ammunition Operations onboard HMC Ships.

d. PO 011 – Manage Shipboard Hazards: was renamed from PO 411 – Administer Risk Management Programs. It was felt that the original title would have been confused with the new NEC PO 002 Conduct Risk Management.

5. At the midway point of the CSE QSP it was determined that it would be impossible to remove the OJPR EO and break the points down into specific EOs as was done with both the NEC and MSE OJPRs. The CSE QSP is constructed in such a way that to dissolve the OJPR EOs and inject them into the various POs, will require a full re-write of the QSP that will take a significant period of time and it was determined that this is beyond the scope of the QSPRB. This was discussed in length between LCdr Tremblett, LCdr Cyr, LCdr Fifield, LCdr Cyr and Capt Prince. During this discussion all avenues to bring the CSE QSP into the same format of the MS/NEC QSPs were explored, but was not achievable in the time allotted to the RB.

6. As directed by DNTE, the number of sea days was reduced from 60 sea days to 40 sea days. This was not a popular dialogue, however, in discussion with the MS QSP members, it was determined that based on the number of available hulls over the next 5 to 7 years, this was the best compromise available to ensure that every trainee would receive a bare minimum number of sea days.

7. Final calculations of the QSP was conducted as per below:

a. Old NCSEAC:
   i. Apps Course: 134.2 days (Training time, Admin Time and Ops Trainer)
   ii. MWBC: 5 days
   iii. DRMIS: 7 days
   iv. OOD Practical: 11 days
   v. Total – 157.2 days

b. New NCSEAC:
   i. Apps Course: 115.5 days (Training Time, Admin Time and Ops Trainer)
   ii. NEC Course: 26.9 days
   iii. MWBC: 5 days
   iv. DRMIS: 7 days
   v. OOD Practical: 11 days
   vi. Total – 165.4 days

c. The difference between both can be accounted for as follows:
   i. 2 days added to NEC for Risk Management;
   ii. 1 day added trip to NDHQ;
   iii. A slight increase of 0.9 days for the inclusion of more ECs;
   iv. 4.2 days accounted for in non-scheduled lab time.

8. There was a discussion on the value of having NCS Eng PHVI trainees conduct their afloat phase on another class of vessel, in particular KIN class. This would provide absolutely no benefit to an NCS Eng as there is very little equipment. A KIN class has a basic RADAR, 40mm BOFORs, basic navigation suite and a basic communication suite. There would be very little ability for the trainee learn the full operational picture or apply the full value of theory training to the equipment.

ASSESSMENT PLAN

9. Annex K of the new format QSP was updated to reflect the amended QSP material from PO 001 to PO 012. The format for assessing the POs was discussed, CFNES Cmdt, OT Div Comd, CFNES QAO and Board SME, it was determined that in order to be fair and provide a true assessment, both the Assessment Guide would need to changed and the weighting of marks will be, in general, 60% for the critical EC, 20% for formative and 20% for ECs. Where there are no ECs in the PO, the marking will be weighted 70% for the critical EC and 30% for formative. Also, it was determined that the minimum mark to pass the CEC is 50% and the minimum mark to pass the formative assessment is 60% and the minimum mark to pass each POs is 60%. The TDO didn't agree with that formula.

10. As part of the overall assessment of a trainee, the wording of the Chapter 3 paragraph of marking allocation was changed to the following:

1. In order to achieve a passing mark for the NCS ENG academic phase (NCSEAC), trainees must pass all modules with a minimum mark of 60%. Trainees must achieve a minimum mark of 50% in all critical ECs. The TDO did not agree with that decision

2. The final mark for each module is calculated using the following formulas:

i. PO 001: 30% for CEC 001 + 30% for CEC 002 + 20% for ECs + 20% for formative work;

ii. PO 002: 70% for CEC 003 + 30% for formative work;

iii. PO 003: 60% for CEC 004 + 20% for ECs + 20% for formative work;

iv. PO 004: 30% for CEC 005 + 30% for CEC 006 + 20% for ECs + 20% for formative work;

v. 005: 60% for CEC 007 + 20% for ECs + 20% for formative;

vi. PO 006: 70% for CEC 008 + 30% for formative work;

vii. PO 007: 15% for CEC 009 + 15% for CEC 10 + 15% for CEC 11 + 15% for CEC 12 + 20% for ECs + 20% for formative work;

viii. PO 008: 70% for CEC 013 + 30% for formative work;

ix. PO 009: 70% for CEC 014 + 30% for formative work;

x. PO 010: No CEC; assessed as part of OJPR;

xi. PO 011: No CEC; assessed as part of OJPR; and

xii. PO 012: No CEC; assessed as part of OJPR.

3. The final overall course mark is an average of all module marks. Full details of the breakdown of the marking scheme for each PO can be found in Annex K of the QSP.

LESSONS LEARNED:

11. Based upon the projected changes to the next NCS Eng BOQ QSP Writing Board, it is necessary to determine the types of jobs that the successful PHVI candidate would be expected to undertake. The future goal is to train for jobs that are not necessarily linked to HoD tours.

12. In the event that the QSP is changed and the goal shifts from training for HoD to training for post PHVI positions, then there must be considerations given to developing a pre-A/HoD and pre-HoD course that would capture the training that is HoD specific. This training could then be stripped out of the PHVI ashore phase, shortening the course, and ultimately providing “just in time” training for A/HoD and HoD.

13. While it is true that basic technical writing is taught at basic training for all military trades, it is very apparent that post PHVI NTOs suffer from the in-ability to draft very basic military documents. It is recommended that some form technical writing refresh be included in any QSP re-write.

14. For any future QSPRB, it is extremely important to clearly establish the scope of the RB prior to commencement of the RB.

FOLLOW-UP ACTION:

SERIAL	ACTION ITEM	TARGET DATE	OPI
001	The reference to the MARCOM Production Books and the MARCOM Personnel Production and Scheduling Procedures in the CSE QSP Chapter One to be confirmed by the TDO		OT DIV TDO
002	Need to determine the maximum number of POs that can be taught at the same time		OT DIV TDO
003	The CFNES OT Div Comd will need to discuss with the CSE Div Comd (MARPAC) that Board results will be by message and not letters		OT DIV Comd
004	MARLANT FATO to develop a MLP for PO 009		MARLANT FATO

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version: 2.0	
Date: 1 July 2023	**Who Initiated Change: HMCS Venture**
Requirement for the Amendment: New format of QSP	
Description of Changes: QSP has been edited to new format. Removal of obsolete equipment out of POs, changing wording to encompass future equipment without needing a new update 1. Add a section to the signature requirements where the candidate is to conduct ten minor system practice boards from the list of all minor systems. These should be selected from the list of minor systems that are not subsystems of a major system or suite, i.e. from the following list: Bathy, Cranes, Horizon Bars (SHRS), NIXIE, PACOM (Main Broadcast), SEOSS, SHINCOM, SPS, Torpedoes, UWT, and Remote Weapon System. System selection is, however, at the discretion of the HOD. The package will be amended to add ten CSEO minor system board presentation signature blocks along with a notation that the candidate is still responsible for the presentation of any of the minor systems on the final board. The candidate is to practice/complete the	

remaining minor systems on their own or with their peers. Substantiation: Verbal comments were received from board chair Cdr that minor system knowledge and corresponding board questions seemed weak across all trainees. This was also noted in some Fleet Pre-Boards. In previous QSPs, all minor systems had “draw and present” components in addition to walkthroughs. The practice boards were removed, likely due to the cumbersome nature of presenting boards to the HOD for approximately 44 minor systems. This will return oversight of the minor system presentations to the HOD while not being overly cumbersome, and ensure that the candidate is aware of the minor system requirement.

2. Add ORO/Director (discussion) and CSEO (presentation) signature blocks for each of the three Layered Defence scenarios. The lack of signatures, including practice boards with the HoD, has led many candidates to inadequately prepare for this question. The candidates will spend a watch or have a discussion with the ORO/Director specifically to discuss the detect-to-engage sequence from an operator’s perspective, as well as conduct a practice board for each of three scenarios. No impact on training time should be expected as the candidates are already required to spend time with the operators and are expected to know and present layered defence. This will verify that the candidate has received feedback from the ORO/Director and CSEO regarding layered defence, increasing the opportunity for success at the final board.
3. Amend Course Completion page (p. 299 of QSP) to require signatures recommending procession to final board from the HoD, XO, and CO. The page currently has a signature block for the HoD only.
4. Amend Qualification section (p. 12 of QSP) to remove requirement for Unit RF Safety Officer Qualification (115868). The requirement is for a Head of Department to hold this qualification, so it would be more appropriate to have the candidate acquire this qualification during, or just immediately prior to, their Assistant Head of Department (A/HoD) phase. It is not required of a Phase VI candidate at this point in his/her career and so is not an effective use of the candidate’s time. FW UNIT RADI
5. Amend the board process to include the trainees drawing the boards, and include these as an OJPR check. The change which removed the requirements to have the trainees draw the major systems on the final qualification board has been noted by senior engineers over the last several years to have a possible negative impact on the amount of knowledge the trainee has at the qualification board.
6. Clarify the difference between Deck Cranes and the Torpedo Handling and Storage System, which includes Bridge Cranes. It has been noted on boards that there is confusion between the two.

7. Removed TACAN, EO 002.10, re-allocated time throughout PO 002 EOs

Small Changes

8. Updates to NAVORDs/NPTORDs/others to reflect new references throughout document
9. Updates to organizational names and references
10. Changed NTDC, NFS(P), CFNES to HMCS Venture throughout.
11. Removal of mentions of RADHAZ – replaced with RF Safety
12. Removal of mentions of ECM – replaced with EA
13. Replaced PO 002 title in chapter 2 to the actual one – Manage Navigation systems
14. Replaced PO 003 title in chapter 2 to the actual one – Manage Communications Systems
15. Fixed PO 005 Performance – changed Above water to Under water
16. Corrected many random line breaks that were adding extra pages
17. Removal of LHR/Operation Honour question – it is no longer being tested on during the board
18. Removed some unused abbreviations from abbreviations table
19. Replaced W Eng Tech (Acoustic) with (Sonar)
20. Edited EO 002.10 to reflect that the board needs to be about navigation suite, not weapons
21. Added in EO 003.02 a TP about security concerns for in service comm systems
22. Edited wording on most OJPR wording when talking about CSE rounds to reflect “all related CSE rounds” to clear up confusion
23. Edited nearly all EOs to change the environment from a boardroom to classroom. We are not teaching in a boardroom
24. Updated some wording to weapons certification
25. Changed TPs in Fire Control System to reflect “in service FC systems” rather then specific systems. Time remains the same
26. “Discuss considerations for CSE rounds” changed to “Discuss considerations for all related CSE rounds” to ensure that all cse rounds are covered for all kit, without implicitly saying each rounds
27. Removed “consecutively” from wording in regards to standing watches for OJPR

Change Tracking number:

Trg Mgr: Lt(N) Newcombe

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. This annex is to be completed at a later date

3. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

4. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

5. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

7. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

8. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

9. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

10. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

11. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

13. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Costing information was not present in original QSP. To be completed.

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Subtotal						
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
Subtotal						
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
	Total Estimated Costs					

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

Allocation of Training Time	Minutes
Apps Course	46200
NEC Course	10760
MWBC	2000
DRMIS	2800
OOD Practical	4400
Sub-total	**66160**
Total mins/400 = # days	
Total Training Days	**165.4**

OPTIMUM TRAINING SCHEDULE

Week 1 **Time**	**Day 1**	**Day 2**	**Day 3**	**Day4**	**Day 5**
Period 1 **0800-0850**	**EO/EC /PC/Admin** Title(s) / Activity				

Week 1 **Time**	**Day 1**	**Day 2**	**Day 3**	**Day4**	**Day 5**
Period 2 **0900-0950**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 3 **1000-1050**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 4 **1100-1150**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 5 **1300-1350**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 6 **1400-1450**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 7 **1500-1550**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 8 **1600-1650**	**EO/EC /PC/Admin** Title(s) / Activity				

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
EC	EO 001.01 – Control Systems	To evaluate candidate's progress in: Defining terms used in controls; and Explaining basic control system concepts.	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help.
		To evaluate candidate's progress in the theory of Control Systems.	Written exam	On completion of EO 001.01, para 7 (1)	50 min.	None.	No textbooks. No notes. No help.	Pass mark: 60% 5% of module EC written test mark	
EC	EO 001.02 – Drive Systems	To evaluate candidate's progress in: Defining terms used in fluid power; Explaining basic fluid power concepts; and Calculating quantities used in fluid power.	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60%, 5% of module formative mark	Trainees will be required to seek out help.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		To evaluate candidate's progress in the theory of Control Systems.	Written exam	On completion of EO 001.02, para 7 (1)	50 min.	None.	No textbooks. No notes. No help.	Pass mark: 60% 5% of module EC written test mark	
CEC 001	EOs 001.01 001.02 - Control and Drive Systems Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory of Control and Drive Systems	Written Exam	On completion of EO 001.01 and EO 001.02	110 min.	None.	No textbooks. No notes. No help.	Pass mark: 50% 30% CEC 001	Trainees will require a re-test. TRB upon failure of re-test
EC	EO 001.03 – Naval Gun Systems	To evaluate candidate's progress in: Defining terms used in Naval Gun Systems; and Explaining basic gun system concepts.	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 3.33% of module formative mark	Trainees will be required to seek out help.
		To evaluate candidate's progress in the theory of	Written Exam	On completion of EO 001.03	50 min.	None.	No textbooks. No notes.	Pass mark: 60% 3.33% of module EC	

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		Naval Gun Systems.					No help.	written test mark	
EC	EO 001.04 – Naval Missile Systems	To evaluate candidate's progress in: Defining terms used in Naval Missile Systems; and Explaining basic missile system concepts.	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 3.33% of module formative mark	Trainees will be required to seek out help.
		To evaluate candidate's progress in the theory of Naval Missile Systems	Written Exam	On completion of EO 001.04	50 min.	None.	No textbooks. No notes. No help.	Pass mark: 60% 3.33% of module EC written test mark	
EC	EO 001.05 – Underwater Systems	To evaluate candidate's progress in:	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 3.33% of module	Trainees will be required to seek out help.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		Defining terms used in Underwater Weapons Systems; and Explaining basic Underwater Weapons Systems concepts.						formative mark	
		To evaluate candidate's progress in the theory Underwater Systems	Written Exam	On completion of EO 001.05	110 min.	None.	No textbooks. No notes. No help.	Pass mark: 60% 3.33% of module EC written test mark	
CEC 002	EOs 001.03 001.04 001.05 Naval Weapon Systems Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory of Naval Weapon Systems.	Written Exam	On completion of: EO 001.03; EO 001.04; and EO 001.05.	110 min.	None.	No textbooks. No notes. No help.	Pass mark: 50% 30% CEC 002	Trainees will require a re-test. TRB upon failure of retest

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
PO 001								Pass mark: 60% 30% CEC 001 + 30% CEC 002 + 20% ECs written test + 20% Formative Work	
EC	EOs 002.01 002.02 002.03 002.04 002.05 002.06 002.07 002.08 Navigation Systems	To evaluate candidate's progress in: Defining terms and explaining basic concepts relating to: Inertial Navigation Systems; Global Positioning Systems;	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 30% of module formative mark	Trainees will be required to seek out help

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		Speed Logs; Navigation Data Distribution Systems; Electronic Chart Systems; Echo Sounders; Stabilized Reference Systems; and TACAN Systems.							
CEC 003	EOs 002.01 002.02 002.03 002.04 002.05 002.06 002.07 002.08 Navigation System Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory and application of Navigation Systems.	Written Exam	On completion of all EOs	110 min.	None.	No textbooks. No notes. No help.	Pass mark: 50% 70% CEC 003	Trainees will require a re-test. TRB upon failure of retest

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
PO 002								Pass mark: 60% 70% CEC 003 + 30% Formative work	
EC	EO 004.01 – Electromagnetics	To evaluate candidate's progress in: Defining terms used in Electromagnetism and EMI; Explaining basic concepts in Electromagnetism; and Solving problems in electromagnetism.	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help.
		To evaluate candidate's progress in the theory	Written Exam	On completion of EO 004.01	50 min.	Calculator.	No textbooks. No notes.	Pass mark: 60%	

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		Electromagnetics consistent within NCS environment.					No help.	5% of module EC written test mark	
EC	EO 004.02 – Antenna Systems	To evaluate candidate's progress in: Defining terms used in Antenna systems; Explaining basic antenna system concepts; and Solving Antenna system problems.	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help.
		To evaluate candidate's progress in the theory Antenna Systems consistent within NCS environment.	Written Exam	On completion of EO 004.02	50 min.	Calculator.	No textbooks. No notes. No help.	Pass mark: 60% 5% of module EC written test mark	
CEC 005	EOs 004.01 004.02	The NCS ENG shall demonstrate a thorough	Written Exam	On completion of EO 004.01	110 min	Calculator	No textbooks. No notes.	Pass mark: 50%	Trainees will require a re-test.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
	Electromagnetics and Antenna Systems Critical EC	understanding of the theory and application of Electromagnetics and Antenna Systems.		and EO 004.02			No help.	30% CEC 005	TRB upon failure of retest
EC	EO 004.03 – RADAR Systems	To evaluate candidate's progress in: Defining terms used in RADAR systems; Explaining basic; RADAR system concepts; and Solving RADAR problems.	Assignment(s); Laboratories;	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help.
		To evaluate candidate's progress in the theory RADAR Systems consistent within NCS environment.	Written Exam	On completion of EO 004.03	110 min.	Calculator.	No textbooks. No notes. No help.	Pass mark: 60% 5% of module EC written test mark	

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
CEC 006	EOs 004.03 004.04 004.05 004.06 004.07 RADAR and Fire Control Systems Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory and application of RADAR and Fire Control Systems.	Written Exam	EO 004.03 EO 004.04 EO 004.05 EO 004.06 EO 004.07	110 min	Calculator.	No textbooks. No notes. No help.	Pass mark: 50% 30% CEC 006	Trainees will require a re-test. TRB upon failure of retest
PO 004								Pass mark: 60% 30% CEC 005 + 30% CEC 006 + 20% EC written test + 20% Formative work	
EC	EO 005.01 – Digital Signal Processing	To evaluate candidate's progress in:	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 10% of module	Trainees will be required to seek out help.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		Defining terms used in Digital Signal Processing; Explaining basic digital signal processing concepts; and Solving digital signal processing problems.						formative mark	
		To evaluate candidate's progress in the theory of Digital Signal Processing.	Written Exam	On completion of EO 005.01	50 min.	Calculator.	No textbooks. No notes. No help.	Pass mark: 60% 10% of module EC written test mark	
EC	EOs 005.02 005.03 005.04 SONAR Systems	To evaluate candidate's progress in: Defining terms used in SONAR systems; Explaining basic SONAR system	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 10% of module formative mark	Trainees will be required to seek out help.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		concepts; and Solving SONAR problems.							
		To evaluate candidate's progress in the theory of SONAR Systems.	Written Exam	On completion of: EO 005.02 EO 005.03 EO 005.04	50 min.	Calculator.	No textbooks. No notes. No help.	Pass mark: 60% 10% of module EC written test mark	
CEC 007	EOs 005.01 005.02 005.03 005.04 DSP and SONAR Systems Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory and application of DSP and SONAR Systems.	Written Exam	EO 005.01 EO 005.02 EO 005.03 EO 005.04	110 min	Calculator.	No textbooks. No notes. No help.	Pass mark: 50% 60% CEC 007	Trainees will require a re-test. TRB upon failure of retest
PO 005								Pass mark: 60% 60% CEC 007 + 20% ECs written test +	

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
								20% Formative work	
EC	EOs 006.01 006.02 006.03 006.04 Command and Control Systems	To evaluate candidate's progress in: Defining terms used in Command and Control Systems; and Explaining basic Command and Control System concepts Defining terms used in Command and Control Systems Hardware; and Explaining Command and Control System Hardware Defining terms used in Command and Control Systems Software; and	Assignment(s); Laboratories.	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 30% of module formative mark	Trainees will be required to seek out help.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		Explaining Command and Control System Software concepts; and Defining terms used in Command and Control Systems Operations.							
CEC 008	EOs 006.01 006.02 006.03 006.04 Command and Control Systems Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory and application of CCS Systems.	Written Exam	On completion of EO 006.01 EO 006.02 EO 006.03 EO 006.04	110 min.	None.	No textbooks. No notes. No help.	Pass mark: 50% 70% CEC 008	Trainees will require a re-test. TRB upon failure of retest
PO 006								Pass mark: 60% 70% CEC 008 +	

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
								30% Formative work	
EC	EO 007.01 – Analog Electronics	To evaluate candidate's progress in: Defining terms used in Analog Electronics; Explaining Analog Electronics concepts; And Solving analog circuits.	Assignment(s); Laboratories	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help
		To evaluate candidate's progress in the theory of Analog Electronics.	Written Exam	Mid-point of EO 007.01	50 min.	Calculator	No textbooks. No notes. No help.	Pass mark: 60% 5% of module formative mark	
CEC 009	EO 007.01 – Analog Electronics Critical EC	The NCS ENG shall demonstrate a thorough	Written Exam	On completion of EO 007.01	50 min.	Calculator	No textbooks. No notes.	Pass mark: 50% 15% CEC 009	Trainees will require a re-test.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		understanding of the theory and application of Analogue Electronics					No help.		TRB upon failure of retest
EC	EO 007.02 – Digital Electronics	To evaluate candidate's progress in: Defining terms used in Digital Electronics; Explaining Digital Electronics concepts; And Solving digital circuits.	Assignment(s); Laboratories	At the end of a phase of instruction	N/A	Textbook	No plagiarism	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help
		To evaluate candidate's progress in the theory of Digital Electronics.	Written Exam	Mid-point of EO 007.02	50 min.	Calculator	No textbooks. No notes. No help.	Pass mark: 60% 5% of module formative mark	

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
CEC 010	EO 007.02 – Digital Electronics Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory and application Digital Electronics.	Written Exam	On completion of EO 007.02	50 min.	Calculator	No textbooks . No notes. No help.	Pass mark: 50% 15% CEC 010	Trainees will require a re-test. TRB upon failure of retest
EC	EO 007.03 - Computer Networks	To evaluate candidate's progress in: Defining terms used in Computer Networks; and Explaining computer networking concepts.	Assignment(s); Laboratories	At the end of a phase of instruction	N/A	Textbook; Personal computer; Network.	No plagiarism	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help
		To evaluate candidate's progress in the theory of Computer Networks.	Written Exam	Mid-point of EO 007.03	50 min.	Calculator	No textbooks. No notes. No help.	Pass mark: 60% 5% of module formative mark	

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
CEC 011	EO 007.03 - Computer Networks Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory and application of Computer Network Systems.	Written Exam	On completion of EO 007.03	50 min.	Calculator	No textbooks. No notes. No help.	Pass mark: 50% 15% CEC 011	Trainees will require a re-test. TRB upon failure of retest
EC	EO 007.04 – Tactical Data Link	To evaluate candidate's progress in: Defining terms used in Tactical Data Link Systems; and Explaining Tactical Data Link Systems concepts.	Assignment(s); Laboratories	At the end of a phase of instruction	N/A	Textbook	None.	Pass mark: 60% 5% of module formative mark	Trainees will be required to seek out help
		To evaluate candidate's progress in the theory of Tactical Data Links.	Written Exam	On completion of EO 007.04	50 min.	None.	No textbooks. No notes. No help.	Pass mark: 60% 5% of module formative mark	
CEC 012	EO 007.04 – Tactical Data Link	The NCS ENG shall	Written Exam	On completion of	50 min.	None.	No textbooks.	Pass mark: 50%	Trainees will require a re-test.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
	Critical EC	demonstrate a thorough understanding of the theory and application of Tactical Data Link Systems.		EO 007.04			No notes. No help.	15% CEC 012	TRB upon failure of retest
PO 007								Pass mark: 60% 15% CEC 009 + 15% CEC 010 + 15% CEC 011 + 15% CEC 012 + 20% EC written test + 20% Formative Work	
EC	EOs 008.01 008.02	To evaluate candidate's progress in:	Assignment(s); Laboratories.	At the end of a phase of	N/A	Textbook	No plagiarism	Pass mark: 60%	Trainees will be

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
	008.03 Electronic Warfare Systems	Defining terms; explaining basic concepts; and calculating quantities relating to: EW Systems; ESM Systems; and EA Systems.		instruction				.	required to seek out help
CEC 013	EOs 008.01 008.02 008.03 Electronic Warfare Systems Critical EC	The NCS ENG shall demonstrate a thorough understanding of the theory and application of: EW Fundamentals; ESM Systems; and EA Systems.	Written Exam	On completion of all EOs in scope.	110 min.	Calculator.	No textbooks. No notes. No help.	Pass mark: 50% 70% CEC 013	Trainees will require a re-test. TRB upon failure of retest

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
PO 008								Pass mark: 60% 70% CEC 013 + 30% Formative work	
EC	EOs 009.01 009.02 009.03 009.04 009.05 Ammunition	To evaluate candidate's progress in: Identifying ammunition types; Locating appropriate references; and Filling in example documentation.	Formative	At the end of a phase of instruction	N/A	Textbook	None.	Pass mark: 60% 30% of module formative mark	Trainees will be required to seek out help.
CEC 014	EOs 009.01 009.02 009.03 009.04 009.05	The NCS ENG shall demonstrate a thorough understanding of the	Written Exam	On completion of all EOs in scope.	50 min.	Textbook. References. Personal Computer.	No help.	Pass mark: 50% 70% CEC 014	Trainees will require a re-test. TRB upon failure of retest

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
	Ammunition Critical EC	shipboard Ammunition Operations.							
PO 009								Pass mark: 60% 70% CEC 014 + 30% Formative work	
	EO 010.01 – Signature Management	To ensure students understand the Ship's Signature Management.	Oral confirmation questions	At the end of a phase of instruction	N/A	N/A	None	N/A	None
	EO 011.01 - Risk Management Programs	To ensure students understand Shipboard Risk Management Programs.	Oral confirmation questions	At the end of a phase of instruction	N/A	N/A	None	N/A	None
	EO 011.02 - Radiation Hazard	To ensure students understand the Shipboard Non-Ionizing Radiation Hazard Safety Program.	Oral confirmation questions	At the end of a phase of instruction	N/A	N/A	None	N/A	None

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
	EO 011.03 - HAZMAT	Practice Hazmat clean up procedures.	Practical test. Students will clean up a variety of Hazmat spills.	At the end of a phase of instruction	240 min.	Damage Control School trainer and associated resources	With supervision	N/A	No pass mark is assigned. Trainees will receive corrections throughout practice.
	EO 012.01 - ERT	To ensure students understand the Emergency Response Team Operations.	Oral confirmation questions	At the end of a phase of instruction	N/A	N/A	None	N/A	None
NCSEAC final mark will be calculated as an average of all module final marks									
Entire QSP	PC 001 NCS BOQ Qualification Board	To evaluate the candidate's technical ability to perform as a Naval Combat Systems Engineer.	Qualification Board (oral board)	On completion of OJPR and NEC OJPR	Up to 300 min, plus up to 60 min prep time	Boardroom. Board Chair (Cdr. NAV ENG Cdr) Examining Officer (LCdr NCS ENG) Board Member (Lt(N) NCS ENG)	No textbooks. No notes. No help. May be stopped if candidate does not achieve at least 60% on board	Pass mark: The candidate shall achieve a minimum overall average mark of 60%, including a minimum of 60% for each of questions	Trainees will require a re-board. TRB upon failure of reboard.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
							questions 1 and 2.	1 and 2 (critical questions).	

ANNEX I – BOARD ASSESSMENT GUIDE

The purpose of the NCS Eng BOQ Board is not simply to assess technical competency, but the ability to assess, analyze and convey complex technical information verbally, demonstrating comprehension beyond memorization**.** Simply put; the candidate shall demonstrate the ability to stand and deliver, conveying engineering fundamentals, technical and military knowledge in a clear, concise, and professional manner.

Technical Efficacy, Demonstrating Technical Knowledge, Technical Currency and being a Technical Advisor are all Facets of PaCE, for all occupations, and the board is to assess these facets with the understanding that if the candidate can learn and understand the complex systems of one class of ship, they can do the same on a new class of ships.

The board consists of five questions total, with the following recommended schedule and timings,

Q	Portion	Time (mins)	Morning Board 0800 Start	Afternoon Board 1300 Start
	Candidate presentation preparation	60	0700-0800	1200-1300
	Candidate and Board Member Introduction	10	0800-0810	1300-1310
1	Major Suite and Major System	60	0810-0910	1310-1410
2	Major System	-	-	-
3	Layered Defense	30	0910-0940	1410-1440
	Break	20	0940-1000	1440-1500
4	Technical Theory	30	1000-1030	1500-1530
5	Divisional Administration	25	1030-1055	1530-1555
	Board Deliberation / Marking	15	1055-1110	1555-1610
	Initial Feedback and Completion	5	1110-1115	1610-1615

INTRODUCTION

1. As noted in Chapter 3, this is the only Performance Check for this qualification. The candidate must successfully challenge the NCS Basic Officer Qualification Board to achieve the AIRX qualification.

EXAMINATION BOARD GUIDANCE

2. **Pre-Examination Preparation.** The candidate shall have 60 minutes prior to the start of the board to draw Q1 and Q2.
3. **Marking.** There shall be a single Scoring/Assessment Checklist/Report submitted for the board by the Chairperson. The Chairperson shall make and record an overall Pass or Fail assessment for the Oral Board, as well as any board comments. The Chairperson shall also provide a numerical score for each questions in accordance with the marking guide provided for the purposes of establishing ranking for NTO Award Selection Boards.
4. The oral examination board is scored with 80% of the mark for each question consisting of correct content and 20% on presentation. The final pass mark for the board is 60%, constituting a combined average/weight of all numerically scored questions. The candidates must attain a passing mark in both Q1 and Q2 separately in order to continue to the next question. Failing either Q1 or Q2 constitutes an immediate failure of the Oral Board.
5. **General Expectations.** All answers to questions are expected to be:
   - *Clear* (e.g. uses appropriate level of language, technical terminology, no colloquialisms/slang, etc.)
   - *Concise* (e.g. answers are brief yet complete)
   - *Logical* (structures briefing points sequentially)
   - *Confident* (answered with conviction)
   - *Accurate* (correctness of response aligns with RCN doctrine and policy)
6. **Board Timings.** The board should last no more than 3 hours from when the board members introduce themselves to the end of candidate debrief.

7. **Board Members.** The responsibilities of each members of the board are as follows:
   a) <u>Chairperson</u>: Provide a fair and unbiased assessment of the candidate's knowledge and competencies IAW the qualification.
   b) <u>Assessing Member</u>: Augment the chairperson's technical knowledge and advice in scoring each questions; asking relevant questions to the candidate, offering own assessment and recommendations to Chairperson.
   c) <u>Examining Officer</u>: Advise the chairperson as whether the candidate is meeting, exceeding or failing to achieve the standard expected for qualification; asking relevant questions to the candidate, offering own assessment and recommendations to Chairperson.

d) Divisional Representative: Clarify discrepancies between the board and the candidate, and advocate for the candidate as required.

8. **Package Preparation.** The question bank for the board will be maintained by HMCS Venture however, the convening authorities will have access IOT compose boards.

Question 1 | Major Suite Drawing & Presentation

60 minutes for question 1 and 2 preparation. 60 Minutes for presentation and follow-on questions

The candidate shall create from memory, a detailed and well laid out drawing of the suite as fitted on their ship. The drawing shall be neat, clear and of sufficient detail to explain the system's operation. The candidate should use supplemental drawings for major components, and use appropriate colours, and scale. The candidate will be marked on the following topics:

- **Purpose.** What is the Design Intent of the system and how does the system contribute to float, move, and fight.

- **Theory**. The candidate shall provide some background on the physics and engineering fundamentals of the system. Include supplemental sketches of the principles of operation, engineering application, building on their post secondary education, NEI, SEMC and CSE Applications training.

- **Overview.** Provide a general overview of the suite and orient the drawing. What spaces is the system in. Is there a port and starboard, primary, and secondary system? Was this part of an upgrade, refit, or as fitted when commissioned? Is this system the same for all ships within class? What colors are used in the drawing and why, and what legend and symbology is used? Standard symbols for pumps, valves, orifice plates, heat exchangers, reservoirs, sumps, etc. shall be used.

- **Major Components.** List and discuss at a minimum, the operation, location, function, and characteristics of all components, building on the theory. Discuss each component's capabilities, limitations, parameters, operating ranges, alarms, and trips.

- **Functional Description.** Describe the normal operation of the complete system. A suggestion is to trace out the flow path for fluid and electrical current. The functional description should start with an operator at a console; entering a value or change in state and end with actuation, feedback, and monitoring of the system.

- **Modes of Operation**. Is it always in operation, only at sea, or only in specific drive modes or missions? Primary and alternate operation modes shall be discussed for systems with local, remote, or primary, backup and emergency operation modes.

- **Interdependencies.** All systems rely on other systems for input, output and interoperability. The relationship, reliance, and resiliency of the topic system to other systems shall be discussed.

- **Trials and Maintenance.** The candidate should be aware of the relevant trials and trial programs for major systems. They are not expected to know the detailed trial agendas or frequency, however a general awareness of who is involved, the duration of the trial, as well as operational and staffing considerations are expected. CSE maintenance routines should be discussed when applicable. On questioning, the candidate should be able to briefly discuss operational limitations, technical bulletins and fleet advisories.

- **Redundancies & Survivability.** The primary, secondary, and tertiary backups and alternate modes shall be discussed. The candidate should cover concepts such as graceful degradation versus catastrophic failure for the system, battle over-ride, lockouts, emergency trips, safety interlocks, and run time on backup systems.

- **Conclusion.** The candidate should review the major system parameters with a brief reiteration of the systems overall purpose and function and provide a formal end and yield to the for further questions.

Follow-up questions shall be limited to the topics above and these additional topics.

- **Preventative Maintenance and Corrective Maintenance routines.** The candidate is not expected to know specific routine frequencies or a complete list of maintenance and repair procedures but should know what is involved in common maintenance operations.

- **Common Failure Modes.** Systems have known weak spots or failure points and quirks of operation. The candidate should be able to discuss the system beyond the academic knowledge of the CFTOs and demonstrate firsthand knowledge of common issues, fleet wide issues or historical failure trends from across the fleet.

- **Technical Administration.** The candidate should be able to speak about the system's history including topical Engineering Changes, refits, upgrades, or lifecycle management. Who the local FMF engineering subject matter experts might be. Relevant portions of SEMS, NEM, NaMMS, and other guiding documents shall be discussed if applicable. The candidate is not expected to have more than a very basic awareness of MBS, Risk Assessment and impact on operational capabilities and failures of this system as they relate to an OPDEF or capabilities. Technical Investigations, Technical Bulletins, operating restrictions, and power reductions could also be discussed here.

- **Trials, specific operating considerations.** The candidate should be aware of the relevant trials and trial programs for major systems. They are not expected

to know the detailed trial agendas or frequency, however a general awareness of who is involved, the duration of the trial, as well as operational and staffing considerations are expected. On questioning, the candidate should be able to briefly discuss operational limitations, technical bulletins and fleet advisories and general directives on best practices.

- **Shore support agencies, Design Authority and Operating Authority, Specialty Tools, And Test Equipment.** The candidate should be able to discuss Field Service Representative support if applicable and mention any specialty tools or procedures for equipment such as the removal of soft patches, crane access, unique Lock Out Tag Out considerations and safety requirements or other considerations, such as docking a ship for access and repair.

Question 2 | Major System

Time concurrent with question 1

The candidate shall create, from memory, a detailed and well laid out drawing of the system as fitted on their ship. The drawing shall be neat, clear and of sufficient detail to explain the system's operation. The candidate should use supplemental drawings for major components, and use an appropriate colour, and scale. The candidate will be marked on the following topics:

- **Purpose.** What is the design intent of the system and how does the system contribute to float, move, and fight.

- **Theory**. The candidate shall provide some background on the physics and engineering fundamentals of the system.

- **Overview.** Provide a general overview of the system and orient the drawing. What spaces is the system in. Is there a port and starboard, primary, and secondary system? Was this part of an upgrade, refit, or as fitted when commissioned? Is this system the same for all ships within class? What colors are used in the drawing and why, and what legend and symbology is used?

- **Major Components.** List and discuss the operation, location, function, and characteristics of all components, building on the theory. Discuss each component's capabilities, limitations, parameters, operating ranges, alarms, and trips.

- **Functional Description.** Describe the normal operation of the complete system. A suggestion is to trace out the flow path for fluid and electrical current. The functional description should start with an operator at a console; entering a value or change in state and end with actuation, feedback, and monitoring of the system. Primary and alternate operation modes shall be discussed for

systems with local, remote, or primary, backup and emergency operation modes.

- **Modes of Operation**. Is it always in operation, only at sea, or only in specific drive modes or missions?

- **Interdependencies.** All systems rely on other systems for input, output and interoperability. The relationship, reliance, and resiliency of the topic system to other systems shall be discussed. Should include all systems that utilise the subject system, and all systems for which the subject system relies upon.

- **Redundancies & Survivability.** The primary, secondary, and tertiary backups and alternate modes shall be discussed. The candidate should talk about graceful versus catastrophic degradation. The candidate should cover battle over-ride, lockouts, emergency trips, safety interlocks, run time on backup systems.

- **Conclusion.** The candidate should review the major system parameters with a brief reiteration of the systems overall purpose and function and provide a formal end and yield to the board members for further questions.

Question 3 | Layered Defense & Threat Response

10 minutes to prepare and 20 minutes to present

This question is to assess the candidate's understanding of the Detect -to-Engage sequence. The candidate will be expected to produce a layered defence/zones of coverage diagram and describe the detect-to-engage sequence for the scenario including the actions of the ship and Ops Team

A threat scenario will be given where an initial contact is made. The detect-to-engage sequence, the relevant equipment capabilities, and sequence of events is to be covered. Rather than individual systems or suites, the candidate is encouraged to think of the engagement in terms of a warfighting vessel and the role these systems and suites play in the entire engagement.

Question 4 | Technical Theory

10 minutes to prepare and 20 minutes to present

The candidate should explain and build upon fundamental scientific concepts that were taught during the CSE applications phase as well as their previous engineering education and experience. The use of equations, basic scientific principles and laws should be explained. Potential topics include deriving and explaining sonar equation, signal processing, synchro signal processing and electromagnetic wave generation concepts.

Question 5 | Divisional Administration

5 minutes preparation, 15 minutes delivery, 5 minutes for follow up questions.

The divisional administration question examines a candidate's ability to act in a wide array of staff, administrative, technical and leadership roles across the entire CAF. The candidate shall demonstrate a working and operationally functional level of knowledge in military administration.

Question five requires a working knowledge of the DAODs, common QR&Os, NCM and Officer Career Management, disciplinary issues, and PaCE. More specific to the RCN and NTO Occupation, the candidate shall be able to discuss Naval specific doctrine, policy, and orders and what is typically covered within CANFORGENs, NAVORDS, SSOs, CSOs, COTMs, EOTMs and ship specific SOPs.

The candidate is not expected to know specific details of specific policies, but rather an emphasis is placed on a broad awareness of what the relevant references are, where to find, and how to properly execute policy to the benefit of members. Ideally a candidate can share their own experience in the topic stemming from professional experience as link in the divisional system chain within the department onboard a ship.

NCS Eng Basic Occupation Qualification (BOQ) Board Marking Sheet	
Candidate:	**Ship:**
Qualification: AIRX	**Date:**
Chairperson:	**Examining Officer:**
Assessing Member:	**Divisional Representative:**

Instructions:
Each question is scored out of 10 for content and presentation which are weighted to create a score out of 10 for each question.[1] Each question is then weighted to create an overall board mark.[2] The candidate shall achieve a minimum overall board mark of 60%, including a minimum of 60% for questions 1 and 2 on both content and presentation (critical questions).

Score	**Content**	Score	**Presentation**[3]
0	Unable to answer any part of the question or even begin to discuss the topic.	0	Many major deficiencies
1	Unable to communicate a basic level of technical knowledge of engineering principles.	1	Some major deficiencies
2	Able to answer minor parts of the main question but unable to answer follow up questions.	2	Many intermediate deficiencies
3	Displays a very low level of technical knowledge of engineering principles.	3	Some intermediate deficiencies
4	Able to answer parts of the main question and a limited number of follow up questions displaying only a restricted level of technical knowledge.	4	Many minor deficiencies
5	Requires assistance answering the main question and requires prodding to answer follow up questions.	5	Some minor deficiencies
6	Displays an acceptable level of technical knowledge of engineering principles and administration requirements.	6	Acceptable
7	Displays an above average level of technical knowledge of engineering principles.	7	Good
8	Displays a high level of technical knowledge of engineering principles.	8	Very Good
9	Generates discussion through the display of excellent technical knowledge of engineering principles.	9	Excellent
10	Displays an exceptional level of technical knowledge answering the main question and follow up questions well beyond the expected level.	10	Exceptional

Note 1	Qualification	Content	Presentation
	NEI	0.8	0.2
	BOQ	0.8	0.2
	HOD	0.6	0.4

Note 2	Question	Weight
	1	5
	2	5
	3	4
	4	3

Note 3	Presentation is an aggregate evaluation of the following: - Dress and Deportment - Professionalism - Use of Proper Terminology - Interaction with Board Members - Eye Contact - Pace and Duration - Volume and Clarity of Voice - Body Language and Movement - Confidence and Conviction in Answers - Organization of the Presentation and Answers to Questions - Clarity of Board Drawings - Use of the Board/Drawing to Assist in Explanations.

NCS Eng Basic Occupation Qualification (BOQ) Board Marking Sheet					
CANDIDATE:			**SHIP:**		
QUALIFICATION: AIRX			**DATE:**		
QUESTION	**CONTENT (80%)**	**PRESENTATION (20%)**	**QUESTION TOTAL**	**WEIGHT FACTOR**	**WEIGHTED SCORE**
1				**5**	
2				**5**	
3				**4**	
4				**3**	

MARK (%): $\frac{\sum Weighted\ Score}{2} =$

RESULT: **PASS / FAIL** (circle one)

COMMENTS:

NOTE: Name and signature are mandatory

Name of Board Chair

Signature

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 2.0	1 Jul 23	New OJPR based on new RCN QSP template

Name and Initials	Rank	Service Number

Date Training Started	
Date of Completion	

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. The NCS Eng trainee shall have a minimum of 40 “sea days” during the afloat period. A “sea day” is considered a day when the ship is away from home-port and under power. The CSEO must take appropriate actions to ensure the trainee receives appropriate opportunities to achieve the minimum number of “sea days” within the designated afloat period.

4. The expected time to complete the OJT is 9 months. If a trainee is required to attend career coursing during the afloat period, the duration of the afloat period is to be extended by the period of time spent on this course. If the trainee has not completed Officer of the Day (OOD) training an additional 3 months will be allotted to complete the OOD OJPR. The candidate’s Commanding Officer (CO) may grant an extension IAW NAVORD 9-28.

5. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

6. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

7. The NCS Eng BOQ Afloat OJPR is designed to be completed concurrently with the NEC Afloat OJPR.

PROGRESS REVIEW

8. Progress review shall be carried out quarterly by the Divisional Officer and HOD.

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
PO 001: Manage Naval Weapon Systems					
EO 001.06	57mm Main Gun (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	AHWCS (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	Mk 48 GMVLS ESSM (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	Close-In Weapon System (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	Mk 46 Torpedo System	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Naval Remote Weapon Station (NRWS)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Additional Requirements	Stand eight (8) watches with W Eng Tech (Armament)			CSEO
		Submit Technical Defect Report (on a single system)			CSEO
PO 002: Manage Navigation Systems					
EO 002.11	Navigation Data Distribution Suite	Observe Operation and Maintenance			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
	(Major System)	Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	INS (MK-49 RLGN)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Global Positioning Systems (GPS)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Doppler Speed Log	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Electronic Chart System (SHINNADS)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Echo Sounder	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Stabilized Horizon Reference System (SHRS)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Automatic Identification System (AIS)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	MET System	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Additional Requirements	Stand four (4) bridge watches with OOW			CSEO
		Submit Technical Defect Report (on a single system)			CSEO
PO 003: Manage Communication Systems					

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
EO 003.07	External Communications Suite (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	SHINCOM	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Trunked Radio	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Main Broadcast	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	HF Tx	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	HF Rx	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	UHF LOS	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	UHF DAMA SATCOM	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Emergency HF Communications	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Emergency UHF Communications	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
	Sub-Net Relay (SNR)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Message Handling and Distribution System (MHDS)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	EMCON Panel	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Helicopter Communications	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	PRF-117F Havequick Radios	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Additional Requirements	Stand six (6) watches with W Eng Tech (Communications)			CSEO
		Stand two (2) watches with NavComms			CSEO
		Submit Technical Defect Report (on a single system			CSEO
PO 004: Manage Above Water Surveillance Systems					
EO 004.08	Radar Suite (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	Fire Control System (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	Navigation Radars	Observe Operation and Maintenance			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
		Walkthrough with Maintainer			CSEO
	Medium Range Search Radar - SG-180	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Long Range Search Radar - SMART-S	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	MK XII IFF	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Optical Sensors (SEOSS)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Additional Requirements	Stand eight (8) watches with W Eng Tech (RADAR)			CSEO
		Stand two (2) watches with SAC/RT			CSEO
		Stand two (2) Watches with SWC/FCS			CSEO
		Submit Technical Defect Report (on a single system			CSEO
PO 005: Manage Underwater Sensor Systems					
EO 005.05	Canadian Towed Array System (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	Hull Mounted Sonar (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
	NIXIE-25A	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Sonobuoy Processing System	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Under Water Telephone	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Bathythermograph System	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Deck Cranes (M111-K2 Hydraulic Cranes)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Torpedo Handling and Storage system (to include the bridge crane)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Additional Requirements	Stand six (6) watches with W Eng Tech (Sonar)			CSEO
		Stand two (2) watches with SONAR Ops			CSEO
		Submit Technical Defect Report (on a single system)			CSEO
PO 006: Manage Command and Control Systems					
EO 006.05	CMS 330 (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	Additional Requirements	Stand one (1) watch with TOW			CSEO
		Stand two (2) watches with ORO / ORS			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
		Submit Technical Defect Report (on a single system			CSEO
PO 007: Manage Data Processing Systems					
EO 007.05	Naval Information System (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	SHF SATCOM (NMT)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Commercial SATCOM (Fleet Broad Band)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	SecLAN (CSNI, CMFP, etc.)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Tactical Data Link Systems (Link 11 / Link 16 / Link 22)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Additional Requirements	Submit Technical Defect Report (on a single system			CSEO
PO 008: Manage Electronic Warfare Systems					
EO 008.04	Electronic Warfare Suite (Major System)	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
		Practice Board			CSEO
	ELISRA	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	SEA SEARCH	Observe Operation and Maintenance			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
		Walkthrough with Maintainer			CSEO
	SRD-504	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	RAMSES	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	MASS	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	SIRIUS	Observe Operation and Maintenance			CSEO
		Walkthrough with Maintainer			CSEO
	Additional Requirements	Stand two (2) watches with EWS			CSEO
		Submit Technical Defect Report (on a single system			CSEO
PO 009: Manage Ship's Ammunition					
EO 009.06	Ammunition Operations	Explain Ammunition Operations			CSEO
		Participate in Ammunition Evolutions			CSEO
PO 010: Manage Ship's Signature					
EO 010.02	Signature Management	Explain Signature Management			CSEO
		Support the Unit Signature Management Officer			CSEO
PO 011: Manage Shipboard Hazards					
EO 011.04	Shipboard Hazards	Explain RF Safety			CSEO
		Explain HAZMAT			CSEO
		Observe HAZMAT Cleanup			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
PO 012: Direct Emergency Response Team (ERT) Operations					
EO 011.04	ERT Operations	Explain ERT			CSEO
		Assist planning an ERT Exercise			CSEO
		Participate in an ERT Exercise			CSEO
		Observe ERT Exercise from HQ1 (DC Perspective)			CSEO
		Participate in Hot Wash-Ups			CSEO
Miscellaneous Requirements					
N/A	Layered Defence	Anti-Air Warfare Discussion with ORO/Director			ORO
		Anti-Air Warfare Practice Board with CSEO			CSEO
		Anti-Surface Warfare Discussion with ORO/Director			ORO
		Anti-Surface Warfare Practice Board with CSEO			CSEO
		Anti-Submarine Warfare Discussion with ORO/Director			ORO
		Anti-Submarine Warfare Practice Board with CSEO			CSEO
N/A	10 Minor System Practice Boards Any 10 of the 44 listed, with preference to those listed below: • Bathythermograph • Deck Cranes (Zodiac crane/RHIB Davit) • Stabilized Horizon Reference System (SHRS) • NIXIE-25A • PACOM (Main Broadcast) • SEOSS • SHINCOM • SPS	Practice Board One			CSEO
		Practice Board Two			CSEO
		Practice Board Three			CSEO
		Practice Board Four			CSEO
		Practice Board Five			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
	• Mk 46 Torpedo System (To include the torpedo handling system) • UWT • Remote Weapon System Note: Candidate still responsible for presentation of any of the minor systems for the final board. Candidate to practice/complete remaining systems on own or with peers.	Practice Board Six			CSEO
		Practice Board Seven			CSEO
		Practice Board Eight			CSEO
		Practice Board Nine			CSEO
		Practice Board Ten			CSEO
NA	Trainee is required to draw each major system and have HOD verify the drawing matches the ship HOD will sign off once every drawing is complete	Drawing Package			CSEO
N/A	Additional Requirements	40 Sea Days			CSEO

OJPR Reviewed By OJT Supervisor (Rank, Position, Name)	Signature	Date
Head of Department		

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Head of Department	Date
Executive Officer	Date
Commanding Officer	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the NCS Eng BOQ QSP and is AIRX qualified.

Commanding Officer	Date

PO 001 - MANAGE NAVAL WEAPON SYSTEMS

EO 001.06 Support Naval Weapons Systems onboard HMC Ships

OJPR Standard: The NCS ENG shall support Naval Weapons Systems as fitted onboard HMC Ships by completing the following:

a. Observe operation and maintenance of shipboard weapon systems to the satisfaction of the CSEO;

1) Operations:
   i. Pre and post firing checks;
   ii. Pre-fire brief;
   iii. Loading and unloading ammunition; and
   iv. Observe all Weapon Systems firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire crew).
2) Tests and Trials:
   i. INO (NDT for guns);
   ii. Initial Velocity (IV) determination;
   iii. Gun Dynamic Tests;
   iv. Weapon Certification;
   v. Missile Readiness Checks/Missile Readiness Inspection;
   vi. Combat System Alignment Tests (CSAT); and
   vii. Torpedo Readiness Checks.
3) Maintenance:
   i. Preventive;
   ii. Corrective;
4) Flash-up/shutdown;
5) Pre-sail checks; and
6) Trouble shooting.

b. Conduct a walkthrough of the weapon systems with the senior maintainer;

1) Identify all system components;
2) Discuss considerations for all related CSE rounds; and
3) Identify MSE systems considerations.

c. Deliver a brief on the major weapon system diagrams to the satisfaction of the CSEO;

1) System description;
2) System overview;
3) Major system components;
4) Operating modes;

5) Capabilities and limitations;
6) Interfaces;
7) Redundancies/survivability;
8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components;
11) Modes of operation;
12) System redundancies;
13) System diagnostic capabilities/tools;
14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Manning requirements;
19) Operating policy and procedures; and
20) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional Requirements:
1) Stand eight (8) watches with the W ENG Tech (Armament); and
2) Submit a Technical Defect Report on a major problem with one of the fitted Naval Weapon Systems selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 002 - MANAGE NAVIGATION SYSTEMS

EO 001.06 Support Navigation Systems onboard HMC Ships

OJPR Standard: The NCS ENG shall support Navigation Systems as fitted onboard by completing the following:

a. Observe operation and maintenance of shipboard navigation systems to the satisfaction of the CSEO;

1) Operations:
   i. Blind pilotage;
   ii. Flying Ops;
   iii. Entering and Exiting Harbour; and
   iv. Weapon Firings.
2) Tests and Trials:
3) Maintenance:
   i. Preventive;
   ii. Corrective;
4) Flash-up/shutdown;
5) Pre-sail checks; and
6) Trouble shooting.

b. Conduct a walk-through of the navigation systems with the senior maintainer;

1) Identify all system components;
2) Discuss considerations for all related CSE rounds; and
3) Identify MSE systems considerations.

c. Deliver a brief on the navigation system suite to the satisfaction of the CSEO;

1) System description;
2) System overview;
3) Major system components;
4) Operating modes;
5) Capabilities and limitations;
6) Interfaces;
7) Redundancies/survivability;
8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components/signal flow/software;
11) Modes of operation;
12) System redundancies;
13) System diagnostic capabilities/tools;

14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Manning requirements;
19) Operating policy and procedures; and
20) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional Requirements:
1) Stand four (4) watches with OOW; and
2) Submit a Technical Defect Report on a major problem with one of the fitted Navigation Systems selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 003 - MANAGE COMMUNICATION SYSTEMS

EO 003.07 Support Communications Systems onboard HMC Ships

OJPR Standard: The NCS ENG shall support Communications Systems as fitted onboard by completing the following:

a. Observe operation and maintenance of shipboard Communication Systems to the satisfaction of the CSEO;

1) Operations:
   i. Set up and monitor/change Comm. plans;
   ii. Set up a radio circuit manually;
   iii. Set up and monitor/change SHINCOM programmable nets;
   iv. Input and transmit/receive messages; and
   v. Crypto change.
2) Tests and Trials.
3) Maintenance:
   i. Preventive;
   ii. Corrective;
4) Flash-up/shutdown;
5) Pre-sail checks; and
6) Trouble shooting.

b. Conduct a walk-through of the communication systems with the senior maintainer;

1) Identify all system components;
2) Discuss considerations for all related CSE rounds; and
3) Identify MSE systems considerations.

c. Deliver a brief on the External Communication Suite to the satisfaction of the CSEO;

1) System description;
2) System overview;
3) Major system components;
4) Operating modes;
5) Capabilities and limitations;
6) Interfaces;
7) Redundancies/survivability;
8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components/signal flow/software;
11) Modes of operation;

12) System redundancies;
13) System diagnostic capabilities/tools;
14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Manning requirement;
19) Operating policy and procedures; and
20) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional requirements:
1) Stand six (6) watches with W ENG TECH (COMM);
2) Stand two (2) watches in CCR with NAVCOMM; and
3) Submit a Technical Defect Report on a major problem with one of the fitted communication systems selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 004 - MANAGE ABOVE WATER SURVEILLANCE SYSTEMS

EO 004.08 Support Naval Above Water Surveillance Systems onboard HMC Ships

OJPR Standard: The NCS ENG shall support Above Water Surveillance Systems as fitted on board by completing the following:

a. Observe operation and maintenance of shipboard Above Water Surveillance Systems to the satisfaction of the CSEO;
   1) Operations:
      i. TRACKEX, ADEX, Encounter Ex;
      ii. Blind pilotage (Ops Room, Bridge);
      iii. Manual Operation of the RDU;
      iv. Manual Interrogation of IFF;
      v. Manipulate RADAR background (AIAT, MIAT, AICAT);
      vi. Manipulate RADAR RCP's;
      vii. SEOSS/OPS;
      viii. Pre and post firing checks;
      ix. Pre-fire brief; and
      x. Observe al Weapons Systems firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire crew).
   2) Tests and Trials:
      i. Naval Electronics Sensor Test Range (NESTRA, NESTRP);
      ii. RADAR Min/Max Ranges;
      iii. Daily/Periodic System Operational (DSOT/PSOT); and
      iv. Weapon Systems certification; and
      v. Missile Readiness Checks/Missile Readiness Inspections
   3) Maintenance:
      i. Preventive;
      ii. Corrective;
   4) Flash-up/shutdown;
   5) Pre-sail Checks; and
   6) Trouble Shooting.

b. Conduct a walk-through of the Above Water Surveillance Systems with the senior maintainer;
   1) Identify all system components;
   2) Discuss considerations for all related CSE rounds; and
   3) Identify MSE systems considerations.

c. Deliver a brief on the Radar Suite, and the Fire Control System to the satisfaction of the CSEO;
   1) System description;

2) System overview;
3) Major system components;
4) Operating modes;
5) Capabilities and limitations;
6) Interfaces;
7) Redundancies/survivability;
8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components/signal flow/software;
11) Modes of operation;
12) System redundancies;
13) System diagnostic capabilities/tools;
14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Manning requirement;
19) Application of the RADAR equation to particular RADARs;
20) Operating policy and procedures; and
21) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional Requirements:
1) Stand eight (8) watches with W.ENG TECH (RADAR);
2) Stand two (2) watches with SAC;
3) Stand two (2) watches with SWC / FCS; and
4) Submit a Technical Defect Report on a major problem with one of the fitted Above Water Surveillance Systems selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 005 - MANAGE UNDER WATER SENSOR SYSTEMS

EO 005.05 Support Underwater Sensor Systems onboard HMC Ships

<u>OJPR Standard:</u> The NCS ENG shall support Under Water Sensor Systems as fitted on board by completing the following:

a. Observe operation and maintenance of shipboard Underwater Sensor Systems to the satisfaction of the CSEO;

1) Operations:
   i. Launches and recoveries of CANTASS/NIXIE/Sonobuoy/Bathythermograph/Probes;
   ii. CASEX;
   iii. Use of countermeasures (configuration/set up/use);
   iv. Launch of Bathythermograph and create SVP on AESS;
   v. Sonobuoy Launch;
   vi. Raising/lowering of transducer (flooding of dome);
   vii. Creation of Sound Velocity Profile (SVP) of AESS; and
   viii. Crane Operations, to include both the Bridge crane as part of the torpedo handling and storage system, and the Torpedo Davit Loading Crane, which also handles the RHIB/Zodiac
2) Tests and Trials:
   i. Winch Certification;
   ii. Wet end;
   iii. Dry end
   iv. SONAR Performance Figure;
   v. C5 Dome Inspection;
   vi. Ranging; and
   vii. Lifting appliance certifications.
3) Maintenance:
   i. Preventive;
      a) OCCAP for CANTASS;
      b) Vibration Analysis (VA); and
      c) ISOPAR-M Replacement;
   ii. Corrective;
      a) Replace Towed Array Module.
4) Flash-up/Shutdown;
5) Pre-sail checks;
6) Trouble shooting; and
7) Emergency operation

b. Conduct a walk-through of the Underwater Sensor Systems with the senior maintainer;

1) Identify all system components;
2) Discuss considerations for all related CSE rounds; and
3) Identify MSE systems considerations.

c. Deliver a brief on the CANTASS and HMS-510 systems to the satisfaction of the CSEO;

1) System description;
2) System overview;
3) Major system components;
4) Operating modes;
5) Capabilities and limitations;
6) Interfaces;
7) Redundancies/survivability;
8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components/signal flow/software;
11) Modes of operation;
12) System redundancies;
13) System diagnostic capabilities/tools;
14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Manning requirements;
19) Operating policy and procedures; and
20) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional Requirements:
1) Stand six (6) watches with W.ENG TECH (SONAR);
2) Stand two (2) watches with SONAR OPS; and
3) Submit a Technical Defect Report on a major problem with one of the fitted Underwater Sensor Systems selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 006 - MANAGE COMMAND AND CONTROL SYSTEMS

EO 006.05 Support CCS onboard HMC Ships

OJPR Standard: The NCS ENG shall support CCS as fitted on board by completing the following:

a. Observe operation and maintenance of shipboard CMS 330 system to the satisfaction of the CSEO;

1) Operations:
   i. All Ops Room personnel at all positions;
   ii. Watch turnover of Ops Room personnel;
   iii. Manipulate SMC pages;
   iv. Detect to engage sequence as SWC; and
   v. History recording/data collection;
2) Tests and Trials:
   i. On-line/offline diagnostics;
   ii. Weapons Certification;
   iii. Combat System Alignment Test (CSAT);
   iv. Combat System Interface Test (CSIT); and
   v. Combat System Operability Test (CSOT);
3) Maintenance:
   i. Preventive; and
   ii. Corrective.
4) Flash-up/Shutdown;
5) Pre-sail checks; and
6) Trouble shooting.

b. Conduct a walk-through of CMS 330 system with the senior maintainer;

1) Identify all system components;
2) Discuss considerations for all related CSE rounds; and
3) Identify MSE systems considerations.

c. Deliver a brief on the CMS 330 system to the satisfaction of the CSEO;

1) System description;
2) System overview;
3) Major system components;
4) Operating modes;
5) Capabilities and limitations;
6) Interfaces;
7) Redundancies/survivability;

8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components/signal flow/software;
11) Modes of operation;
12) System redundancies;
13) System diagnostic capabilities/tools;
14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Manning requirements;
19) Operating policy and procedures; and
20) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional Requirements:
1) Stand one (1) watch with the CSE TOW;
2) Stand two (2) watches with the ORO/ORS; and
3) Submit a Technical Defect Report on a major problem with the CMS 330 selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 007 - MANAGE DATA PROCESSING SYSTEMS

EO 007.05 Support Data Processing Systems onboard HMC Ships

OJPR Standard: The NCS ENG shall support Data Processing Systems as fitted on board by completing the following:

a. Observe operation and maintenance of shipboard Data Processing Systems to the satisfaction of the CSEO;

1) Operations:
   i. Setup of network;
   ii. Monitor network;
   iii. Administration of network; and
   iv. Usage of network.
2) Tests and Trials:
   i. POFA;
   ii. Ping; and
   iii. Technical COMSEC Inspection.
3) Maintenance:
   i. Preventive; and
   ii. Corrective.
4) Flash-up/shutdown;
5) Pre-sail checks; and
6) Trouble shooting.

b. Conduct a walk-through of Data Processing Systems with the senior maintainer;

1) Identify all system components;
2) Discuss considerations for all related CSE rounds; and
3) Identify MSE systems considerations.

c. Deliver a brief on the Naval Information System (NavIS) to the satisfaction of the CSEO;

1) System description;
2) System overview;
3) Major system components;
4) Operating modes;
5) Capabilities and limitations;
6) Interfaces;
7) Redundancies/survivability;
8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components/signal flow/software;

11) Modes of operation;
12) System redundancies;
13) System diagnostic capabilities/tools;
14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Operating policy and procedures; and
19) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional Requirements:

1) Submit a Technical Defect Report on a major problem with the Data Processing System selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 008 - MANAGE ELECTRONIC WARFARE SYSTEMS

EO 008.04 Support EW Systems onboard HMC Ships

OJPR Standard: The NCS ENG shall support EW Systems as fitted on board by completing the following:

a. Observe operation and maintenance of shipboard EW Suite to the satisfaction of the CSEO;

1) Operations:
   i. EA Pre and post firing checks;
   ii. EA Pre-fire brief;
   iii. Loading and unloading ammunition;
   iv. Observe EA System firings (simulated or actual) from all operational/maintenance positions (i.e. Bridge, Ops, Local, Misfire crew);
   v. Jamming/Deception;
   vi. ADEX, EWEX, Encounter EX;
   vii. Direction finding; and
   viii. Communications Intercept.

2) Tests and Trials:
   i. NESTR(A/P) Trial; and
   ii. Daily/Periodic System Operational Tests (DSOT/PSOT).
3) Maintenance:
   i. Preventive; and
   ii. Corrective.
4) Flash-up/Shutdown;
5) Pre-sail checks; and
6) Trouble shooting.

b. Conduct a walk-through of all EW Systems with the senior maintainer;

1) Identify all system components;
2) Discuss considerations for all applicable CSE rounds; and
3) Identify MSE systems considerations.

c. Deliver a brief on the EW Suite to the satisfaction of the CSEO;

1) System description;
2) System overview;
3) Major system components;
4) Operating modes;
5) Capabilities and limitations;

6) Interfaces;
7) Redundancies/survivability;
8) Combat/CSE liaison considerations;
9) Performance characteristics, capabilities and limitations;
10) System components/signal flow/software;
11) Modes of operation;
12) System redundancies;
13) System diagnostic capabilities/tools;
14) Interfacing/integration with other systems;
15) Maintenance requirements;
16) System trials/alignments;
17) Safety considerations;
18) Manning requirement;
19) Operating policy and procedures; and
20) CSEO should confirm knowledge by requesting impact of component failure (Components to be selected by CSEO).

d. Additional Requirements:
1) Stand two (2) watches with the EWS;
2) Submit a Technical Defect Report on a major problem with an EW Suite selected by the CSEO:
   i. Identification and discussion of the problem or nature of the failure;
   ii. Operational impact of the problem or failure;
   iii. Options to resolve the problem; and
   iv. Actual/recommended solution to the problem.

PO 009 - MANAGE AMMUNITION

EO 009.06 Participate in Ammunition Operations onboard HMC Ships

<u>OJPR Standard:</u> The NCS ENG shall participate in Ammunition Operations on board by completing the following:

a. Explain Ammunition Operations to include;
   1) Ammunition Depot; and
   2) Formation Ammunition Technical Authority; and
      i. Formation Ammunition Technical Officer (FATO); and
      ii. Formation Ammunition Inspector.
   3) Fleet/Formation Staff;
   4) Ammunition Orders; and
   5) Ammunition Check-list.

b. Participate in Ammunition Evolutions;
   1) Embarking/Disembarking and internal transfer of Ammunition;
      i. Alongside in Home port;
         i. Operational necessity; and
         ii. Jetty limits.
      ii. Alongside in Foreign Port; and
         i. Local Regulations; and
         ii. Ammunition Depot or buoy.
      iii. At sea ammo transfers.
   2) Ammunition Rounds and Inspections;
      i. Annual;
      ii. Adjacent compartment rounds;
      iii. Magazine rounds; and
      iv. Free from ammunition certification.
   3) Respond to Ammunition Accidents and Incidents under supervision;
      i. Definitions
         a. Accident;
         b. Incident;
         c. Defect; and
         d. Malfunction.
      ii. Reporting and investigation requirements;
      iii. Damaged ammunition;
         a. Retaining;
         b. Jettisoning; and
         c. Required reports.
   4) Manage work in proximity to ammunition / magazines under supervision.
      i. Docking Requirements;
         a. Routine; and
         b. Emergency.

ii. Forces protection ammunition;

iii. Burning and welding;

a. Definition; and

b. Max working distance.

iv. Waivers

PO 010 - MANAGE SHIP'S SIGNATURE

EO 010.02 Support Ship's Signature Management onboard HMC Ships

<u>OJPR Standard:</u> The NCS ENG shall support Ship's Signature Management on board by completing the following:

a. Explain Signature Management;
    1) Explain Signature Management Participate in the Signature Management Organization (SMO);
        i. Review and interpret trial results;
        ii. Observe and participate in the planning and conduct of ship's signature ranging;
        iii. Identify and understand Ship's Signature Logs;
        iv. Advise Command on ship's signature;
        v. Participate in the conduct of Noise Rounds; and
        vi. Participate in completion of ship's signature documentation (SOPs/reports).
    2) Demonstrate a detailed understanding of Signature Management to the satisfaction of the CSEO;
        i. Role and responsibility of the Signature Management Organization;
        ii. Types of signatures;
        iii. Signature reduction methods; and
        iv. Range resources co-ordination.
b. Support the Unit Signature Management Officer in daily activities.

PO 011 - MANAGE SHIPBOARD HAZARDS

EO 011.04 Participate in Risk Management Programs onboard HMC Ships

<u>OJPR Standard:</u> The NCS ENG shall, to the satisfaction of the CSEO and under supervision:

a. Explain RF Safety;
    1) Present general RF Safety brief to ship's company;
    2) Present a man aloft brief to designated personnel;
    3) Co-ordinate man aloft/RF Safety evolutions during a DMP/SWP/DWP;
    4) Conduct leak tests; and
    5) Observe RF Safety surveys.
b. Explain HAZMAT and HAZMAT Spill Cleanup Procedures;
c. Explain the difference of CSE and MSE responsibilities in regards to spill responses;
d. Support URFSO in execution of daily duties;
e. Observe the spill clean-up team.
    1) Dressing area;
    2) Spill area; and
    3) Command area.

PO 012 - DIRECT EMERGENCY RESPONSE TEAM (ERT) OPERATIONS

EO 012.02 Participate in ERT Operations onboard HMC Ships

<u>OJPR Standard:</u> The NCS ENG shall participate in ERT Operations on board by completing the following:

a. Explain ERT procedures and policies to include;
   1) Key entities and components to include;
      i. ERT Ops;
      ii. ERT HQ Communicator;
      iii. FWD or AFT ERT (communicator and roundsman)
      iv. HQ1; and
      v. Command (Bridge).

   2) Operational description of the 60 and 400Hz power;
   3) Typical distribution from Main Switchboards to equipment;
   4) Modes of operation of the 60 and 400Hz power;
   5) Control and Monitoring;
   6) System Inter-relationships; and
   7) Redundancy and survivability of the 60 and 400Hz power;

b. Assist with the planning of an ERT exercise
c. Participate in an ERT Exercise;
d. Observe an ERT exercise from HQ1 perspective;
e. Participate in a Hot Wash-up
f. Draw system level diagrams for both the 60 Hz and 400 Hz Power Generation and Distribution Systems (PG&D) and verbally explain them to the CSEO.

ANNEX K – TT PLAN

With the new format of the QSP, this information was not found in the previous version of the QSP. The table has been left here to be completed at a later date.

EO	**COA 1** **TEL Delivery**	**COA 2** **Interim Delivery**	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

2. With the new format of the QSP, this information was not found in the previous QSP. The qualification standard record of decisions has some specific about the TPWB.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version: 2.0	
Date: 1 July 2023	**Who Initiated Change:** HMCS Venture
Requirement for the Amendment: New format of QSP	
Description of Changes: QSP has been edited to new format. Removal of obsolete equipment out of POs, changing wording to encompass future equipment without needing a new update 1. Add a section to the signature requirements where the candidate is to conduct ten minor system practice boards from the list of all minor systems. These should be selected from the list of minor systems that are not subsystems of a major system or suite, i.e. from the following list: Bathy, Cranes, Horizon Bars (SHRS), NIXIE, PACOM (Main Broadcast), SEOSS, SHINCOM, SPS, Torpedoes, UWT, and Remote Weapon System. System selection is, however, at the discretion of the HOD. The package will be amended to add ten CSEO minor system board presentation signature blocks along with a notation that the candidate is still responsible for the presentation of any of the minor systems on the final board. The candidate is to practice/complete the remaining minor systems on their own or with their peers. Substantiation: Verbal comments were received from board chair Cdr that minor system knowledge and corresponding board questions seemed weak across all trainees. This was also noted in some Fleet Pre-Boards. In previous QSPs, all minor systems had “draw and present” components in addition to walkthroughs. The practice boards were removed, likely due to the cumbersome nature of presenting boards to the HOD for approximately 44 minor systems. This will return oversight of the minor system presentations to the HOD while not being overly cumbersome, and ensure that the candidate is aware of the minor system requirement.	

2. Add ORO/Director (discussion) and CSEO (presentation) signature blocks for each of the three Layered Defence scenarios. The lack of signatures, including practice boards with the HoD, has led many candidates to inadequately prepare for this question. The candidates will spend a watch or have a discussion with the ORO/Director specifically to discuss the detect-to-engage sequence from an operator's perspective, as well as conduct a practice board for each of three scenarios. No impact on training time should be expected as the candidates are already required to spend time with the operators and are expected to know and present layered defence. This will verify that the candidate has received feedback from the ORO/Director and CSEO regarding layered defence, increasing the opportunity for success at the final board.
3. Amend Course Completion page (p. 299 of QSP) to require signatures recommending procession to final board from the HoD, XO, and CO. The page currently has a signature block for the HoD only.
4. Amend Qualification section (p. 12 of QSP) to remove requirement for Unit RF Safety Officer Qualification (115868). The requirement is for a Head of Department to hold this qualification, so it would be more appropriate to have the candidate acquire this qualification during, or just immediately prior to, their Assistant Head of Department (A/HoD) phase. It is not required of a Phase VI candidate at this point in his/her career and so is not an effective use of the candidate's time. FW UNIT RADI
5. Amend the board process to include the trainees drawing the boards, and include these as an OJPR check. The change which removed the requirements to have the trainees draw the major systems on the final qualification board has been noted by senior engineers over the last several years to have a possible negative impact on the amount of knowledge the trainee has at the qualification board.
6. Clarify the difference between Deck Cranes and the Torpedo Handling and Storage System, which includes Bridge Cranes. It has been noted on boards that there is confusion between the two.
7. Removed TACAN, EO 002.10, re-allocated time throughout PO 002 EOs

Small Changes

8. Updates to NAVORDs/NPTORDs/others to reflect new references throughout document
9. Updates to organizational names and references
10. Changed NTDC, NFS(P), CFNES to HMCS Venture throughout.
11. Removal of mentions of RADHAZ – replaced with RF Safety

12. Removal of mentions of ECM – replaced with EA
13. Replaced PO 002 title in chapter 2 to the actual one – Manage Navigation systems
14. Replaced PO 003 title in chapter 2 to the actual one – Manage Communications Systems
15. Fixed PO 005 Performance – changed Above water to Under water
16. Corrected many random line breaks that were adding extra pages
17. Removal of LHR/Operation Honour question – it is no longer being tested on during the board
18. Removed some unused abbreviations from abbreviations table
19. Replaced W Eng Tech (Acoustic) with (Sonar)
20. Edited EO 002.10 to reflect that the board needs to be about navigation suite, not weapons
21. Added in EO 003.02 a TP about security concerns for in service comm systems
22. Edited wording on most OJPR wording when talking about CSE rounds to reflect “all related CSE rounds” to clear up confusion
23. Edited nearly all EOs to change the environment from a boardroom to classroom. We are not teaching in a boardroom
24. Updated some wording to weapons certification
25. Changed TPs in Fire Control System to reflect “in service FC systems” rather then specific systems. Time remains the same
26. “Discuss considerations for CSE rounds” changed to “Discuss considerations for all related CSE rounds” to ensure that all cse rounds are covered for all kit, without implicitly saying each rounds
27. Removed “consecutively” from wording in regards to standing watches for OJPR

Change Tracking number:

Trg Mgr: Lt(N) Newcombe

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

ACRONYM / ABBREVIATION	TERM/DEFINTIONS
ALARP	As Low as Reasonably Practicable
BOI	Board of Inquiry
BOQ	Basic Officer Qualification
CANAVMOD	Canadian Naval Modification
CCA	Curriculum Control Authority
CFQAR	Canadian Forces Quality Assurance
CRCN	Commander Royal Canadian Navy
CRR	Combat Readiness Requirements
CNPTG	Commander Naval Personnel Training Group
DGMEPM	Director General Materiel Engineering Procurement Management
DGMFD	Director General Maritime Force Development
DMCM	Director Maritime Class Management
DMFM	Director Maritime Fleet Management
DMP	Directed Maintenance Period
DMS	Defence Management System
DMSS	Director Maritime Ship Support
DRDC	Defence Research Development Canada
DRMIS	Defence Resource Management Information System
DRP	Directed Repair Period
DSP	Defence Services Program
EC	Engineering Change
EHM	Equipment Health Monitoring
EMREQ	Emergency Requirement
ESL	Equipment Support Lists
EWP	Extended Work Period
FEL	Fitted Equipment List
FMF	Fleet Maintenance Facility
FSR	Fleet Support Representative
FTA	Fleet Technical Authority
HPR	High Priority Request
HRI	Harbour Readiness Inspections
ILS	Integrated Logistic Support
LCMM	Life Cycle Materiel Manager
MATREQ	Materiel Requirement
MMPT	Main Machinery Power Trial
NaMMs	Naval Materiel Management System

ACRONYM / ABBREVIATION	TERM/DEFINTIONS
NAVORDs	Naval Orders
NDID	Naval Defence Identification Directory
NEC	Naval Engineering Common
NEM	Naval Engineering Manual
NESTR	Naval Electronics Test Range
NETE	Naval Engineering Test Establishment
NFP	Notice For Power
NFS(A / P)	Naval Fleet School (Atlantic / Pacific)
NPTG	Naval Personnel Training Group
OCCAP	Oil and Coolant Condition Analysis Program
OEM	Original Equipment Manufacturer
OJPR	On-Job Performance Requirements
OPDEF	Operational Deficiencies
OPEVAL	Operational Evaluation
OPSKED	Operational Schedule
OT Div	Officer Training Division
PAYE	Payable at Year End
PEL	Periodic Engineering Letter
PIF	Pre Installation Failure
PLAR	Prior Learning Assessment and Recognition
PWGSC	Public Works Government Services Canada
QETE	Quality Engineering Test Establishment
QSP	Qualification Standard and Plan
RRI	Reduced Readiness Inspections
RSP	Readiness and Sustainment Program
SEMS	Safety and Environmental Management System
SHIPALT	Ship Alteration
SHIPMOD	Ship Modification
SI	Summary Investigation
SITREP	Situation Report
SMP	Self Maintenance Period
SOCD	Statement of Capability Deficiency
SWP	Short Work Period
TDP	Technical Data Package
TECHVAL	Technical Evaluation
TI	Technical Investigation
TRANREQ	Transfer Request
TRB	Training Review Board
TRP	Tiered Readiness Program
UCR	Unsatisfactory Condition Report
VA	Vibration Analysis

ANNEX N – DESIGNATED TRAINING ESTABLISHMENTS AND EOs SPLIT BY MITE

MITE	109385	109386	109387
TE	HMCS Venture / NFS(A)	HMC Ships	HMCS Venture / NFS(A)
Phase	Ashore Phase	Afloat Phase	Qualification Board
PO	EOs covered in each MITE		
001	001.01, 001.02, 001.03, 001.04, 001.05	001.06	
002	002.01, 002.02, 002.03, 002.04, 002.05, 002.06, 002.07, 002.08, 002.09	002.10	
003	003.01, 003.02, 003.03, 003.04, 003.05, 003.06	003.07	
004	004.01, 004.02, 004.03, 004.04, 004.05, 004.06	004.07	
005	005.01, 005.02, 005.03, 005.04	005.05	
006	006.01, 006.02, 006.03, 006.04	006.05	
007	007.01, 007.02, 007.03, 007.04	007.05	
008	008.01, 008.02, 008.03	008.04	
009	009.01, 009.02, 009.03, 009.04, 009.05	009.06	
010	010.01	010.02	
011	011.01, 011.02, 011.03	011.04	
012	012.01	012.02	
PCs			001